

Topology Optimization of Flow Configurations Operating in the Turbulent Regime

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Preface

This thesis marks the completion of my Master of Science in Mechanical Engineering at KU Leuven. What began as a compact mathematical problem in turbulent-flow topology optimization quickly became a broader exercise in connecting models, numerical methods, and engineering judgment. The work moved between deriving weak forms, building FEniCS solvers, preparing wall-resolved meshes, running optimization cases, and checking the resulting geometries with independent CFD. Many of the most useful lessons came from failed runs, unexpected discrepancies, and the effort required to make each modeling assumption explicit.

I am sincerely grateful to my supervisors, Prof. Dr. Ir. Niels Horsten and Prof. Dr. Ir. Maarten Blommaert, for giving me the opportunity to work on this topic and for their guidance throughout the project. Their feedback helped keep the research direction focused and the conclusions grounded. I am especially thankful to my assistant-supervisor, Ir. Esteban Foronda Obando, for the frequent technical discussions, practical FEniCS advice, and patient support while working through difficult numerical issues.

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Abstract

As compact electronic, automotive, and energy systems become more power-dense, thermal-management hardware increasingly depends on low-loss internal-flow paths. This thesis addresses the hydraulic part of that problem: turbulent flow-only topology optimization for bends, U-bends, and diffusers before conjugate heat transfer is added. A density-based FEniCS framework uses Brinkman penalization, wall-resolved Spalart-Allmaras (SA) turbulence modeling, a relaxed Eikonal wall-distance calculation, and a frozen-turbulence adjoint that fixes eddy-viscosity and wall-distance fields during sensitivity assembly.

The work first defines the modeling limits of SA for curved and separated internal flows, demonstrating its viability for moderate curvature while acknowledging its deterioration in strongly curved or separated regimes. The custom FEniCS-SA solver is successfully verified against Ansys Fluent for a turbulent channel and a two-dimensional U-bend, showing excellent cross-code agreement in macroscopic pressure drop and local velocity profiles.

When evaluated on turbulent benchmark geometries, the framework demonstrates that the reliability of an SA-optimized design depends fundamentally on the target flow physics. The pipe bend represents a highly successful transfer case for attached, moderately curved flow. The extracted turbulent geometry remains robust under independent Ansys CFD validation; replacing the SA closure with the more separation-sensitive SST $k-\omega$ model yields only modest variations in predicted losses. Under identical turbulent operating conditions, the turbulent-optimized pipe bend decisively outperforms a laminar-optimized reference, reducing pressure drop by 72.0% and viscous dissipation by 47.0%.

Conversely, the U-bend and diffuser benchmarks expose the physical limitations of relying on a one-equation closure for strongly separated flows. Rather than producing universally superior layouts, the optimizer heavily tunes these geometries to the SA model's specific, overly forgiving prediction of adverse pressure gradients and recirculation. Consequently, cross-examining these extracted designs with SST $k-\omega$ reveals significant performance deterioration and ranking reversals compared to laminar baselines.

Overall, frozen-SA topology optimization yields credible, low-loss turbulent design candidates for attached or moderately curved flows, provided assumptions are explicit and final geometries are independently validated. Primary limitations remain the frozen adjoint formulation, two-dimensional constraints, diffuse-to-sharp geometry extraction, and strong closure sensitivity in separated regimes.

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List of Abbreviations and Symbols

Abbreviations

AMG	Algebraic Multigrid
BiCGSTAB	Bi-Conjugate Gradient Stabilized method
CAD	Computer-Aided Design
CEV	Constant Eddy Viscosity
CFD	Computational Fluid Dynamics
CG1	Continuous Galerkin (Linear Basis Functions)
CG2	Continuous Galerkin (Quadratic Basis Functions)
CHT	Conjugate Heat Transfer
DG0	Discontinuous Galerkin (Piecewise Constant Elements)
DNS	Direct Numerical Simulation
DOF	Degree of Freedom
FEM	Finite Element Method
FD	Finite Difference
FVM	Finite Volume Method
GMRES	Generalized Minimal Residual method
IPCS	Incremental Pressure Correction Scheme
ILU	Incomplete LU factorization
MMA	Method of Moving Asymptotes
MUMPS	Multifrontal Massively Parallel Sparse direct Solver
NS-SA	Navier-Stokes-Spalart-Allmaras coupling
PDE	Partial Differential Equation
RAMP	Rational Approximation of Material Properties
RANS	Reynolds-Averaged Navier-Stokes
SA	Spalart-Allmaras
SIMP	Solid Isotropic Material with Penalization
SNES	Scalable Nonlinear Equations Solver
SUPG	Streamline-Upwind/Petrov-Galerkin stabilization
SST $k - \omega$	Shear Stress Transport $k - \omega$ model
TMR	NASA Turbulence Modeling Resource
TO	Topology Optimization
UFL	Unified Form Language
VMFL	Ansys Fluent verification benchmark case
XFEM	Extended Finite Element Method

Symbols

Fluid Dynamics & Turbulence

D, D_h	Pipe diameter and hydraulic diameter [m]
De	Dean number [-]
H, L	Channel height and characteristic length [m]
k	Turbulent kinetic energy [m^2/s^2]
$p, \bar{p}, p'$	Instantaneous, mean, and fluctuating static pressure [Pa]
ΔP	Macroscopic total pressure drop [Pa]
R_c	Radius of curvature [m]
Re	Bulk Reynolds number [-]
$S, \tilde{S}$	Scalar vorticity magnitude and modified SA vorticity magnitude [1/s]
t	Time / Pseudo-time [s]
τ_w	Wall shear stress [Pa]
$\mathbf{u}, \bar{u}_i$	Velocity vector / Mean velocity component [m/s]
u'_i	Fluctuating velocity component [m/s]
$U_{\text{ref}}, U_{\text{bulk}}, U_{\text{max}}$	Reference, bulk, and maximum velocity [m/s]
u^+	Non-dimensional mean velocity [-]
u_τ	Friction velocity [m/s]
y, d_w	Distance to the nearest wall [m]
y^+	Non-dimensional wall distance [-]
C_f	Skin-friction coefficient [-]
c_{b1}, c_{b2}, c_{v1}	Spalart-Allmaras production and viscosity-ratio constants [-]
c_{w1}, c_{w2}, c_{w3}	Spalart-Allmaras wall-destruction constants [-]
f_{v1}, f_{v2}, f_w	Spalart-Allmaras damping and wall-destruction functions [-]
$r_{\text{SA}}, g_{\text{SA}}$	Spalart-Allmaras wall-destruction auxiliary variables [-]
δ_{ij}	Kronecker delta [-]
μ, μ_t	Dynamic and turbulent eddy viscosity [Pa·s]
ν, ν_t	Kinematic and turbulent eddy kinematic viscosity [m^2/s]
$\tilde{\nu}$	Spalart-Allmaras working variable [m^2/s]
ρ	Fluid density [kg/m^3]
ω	Scalar vorticity in two-dimensional flow [1/s]
Ω_{ij}	Vorticity tensor [1/s]
χ	Spalart-Allmaras viscosity ratio, $\tilde{\nu}/\nu$ [-]
κ	von Kármán constant [-]
σ	Spalart-Allmaras turbulent Prandtl number [-]

Numerical Implementation of the Spalart-Allmaras Model

G, G_0	Relaxed Eikonal field variable and boundary value [1/m]
h, h_{avg}	Finite element cell diameter and average cell size [m]
p_k	Kinematic pressure [m ² /s ²]
s	Arlength coordinate along a profile line [m]
e_i	Pointwise profile error or design-space basis vector, depending on context [-]
E_∞, E_{L_2}	Maximum and relative L_2 profile discrepancies [-]
C_{SUPG}	User-selected SUPG stabilization factor [-]
Δh	Finite-difference design perturbation [-]
$\Delta t_{\tilde{\nu}}$	Pseudo-time step used for optional SA transport damping [s]
S_i^{FD}, S_i^{fr}	Finite-difference and frozen-adjoint sensitivity of sampled design variable i
ε_i^{fr}	Relative frozen-adjoint sensitivity discrepancy [-]
$\ \cdot\ _{L_2}$	L_2 (Euclidean) norm [-]
$[\![\cdot]\!]$	Jump operator across an internal element facet [-]
$\Gamma_{in}, \Gamma_{out}, \Gamma_D$	Inlet, outlet, and profile-extraction boundaries
$\mathbf{n}$	Outward-pointing normal vector [-]
$\mathcal{R}_{\tilde{\nu}}$	Strong residual of the Picard-linearized SA transport equation [-]
T_h, V_h, Q_h	Discrete turbulence, velocity, and pressure finite-element spaces
$\tau_{\tilde{\nu}}$	SUPG stabilization parameter for SA working-variable transport [s]
v, ξ	Test functions for pressure and turbulence equations [-]
$\alpha_u, \alpha_{\tilde{\nu}}$	Relaxation factors for velocity and turbulence [-]
Δt	Pseudo-time step [s]
ϵ_{tol}	Convergence tolerance [-]
ϕ	General scalar field variable [-]
σ_w	Eikonal relaxation constant [-]

Topology Optimization

$\alpha(\gamma)$	Continuous Brinkman penalization term / inverse permeability
$\alpha_{solid}, \alpha_{fluid}$	Maximum solid and minimum fluid Brinkman penalties
α_{dg}	Artificial diffusion scaling factor for DG0 filter [-]
β	Heaviside projection steepness parameter [-]
γ, γ_{raw}	Continuous and raw design variable (1 for fluid, 0 for solid) [-]
$\tilde{\gamma}, \bar{\gamma}$	Filtered and Heaviside-projected design variables [-]
η	Heaviside projection threshold [-]
Π_β	Heaviside projection operator [-]
$\mathcal{H}$	Helmholtz filter operator [-]
M	Active design mask [-]
J	Generic objective function used in the optimization problem [problem dependent]
J_D	Dissipation objective, evaluated from viscous dissipation [W, or W/m for two-dimensional unit-depth cases]
J_p	Average inlet-pressure objective [Pa]
Φ_D	Sharp-geometry viscous dissipation metric [W, or W/m for two-dimensional unit-depth cases]
q	RAMP penalization tuning parameter [-]
r	Continuous Helmholtz filter radius [m]
$r_V^{(k)}$	Volume residual at optimization iteration k [-]
$v_f^{(k)}$	Realized fluid volume fraction at optimization iteration k [-]
$\mathbf{s}, \mathbf{s}^*$	Monolithic forward state and frozen-flow adjoint state vectors
V_f, V_{domain}	Physical fluid volume and total design domain volume [m ³]
V_{mask}	Volume-mask measure [m ³]
V_{max}	Maximum allowable fluid volume fraction constraint [-]
$\Omega, \partial\Omega, \Gamma_{int}$	Computational domain, domain boundary, and internal element facets
Ω_D, Ω_V	Objective-integration and volume-integration regions

Chapter 1

Introduction

1.1 Background and Motivation

Many thermal and energy systems first face a hydraulic design problem: the working fluid has to be routed through a small available volume without wasting too much pumping power. This issue appears in liquid-cooled electronics, battery and power-electronics modules, fuel-cell and electrolyzer channels, compact heat exchangers, and coolant distribution hardware for automotive and aerospace systems. Even when a geometry offers good local heat-transfer potential, it can be unusable if the flow path creates a large pressure drop, poor distribution, or separated regions that undermine the intended operating point.

The broader motivation is thermal-management design, but the model problem treated here is intentionally hydraulic. The thesis does not solve the energy equation, does not perform conjugate heat-transfer optimization, and does not minimize temperature. It instead considers flow-only bends, U-bends, and diffusers, where the relevant behavior is pressure loss, viscous dissipation, separation, and the response of the turbulence closure. Establishing that hydraulic baseline is necessary before the same optimization framework can be trusted for turbulent heat-transfer problems.

Jury question: Why did you leave out heat transfer if the motivation is thermal management?

Answer: Because the hydraulic turbulent TO problem is already the unstable part of the later CHT problem: the solver must handle RANS-SA coupling, evolving walls, Brinkman penalization, and adjoint sensitivities. Adding the energy equation before verifying that baseline would mix thermal errors with unresolved momentum-model errors.

Topology optimization is attractive for this setting because the designer does not prescribe the final channel layout beforehand. The method has become an established computational design approach for fluid and thermal-management applications [26]. For a coupled thermal device, an objective could involve heat-transfer rate, maximum temperature, or thermal resistance. In the present flow-only setting, the objective is instead tied to pressure drop or dissipated mechanical energy. This scope reduction still leaves a difficult problem: in turbulent density-based topology

optimization, the flow solver, wall treatment, turbulence model, geometry representation, and adjoint sensitivity calculation all change together as the density field evolves.

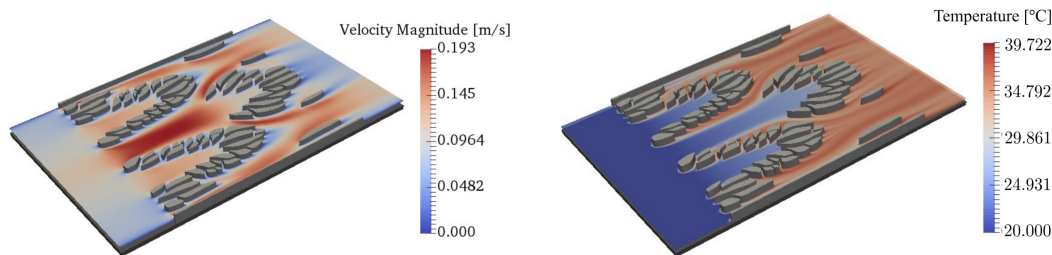


FIGURE 1.1: Topology-optimized heat sink from Barroca et al. [12].

The heat-sink example in Figure 1.1 illustrates the type of hydraulic layout that motivates the present work: the optimized fin structure guides the velocity field while the thermal field responds to that flow distribution. The central gap addressed in this thesis is therefore that density-based fluid topology optimization is far more mature for laminar flows than for turbulent internal flows. Laminar assumptions are attractive because they simplify both the governing equations and the associated adjoint sensitivities, but they are often not representative of compact engineering flow paths operating at industrial Reynolds numbers. In turbulent internal flows, the chosen Reynolds-Averaged Navier-Stokes (RANS) closure, near-wall treatment, and response to curvature or separation can influence the optimized topology itself. Before turbulent conjugate heat-transfer optimization can be treated as a reliable design tool, the underlying turbulent flow-only optimization problem must first be stabilized, verified, validated, and interpreted critically.

1.2 Conceptual Overview of Density-Based Topology Optimization

The distinction between topology optimization and more classical design optimization is important for the turbulent-flow problem. Size optimization changes parameters such as member thicknesses, while shape optimization moves the boundary of an already chosen layout. Both can be effective, but both inherit the connectivity of the initial design. A shape optimizer, for example, can refine an existing bend but cannot freely decide to split one passage into two or create a new opening.

Density-based topology optimization, pioneered by Bendsoe and Kikuchi in 1988 for solid mechanics [14], approaches the problem differently. Instead of moving boundaries, the design space is discretized into a fixed grid of finite elements. The algorithm then determines the optimal distribution of material across this grid. In fluid problems, this is achieved by assigning a continuous design variable, $\gamma \in [0, 1]$, to every element in the domain. In the convention adopted for this thesis, $\gamma = 1$ represents a pure fluid channel, while $\gamma = 0$ represents a solid, impermeable structure.

For gradient-based optimization, the binary fluid–solid choice is relaxed to a continuous field. Intermediate values are then interpreted through a porous-medium model: the momentum equation receives a Brinkman-type resistance whose magnitude depends on the local density [15, 29]. The optimizer updates this field to reduce a chosen objective, such as pressure loss or dissipated power, while satisfying constraints such as a prescribed fluid volume fraction. Figure 1.2 illustrates the typical path from an initially uniform field, through a smooth intermediate design, toward a nearly discrete topology. In this introductory figure, the displayed quantity should be read as the physical projected density used by the flow equations; Chapter 5 later separates this from the raw design variable and the filtered density.

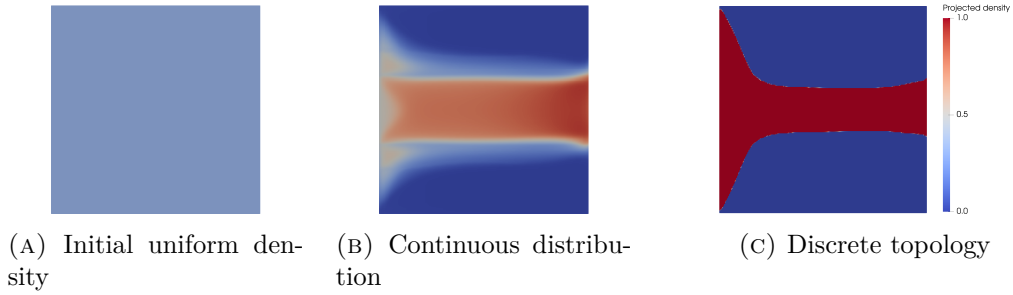


FIGURE 1.2: Density-field evolution during topology optimization.

1.3 From Laminar to Turbulent Fluid Topology Optimization

While density-based topology optimization became a standard tool in structural engineering in the late 1990s, its application to fluid dynamics is relatively recent. The optimization of internal-flow layouts remains an active and highly challenging area of research [26]. The genesis of fluid topology optimization is widely attributed to Borrvall and Petersson (2003), who successfully applied the technique to minimize power dissipation in creeping Stokes flow [15]. This was quickly followed by Gersborg-Hansen et al. (2005), who extended the framework to steady, laminar Navier-Stokes flows [29]. Over the past decade, the literature surrounding laminar flow optimization expanded extensively, adapting the methodology to standard incompressible Navier-Stokes equations [21, 4, 64].

However, the assumption of laminar flow significantly simplifies both the governing equations and the derivation of adjoint sensitivities required for gradient-based optimization. Unfortunately, this assumption prevents the application of these methodologies to many industrial and high-performance applications, where flow velocities are high and turbulence dominates the flow physics. Only recently has the computational mechanics community begun to tackle the severe non-linearities and numerical instabilities associated with optimizing geometries directly in the turbulent regime [61, 22, 62].

Extending topology optimization to the turbulent regime is highly non-trivial. It requires coupling optimization algorithms with Reynolds-Averaged Navier-Stokes (RANS) turbulence models. These models introduce steep velocity gradients, highly non-linear source terms, near-wall resolution requirements, and severe numerical instabilities during the iterative optimization process.

To mitigate this, gradient-based turbulent optimizations frequently rely on the “frozen turbulence” assumption, where the turbulent viscosity field is updated in the forward solve but treated as fixed during sensitivity assembly.

Jury question: Is frozen turbulence just a convenience that makes the gradients physically wrong?

Answer: It is a controlled approximation, not a claim of full RANS sensitivity. The forward solve still updates turbulence for every design; the adjoint omits the infinitesimal derivative of ν_t and wall distance. The thesis therefore verifies the frozen derivative internally and validates extracted designs independently instead of presenting the gradient as a fully coupled turbulent adjoint.

While this assumption provides necessary solver stability and has driven success in industrial adjoint shape optimization for automotive aerodynamics [43], advancing toward fully coupled turbulent sensitivities in topology optimization remains an ongoing mathematical challenge.

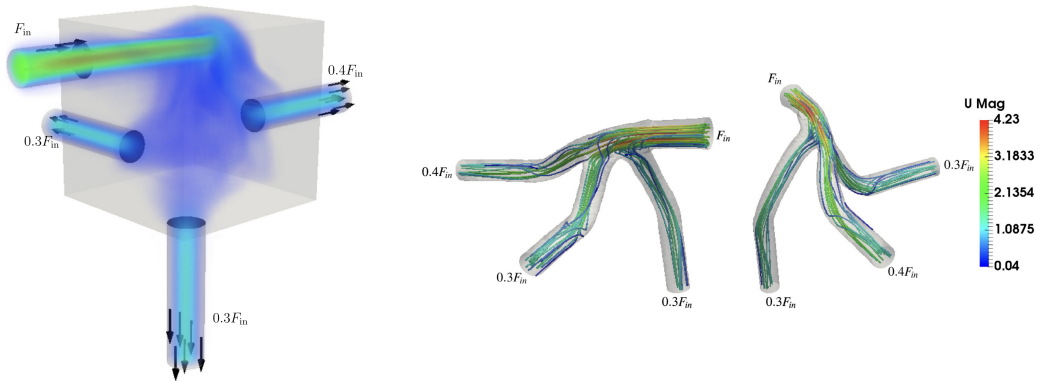


FIGURE 1.3: Turbulent-flow topology optimization example. Adapted from Dilgen et al. [22].

Pioneering works by Yoon (2016) [61] and Dilgen et al. (2018) [22] have pushed the boundaries of turbulent frameworks. As seen in Figure 1.3, integrating turbulence models allows the optimizer to generate complex internal-flow layouts tailored to guide turbulent flow distributions efficiently.

1.4 Objectives and Research Questions

The objective of this Master’s thesis is to build and assess a density-based topology-optimization framework for turbulent internal-flow configurations. The title deliberately uses “flow configurations” because the work addresses the hydraulic part

of energy and thermal-management devices rather than the complete heat-transfer device. The framework combines incompressible flow, RANS turbulence modeling, Brinkman topology penalization, and gradient-based optimization in a finite-element implementation. By separating the fluid mechanics from heat transfer, the thesis focuses on a baseline that can be verified and later extended.

The core implementation is written in **FEniCS**, an open-source finite-element platform for partial differential equations [5]. This choice keeps the weak forms and residual derivatives explicit, which is useful for implementing the Spalart-Allmaras (SA) turbulence equation and for assembling the frozen-turbulence adjoint used later in the optimization loop.

At this stage, it is important to distinguish two different comparisons that appear later in the thesis. A comparison between laminar-optimized and turbulent-optimized layouts is useful as supporting design context, because it shows how inertia and turbulence alter the preferred topology. However, this comparison is not treated as the principal CFD validation step. The decisive assessment instead consists of validating the final SA-optimized geometries in Ansys Fluent, first with the same SA model and then with the SST $k - \omega$ closure.

The work is organized around three research questions:

1. **How accurately does the Spalart-Allmaras turbulence model predict curved and separated internal turbulent flows, and where do its physical limitations appear?**

Chapter 2 answers this question by combining turbulence-model theory, curved-pipe and separated-flow literature evidence, and Ansys Fluent comparisons on curved internal-flow benchmarks. The Dean number is used only as a rough physical scale for the bend cases, not as a universal validity threshold for SA.

2. **Does the custom FEniCS implementation of the Spalart-Allmaras model reproduce Ansys Fluent predictions for the selected turbulent internal-flow benchmarks?**

Chapters 3 and 4 answer this question by presenting the weak forms, implementing the standalone SA solver, and comparing it against Ansys Fluent on a turbulent channel and a U-bend benchmark.

3. **To what extent do SA-driven topology optimizations produce final designs whose predicted performance remains consistent under independent Ansys Fluent validation with SA and SST $k - \omega$, and how do they compare with laminar-optimized designs under identical turbulent operating conditions?**

Chapter 6 answers this question using the pipe-bend, U-bend, and diffuser benchmarks. It reports the FEniCS optimizations, sharp-geometry Ansys-SA validation, SST $k - \omega$ validation, and laminar-versus-turbulent design comparisons. Appendix E contains the supporting frozen-sensitivity verification.

1.5 Outline of the Thesis

The remainder of this thesis is structured into the following six chapters:

- **Chapter 2 (Turbulence Modeling):** Provides the theoretical background on fluid dynamics and turbulence. It discusses the Navier-Stokes equations, RANS modeling, near-wall treatment, the physical characteristics of the Spalart-Allmaras model, and the curvature and separation limits relevant to this thesis.
- **Chapter 3 (Implementation of the Spalart-Allmaras Model in FEniCS):** Details the numerical implementation of the standalone SA solver in FEniCS, including weak forms, IPCS flow solves, Picard coupling, and wall-distance computation.
- **Chapter 4 (Verification of the Spalart-Allmaras Solver for Turbulent Internal Flows):** Verifies the standalone FEniCS-SA forward solver against Ansys Fluent on turbulent channel-flow and U-bend benchmarks.
- **Chapter 5 (Turbulent Topology Optimization Methodology):** Introduces the density-based optimization framework. It details Brinkman momentum penalization, topology penalties, reciprocal wall-distance penalization, filtering, projection, MMA, and the frozen-turbulence adjoint.
- **Chapter 6 (Turbulent Topology Optimization Results):** Presents the turbulent topology optimization results for the pipe-bend and U-bend benchmarks from Bayat et al. [13] and the diffuser benchmark from Yoon [61], and evaluates extracted designs with both SA and SST $k - \omega$ turbulence models. Appendix E documents the pipe-bend sensitivity verification and numerical support material.
- **Chapter 7 (Conclusion and Future Work):** Summarizes the main findings, answers the research questions conservatively, and outlines remaining limitations and future extensions toward three-dimensional and conjugate heat-transfer applications.

Chapter 2

Turbulence Modeling

2.1 Introduction to Fluid Dynamics and Turbulence

The starting point for the flow model is the conservation of mass and momentum for a viscous continuum fluid, written in Navier-Stokes form [59]. This chapter first uses index notation because it makes the tensor components in the turbulence model explicit. The indices i and j refer to spatial directions, and repeated indices imply summation. Later chapters switch to vector notation, matching the FEniCS Unified Form Language expressions used in the implementation.

Mass conservation gives the continuity equation. For a compressible fluid, it is written as

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_i)}{\partial x_i} = 0 \quad (2.1)$$

where ρ denotes density. The liquid and low-speed internal-flow cases considered in this thesis are modeled as incompressible, so ρ is constant and the continuity equation becomes

$$\frac{\partial u_i}{\partial x_i} = 0. \quad (2.2)$$

For a Newtonian fluid with constant dynamic viscosity μ , momentum conservation reduces to

$$\rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = - \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial u_i}{\partial x_j} \right) \quad (2.3)$$

where p is the static pressure. Direct Numerical Simulation (DNS) would solve Eq. 2.3 without modeling the turbulent fluctuations, but that becomes impractical once the Reynolds number is high enough for a wide range of turbulent length and time scales to appear [45]. The meshes and time steps needed to resolve the smallest eddies are far beyond what is reasonable for the design loops considered here. The thesis therefore uses turbulence modeling to represent the mean effect of the unresolved fluctuations [57, 6].

2.2 Reynolds-Averaged Navier-Stokes (RANS) Equations

The Reynolds-Averaged Navier-Stokes (RANS) framework is the engineering turbulence model class used in this thesis. It follows the Reynolds decomposition: each instantaneous quantity is split into a mean part, $(\bar{u}_i, \bar{p})$, and a fluctuation, (u'_i, p') [45]:

$$u_i = \bar{u}_i + u'_i \quad \text{and} \quad p = \bar{p} + p'. \quad (2.4)$$

Figure 2.1 illustrates this separation for velocity and pressure signals measured at a fixed point in a turbulent flow. The instantaneous signal contains rapid turbulent fluctuations, while the dashed mean line represents the slowly varying quantity retained by the RANS equations. By construction, the mean of the fluctuating component is zero, but products of fluctuations, such as $\overline{u'_i u'_j}$, remain non-zero and generate the Reynolds-stress terms. The schematic is adapted from the Fluid Mechanics on the Web Chapter 6 material based on White's viscous-flow treatment [42].

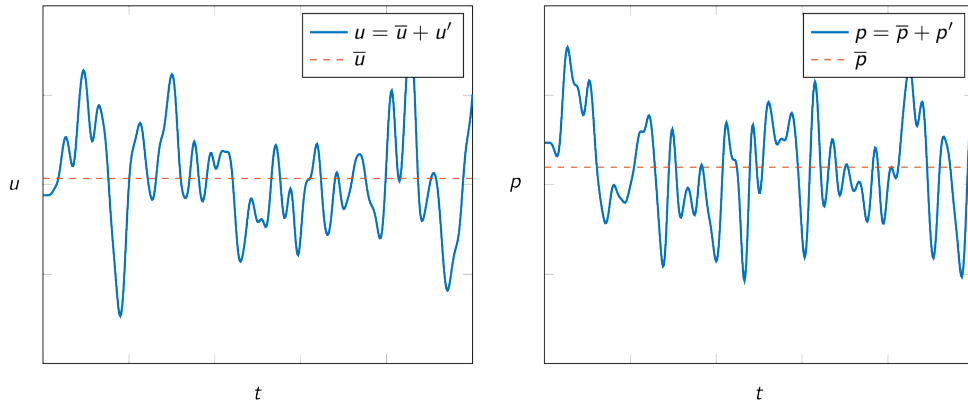


FIGURE 2.1: Reynolds decomposition of velocity and pressure signals. Adapted from [42].

Substituting these decomposed variables into the standard Navier-Stokes equations and taking the time average yields the RANS momentum equations:

$$\rho \left(\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} \right) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial \bar{u}_i}{\partial x_j} - \rho \overline{u'_i u'_j} \right). \quad (2.5)$$

The additional tensor $-\rho \overline{u'_i u'_j}$ is the Reynolds stress. It carries the effect of turbulent momentum transport on the mean flow. Because this tensor is not known from the mean velocity and pressure alone, Eq. 2.5 is not closed without an additional turbulence model [57].

2.2.1 The Boussinesq Hypothesis and the Isotropic Assumption

Most RANS closures used in practical CFD adopt the Boussinesq eddy-viscosity hypothesis [45]. The hypothesis models turbulent momentum transport by analogy with viscous diffusion and introduces a scalar turbulent viscosity μ_t :

$$-\rho \overline{u'_i u'_j} = \mu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{2}{3} \rho k \delta_{ij} \quad (2.6)$$

where k is the turbulent kinetic energy and δ_{ij} is the Kronecker delta. A Boussinesq-based model therefore mainly supplies a local value for μ_t .

Equation 2.6 also defines an important limitation for the present work. Since μ_t is scalar, the modeled stress is aligned with the mean strain-rate tensor and cannot independently represent Reynolds-stress anisotropy. This is acceptable for many attached shear flows, but it becomes weaker for flows with strong curvature, rotation, or separation. Curved ducts and bends are therefore critical test cases because centrifugal effects redistribute momentum across the cross-section and can create secondary motion that a scalar eddy-viscosity closure cannot reproduce exactly [48, 47].

Jury question: If scalar eddy viscosity is known to fail in anisotropic flows, why use it at all?

Answer: Because the thesis uses SA as a practical optimization model inside a stated envelope, not as closure-independent truth. A Reynolds-stress model or LES would be harder to stabilize inside density-based TO and much more expensive. The defense is the evidence chain: identify the limitation, verify the implementation, and cross-check final geometries with SST k-omega.

2.3 Boundary Layer Theory and Near-Wall Treatment

The wall treatment must be chosen together with the turbulence model, because μ_t depends strongly on the near-wall region. Wall-bounded turbulent flows are commonly discussed using the non-dimensional wall coordinate y^+ and velocity u^+ :

$$y^+ = \frac{y u_\tau}{\nu}, \quad u^+ = \frac{\bar{u}}{u_\tau} \quad (2.7)$$

Here y is the distance to the wall, ν is the kinematic viscosity, and $u_\tau = \sqrt{\tau_w / \rho}$ is the friction velocity defined from the wall shear stress τ_w [45].

Figure 2.2, adapted from Altair's AcuSolve near-wall modeling documentation [7], summarizes the wall-unit regions used when discussing RANS wall treatment. The important point for this thesis is not that one near-wall strategy is universally required for topology optimization, but that the chosen turbulence closure and wall treatment must remain stable while the solid-fluid boundary evolves.

1. **The viscous sublayer ($y^+ < 5$):** Immediately adjacent to the wall, turbulent fluctuations are damped by kinematic constraints. Viscous shear dominates

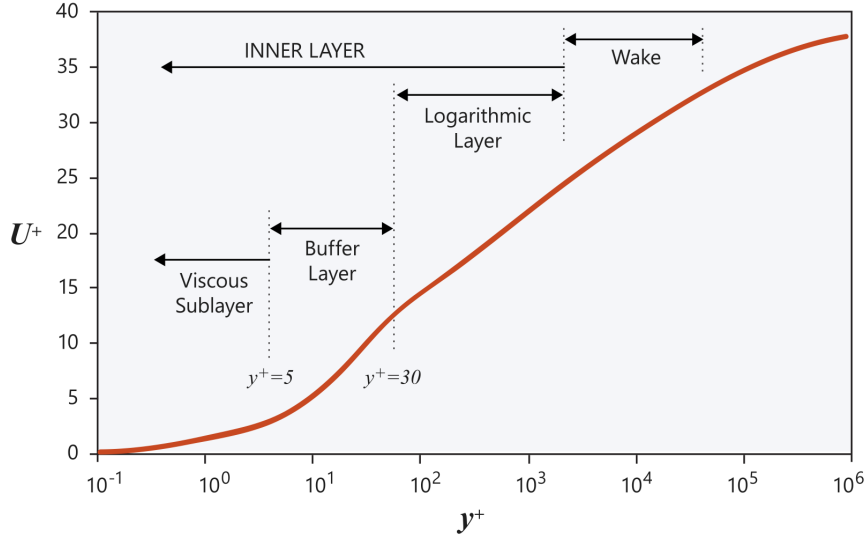


FIGURE 2.2: Near-wall boundary-layer regions. Adapted from [7].

and the velocity profile is approximately linear, $u^+ = y^+$. Low-Reynolds wall-resolved RANS calculations place the first cell around $y^+ \approx 1$ so this region is resolved rather than modeled.

2. **The buffer layer ($5 < y^+ < 30$):** This is a transition region where viscous and turbulent shear stresses are both important. It is generally an undesirable location for the first cell center because it is neither cleanly wall-resolved nor safely inside the log-law range used by wall functions.
3. **The log-law and outer regions ($y^+ > 30$):** Farther from the wall, turbulence dominates and the velocity follows an approximate logarithmic distribution controlled by the von Karman constant. High-Reynolds wall-function methods place the first cell in this range and model the inner layer empirically, while the outer wake region becomes increasingly sensitive to pressure-gradient and outer-flow effects [57, 51, 7].

High-Reynolds models, such as the standard $k - \epsilon$ model, use empirical wall functions to bridge the gap between the wall and the log-law region, typically requiring the first computational cell center to be placed at $y^+ > 30$. Wall functions are not fundamentally incompatible with turbulent topology optimization; Bayat et al. [13], for example, use a wall-function formulation with an implicit wall penalty. For this thesis, however, a wall-resolved $y^+ \approx 1$ strategy is more convenient because it avoids introducing a moving empirical wall-function interface into the FEniCS density-based formulation. The meshes used here are therefore generated as wall-resolved meshes, and the SA model is integrated through the viscous sublayer.

2.4 The Spalart-Allmaras One-Equation Model

Two-equation models such as $k - \epsilon$ and $k - \omega$ transport both a turbulence scale and a length- or frequency-scale variable. That extra physics is useful, but it also introduces nonlinear source terms that complicate robust sensitivity calculations in a topology-optimization loop. This thesis therefore uses the one-equation Spalart-Allmaras (SA) model, originally developed for aerodynamic flows with adverse pressure gradients [49]. The model transports a working variable $\tilde{\nu}$ and converts it algebraically to the turbulent kinematic eddy viscosity, $\nu_t = \mu_t/\rho$ [50].

The transport equation for the standard SA model is given by:

$$\frac{\partial \tilde{\nu}}{\partial t} + \bar{u}_j \frac{\partial \tilde{\nu}}{\partial x_j} = c_{b1} \tilde{S} \tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{d} \right)^2 + \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left((\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + c_{b2} \left(\frac{\partial \tilde{\nu}}{\partial x_j} \right)^2 \right]. \quad (2.8)$$

The right-hand side contains production, wall destruction, and diffusion of the working variable. Production is controlled by the modified vorticity $\tilde{S}$, while destruction depends on the wall distance d . The model can therefore be run in a wall-resolved low-Reynolds-number mode, integrating through the viscous sublayer without introducing a separate empirical wall function or a second transported length-scale equation [46].

At this stage of the thesis, the SA model is treated as a turbulence closure rather than an implementation object. Therefore, Chapter 2 only states the main transport equation and its physical role. Detailed constants, damping functions, linearization choices, and weak forms are deferred to Chapter 3, where the FEniCS implementation is discussed.

2.5 Evaluation of Curvature Effects and the Dean Number

Despite its computational advantages for topology optimization, the standard SA model relies on a single length scale tied to the nearest wall distance. It lacks an explicit rotation/curvature correction and remains constrained by the Boussinesq eddy-viscosity assumption. It can therefore struggle to capture the anisotropic physics of streamline curvature.

When fluid passes through a bend, centrifugal forces induce secondary motion perpendicular to the main flow direction. In curved-pipe literature, the dimensionless Dean number (De) is often used to report the combined Reynolds-number and curvature scale:

$$De = Re_d \sqrt{\frac{d}{2R_c}} \quad (2.9)$$

where d is the characteristic length of the cross-section, such as the pipe diameter D or hydraulic diameter D_h , R_c is the radius of curvature, and Re_d is the bulk Reynolds number based on that same characteristic length [20]. A higher Dean number indicates a stronger cross-sectional influence of centrifugal forces.

The Dean number is academically relevant here because it separates curvature severity from Reynolds number alone. However, it should not be interpreted as a universal threshold for SA validity. The breakdown of an eddy-viscosity model depends on geometry, separation, mesh resolution, boundary conditions, and the validation quantity being compared. In this thesis, the Dean number is used only as a rough physical descriptor of the bend cases; the final judgment still comes from profile comparisons and independent CFD validation.

Jury question: Can you really defend the De approximately 15000 and 18000 distinction?

Answer: Only as case-specific evidence. The thesis does not claim a universal Dean-number cutoff. The point is that two similar-Re Ansys verification cases show different SA behavior when curvature changes, which warns that curvature severity matters and must be checked by profiles and validation metrics.

For mildly curved flows, the SA model can perform relatively well. Apalowo [10] evaluated a 90° circular pipe bend with a Dean number of $De \approx 9,000$ at $Re_D \approx 34,000$. Under these moderate conditions, where $R_c = 7D$, the SA result in the bottom panel of Figure 2.3 follows the measured streamwise-velocity profile about as well as the two $k-\epsilon$ variants. This can be seen from the similar profile shape and from the reported error values, which are slightly lower for the SA panel than for the two panels above it. The comparison is therefore used as evidence that mild curvature does not automatically invalidate SA.

At stronger bend curvature and higher Dean number, the limitations of the isotropic Boussinesq assumption become more visible. Abuhatira et al. [2] evaluated turbulent flow through a 90° pipe bend with $R_c = 2.0D$ and $Re_D = 60,000$, corresponding to $De \approx 30,000$. Figure 2.4 shows the velocity profile just outside the bend, at the beginning of the straight outlet section. None of the RANS models perfectly follows the experimental data, including SST $k-\omega$, which is a useful warning in itself. The yellow SA curve is nevertheless the worst of the compared closures near the inner-wall side of the profile, where it smooths the velocity redistribution more strongly than the other models. This provides supporting evidence that strong curvature can expose the limitations of the standard SA closure, but it is not treated as proof of a universal failure threshold.

The literature therefore provides useful rough-scale evidence: SA can be acceptable for mild curvature and can deteriorate for stronger curvature. The thesis-specific evidence must come from the Ansys comparisons below, which evaluate the geometries and quantities most relevant to the present optimization study.

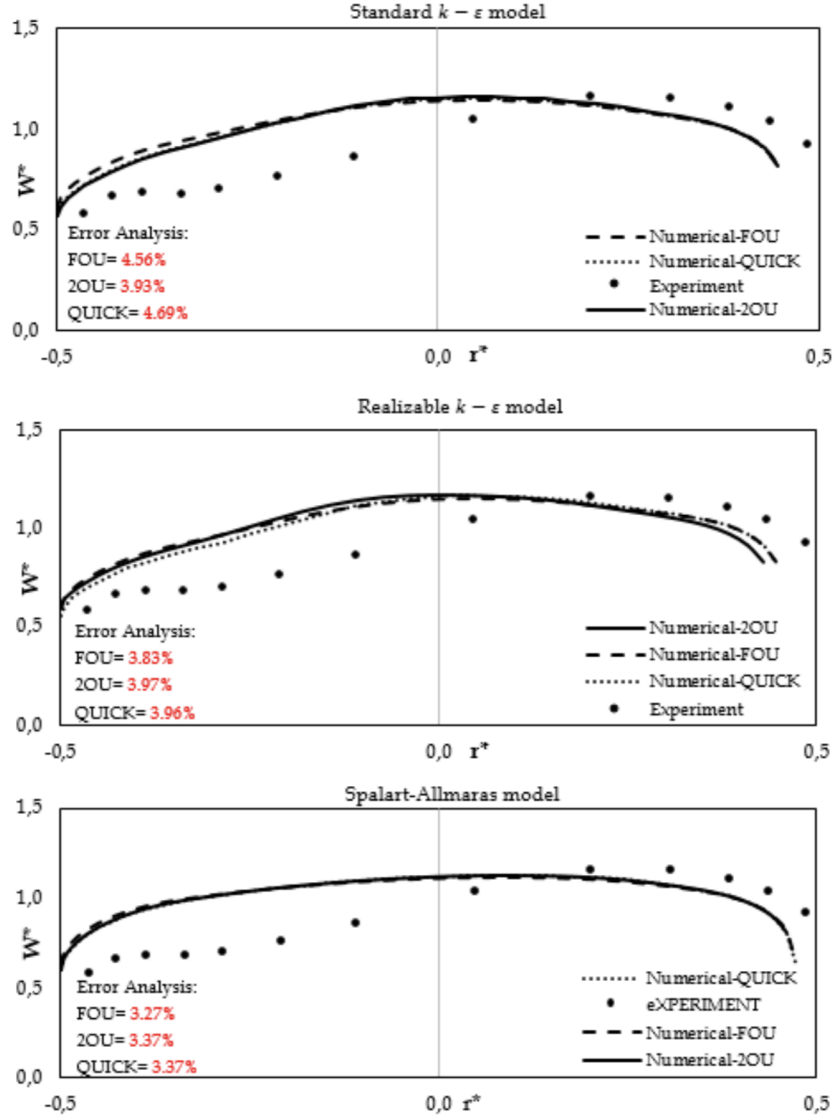


FIGURE 2.3: Streamwise velocity profiles in a mildly curved pipe bend. Reprinted from Apalowo [10].

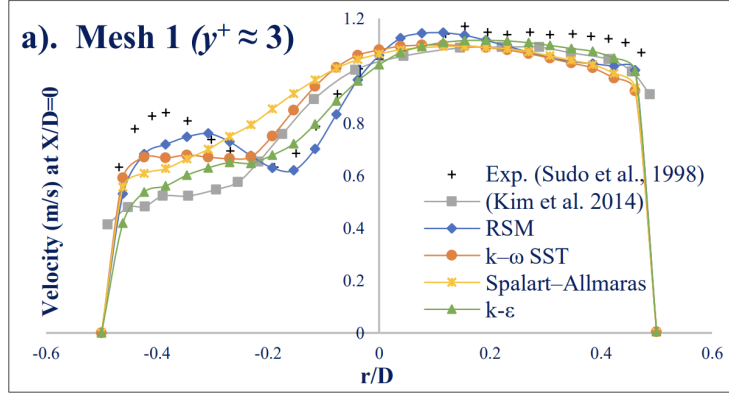


FIGURE 2.4: Turbulence-model predictions downstream of a strongly curved pipe bend. Adapted from Abuhatira et al. [2].

2.6 Separated Flow Without Curvature

Curvature is not the only mechanism that can expose the limitations of a scalar eddy-viscosity model. A fully developed channel is an attached wall-bounded flow and is therefore not treated as a model-limitation case in this chapter; it is used later as an implementation-verification benchmark. A backward-facing step is more relevant here because it removes bend curvature while introducing a sudden expansion, separated shear layer, recirculation region, and downstream reattachment [24, 33]. This resembles the local separation that can occur in topology-optimized internal-flow paths and diffuser-like designs even when the geometry is not dominated by pipe-bend curvature. Because there is no curvature radius, the Dean number is not meaningful for this case; the standard integral metric is the normalized reattachment length x_r/h , with h the step height.

The modeling difficulty is that separation breaks the near-wall local-equilibrium picture on which simple wall-resolved closures often behave well. Turbulence generated in the detached shear layer is transported downstream before it affects the mean flow, and reattachment depends on shear-layer growth, entrainment, pressure recovery, and stress redistribution. Linear eddy-viscosity models such as SA and SST $k - \omega$ can represent extra turbulent mixing through an effective scalar viscosity, but they cannot fully represent stress anisotropy or coherent unsteadiness during reattachment [45, 57]. For topology optimization, this matters because a locally separated design may appear favorable if the chosen closure overpredicts reattachment or underpredicts separation losses.

The Driver and Seegmiller backward-facing-step experiment is a canonical validation case for separated internal flow [23]. In the NASA Turbulence Modeling Resource comparison shown in Figure 2.5, the experimental reattachment length is $x_r/h = 6.26 \pm 0.10$ at $Re_H \approx 36,000$. Representative SA computations predict reattachment near $x_r/h \approx 6.1$, while the SST $k - \omega$ results reported by NASA as SSTm predict reattachment near $x_r/h \approx 6.5$ [40]. Judged only by this integral metric, SA does not simply fail. The velocity profiles in panels (c) and (d), however, show the

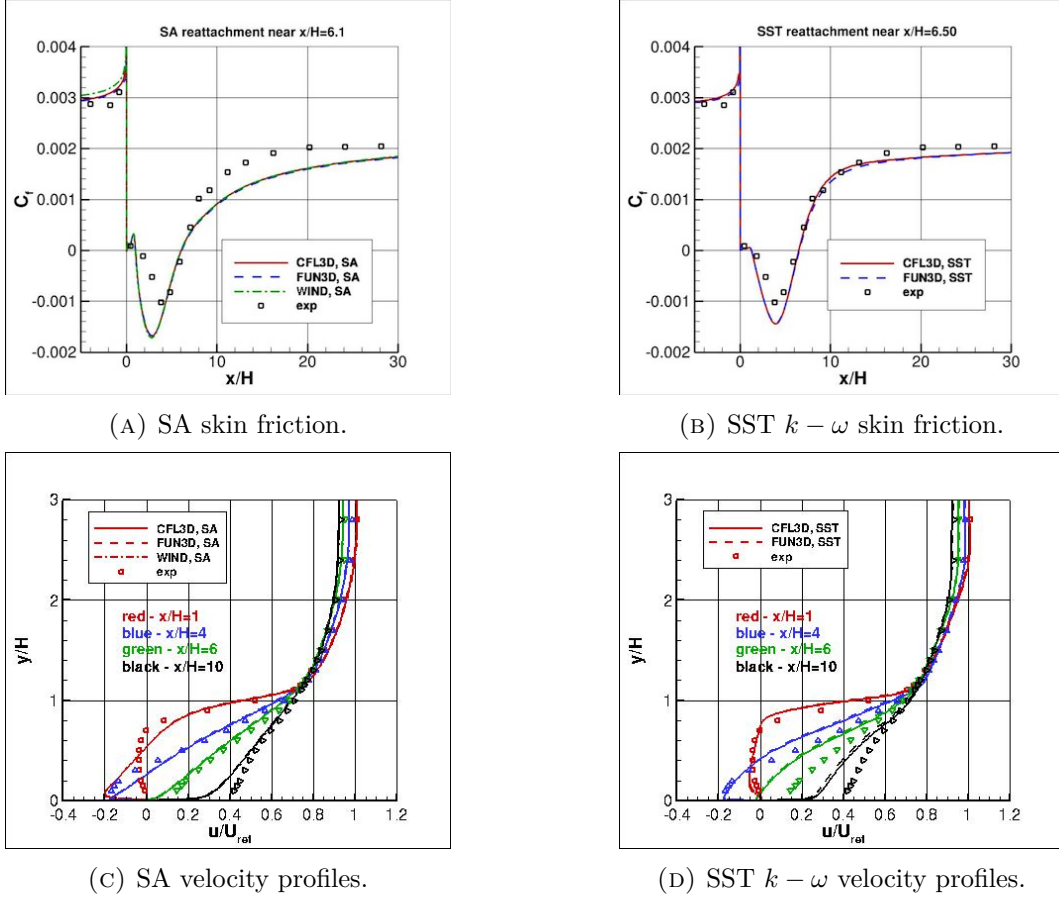


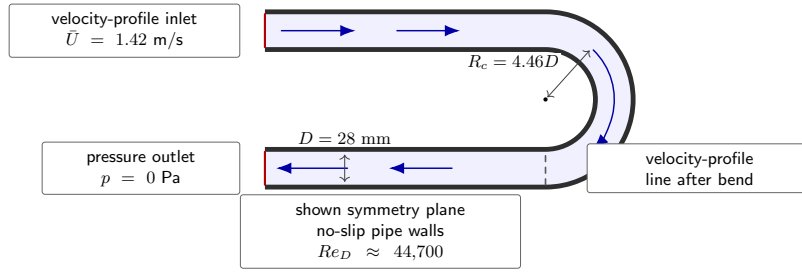
FIGURE 2.5: NASA TMR comparison for the Driver–Seegmiller backward-facing-step experiment [23, 40].

more relevant limitation: SST $k - \omega$ follows the measured separated-flow recovery more closely than SA. Since topology optimization is sensitive to local velocity recovery, wall shear, and loss generation, the backward-facing-step evidence supports a cautious conclusion: separated internal flows are strongly turbulence-model dependent, and SA-based predictions should be checked with an independent closure when local separation is present.

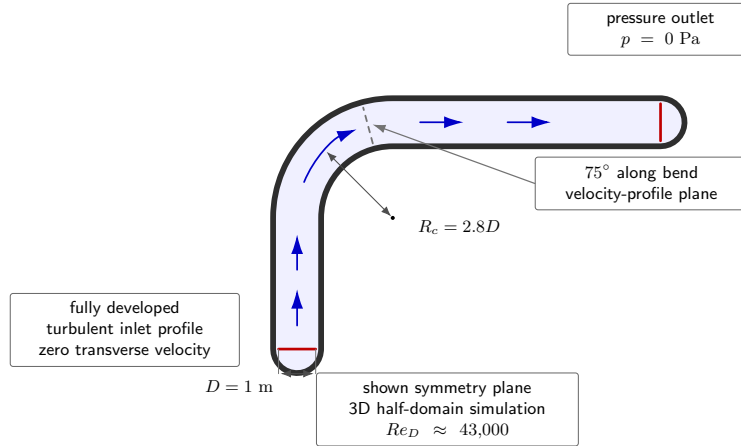
2.7 Benchmark Studies in Ansys Fluent

To investigate where the SA model’s accuracy deteriorates for the present study, benchmark simulations were performed using **Ansys Fluent** 2024 R2 [8]. Three internal-flow benchmarks were considered. The first two are fully three-dimensional curved-flow cases: a 180° U-bend and a tighter 90° pipe bend. To ensure a standardized numerical baseline for these curved cases, the computational meshes, fluid properties, and boundary conditions were adopted from the official Ansys Fluid

Dynamics Verification Manual [9]. Specifically, the 3D 90° bend corresponds to verification case VMFL030, while the 3D U-bend corresponds to case VMFL048. The third benchmark, the backward-facing step, is treated differently: it is not used as a new validation case, because the literature-supported validation evidence is already given in Section 2.6. It is instead used as a separated-flow model-sensitivity benchmark in Appendix A, comparing SA against SST $k - \omega$ [39] on the local backward-facing-step geometry used in the repository and in the corresponding thesis work [37]. The Ansys Fluent convergence monitors and final-window criteria are reported in Appendix A.1, so the main chapter can focus on the physical comparison between SA and SST $k - \omega$.



(A) VMFL048 3D U-bend.



(B) VMFL030 3D 90° bend.

FIGURE 2.6: Ansys Fluent curved-flow benchmark geometries from the verification manual [9] (ChatGPT, 2026).

2.7.1 Mild Curvature: The U-Bend Case

The first benchmark evaluated a 3D U-bend geometry characterized by a moderate curvature ratio of $R_c \approx 4.46D$. The Ansys Fluent setup is summarized in Figure 2.6a: the VMFL048 case uses a velocity-profile inlet, a zero-gauge pressure outlet, no-slip pipe walls, and a symmetry boundary on one side of the 3D domain.

Operating at a bulk Reynolds number of $Re_D \approx 44,700$, this geometry resulted in a Dean number of $De \approx 15,000$. The comparison below combines velocity contours, the bend-exit velocity profile, and turbulent-viscosity-ratio contours so that both the mean-flow prediction and the modeled turbulent diffusion can be assessed.

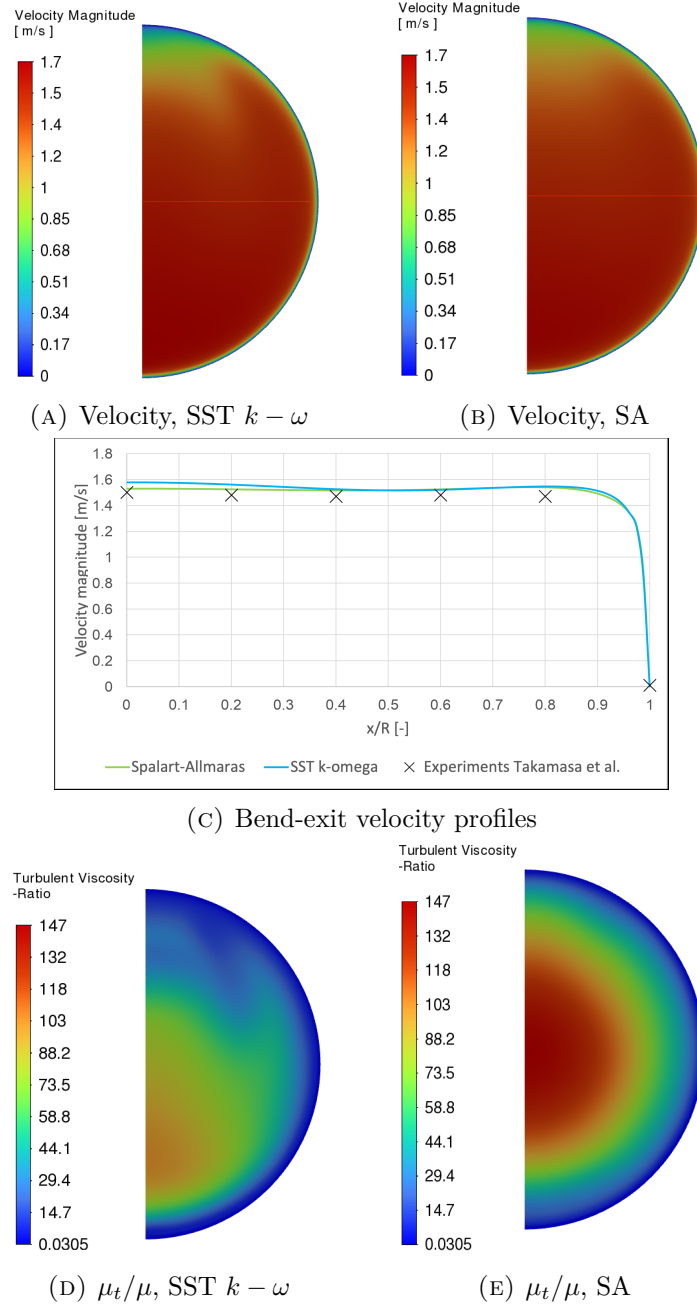


FIGURE 2.7: VMFL048 U-bend RANS comparison. Panels (d) and (e) use the same color scale for μ_t/μ because the SST $k - \omega$ and SA ranges are close, so the color intensity can be compared directly.

Consistent with the moderate-curvature literature evidence, the SA model performed well under these conditions. As shown in Figure 2.7, the qualitative velocity contours of the SA model reproduce the bulk flow distribution of the SST $k - \omega$ model. In panels (a) and (b), the line drawn on the exit-plane contour is the radius along which the VMFL048 velocity profiles in panel (c) are extracted. The profile coordinate follows this radius from left to right, i.e. from the inner part of the pipe at $x/R = 0$ toward the pipe wall at $x/R = 1$. This links the one-dimensional profile comparison directly to the displayed exit-plane flow field. The quantitative velocity profiles agree well with the experimental data from Takamasa and Kondo [55].

The turbulent-viscosity panels in Figure 2.7 clarify why the VMFL048 velocity agreement is good but not model-identical. Both closures damp μ_t/μ to low values at the no-slip wall and generate the largest eddy viscosity away from the wall where mean shear and turbulent mixing are strongest. The SA contour is more radially uniform than the SST $k - \omega$ contour and its scale reaches a slightly higher maximum, which is consistent with a mild tendency to overpredict eddy viscosity and smooth gradients. In this moderately curved U-bend, that extra diffusion remains spatially aligned with the main shear region, so it does not strongly distort the extracted velocity profile.

2.7.2 Strong Curvature: The 90° Bend Case

The second benchmark evaluated a tighter 90° pipe bend characterized by $R_c = 2.8D$. The corresponding VMFL030 setup is shown in Figure 2.6b, including the fully developed turbulent inlet profile, pressure outlet, no-slip pipe walls, half-domain symmetry, and the 75° velocity-profile plane used for comparison. Despite operating at a similar bulk Reynolds number of $Re_D \approx 43,000$, this tighter bend radius pushed the Dean number to $De \approx 18,000$. Because the Reynolds numbers are similar between the two benchmark cases, the comparison isolates the geometric effect of streamline curvature more clearly than a Reynolds-number-only comparison. As for VMFL048, the figure combines velocity contours, the extracted profile, and turbulent-viscosity-ratio contours.

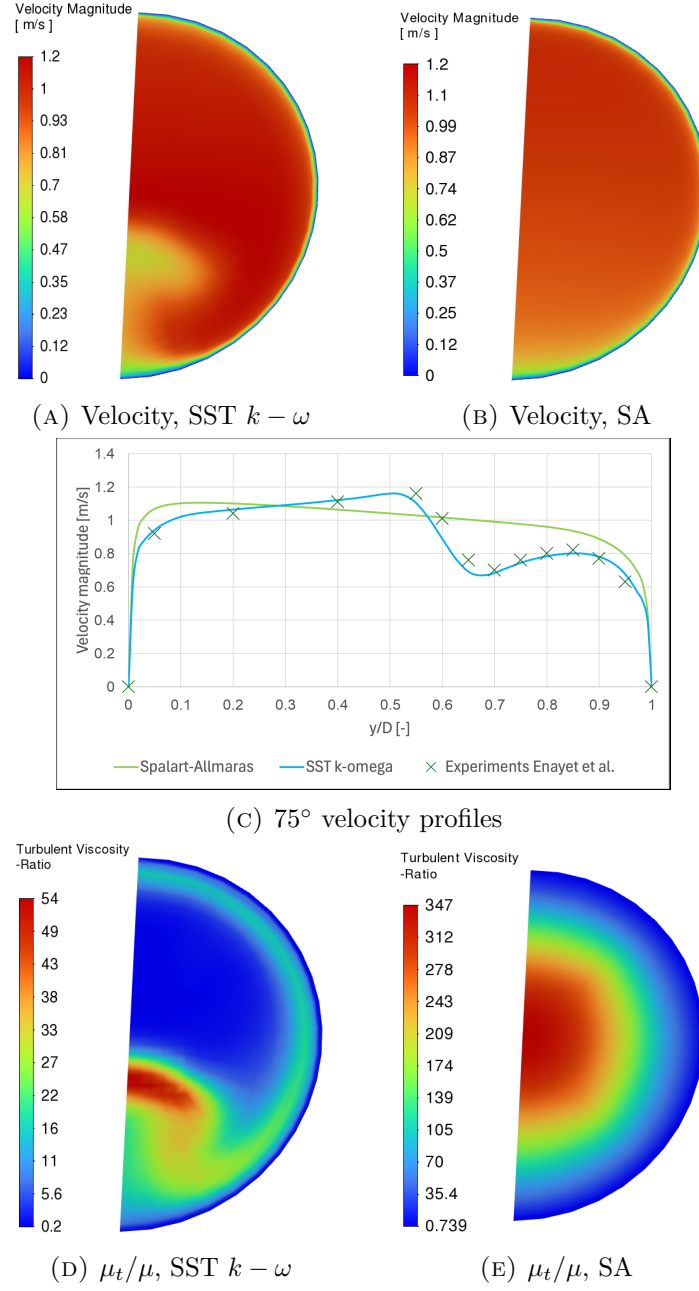


FIGURE 2.8: VMFL030 90° pipe-bend RANS comparison. Panels (d) and (e) use different color scales for μ_t/μ because the SST $k - \omega$ and SA ranges differ strongly; the colorbar values should therefore be checked rather than comparing color intensity alone.

As shown in Figure 2.8, the velocity profile at the 75° plane along the bend demonstrates clear deterioration of the SA prediction. The profile in panel (c) is extracted along the pipe diameter. In the contour views of panels (a) and (b), this diameter coincides with the left edge of the plotted cross-section, so an additional guide line cannot be drawn without hiding the contour boundary. The coordinate follows the diameter from top to bottom, with $y/D = 0$ at the top and $y/D = 1$ at the bottom. Driven by the tighter bend radius and stronger anisotropic centrifugal effects, the SA model over-predicts the turbulent viscosity. This causes it to over-smooth the velocity profile, failing to capture the sharp velocity gradients near the inner wall when compared against both the SST $k - \omega$ prediction and the experimental data from Enayet et al. [25].

The turbulent-viscosity panels in Figure 2.8 explain the poorer SA velocity prediction in the tighter bend. The SST $k - \omega$ result keeps the high μ_t/μ region more localized, with a low-viscosity core where the curvature-induced redistribution is not simply equivalent to local shear production. The SA result instead produces a broad high-viscosity region with a peak several times larger than the SST $k - \omega$ scale. Physically, the strong bend curvature drives secondary motion and Reynolds-stress anisotropy, while the standard SA closure still represents turbulence through a scalar eddy viscosity based on wall distance and a modified vorticity scale. In this case SA assigns too much eddy viscosity to too much of the bend cross-section, so the modeled turbulent diffusion smooths velocity gradients that remain visible in both the SST $k - \omega$ result and the experiment. This behavior is much more severe than in VMFL048: in the U-bend the SA overprediction is moderate and spatially consistent with the SST $k - \omega$ field, whereas in VMFL030 the eddy-viscosity magnitude and distribution are both wrong enough to wash out the measured velocity-gradient structure.

2.7.3 Sudden Expansion: The Backward-Facing Step Case

The backward-facing-step benchmark is retained as a separated-flow model-sensitivity check. Its setup, mesh-sensitivity table, Ansys Fluent contours, and downstream profiles are reported in Appendix A. The main argument relies on the literature comparison in Figure 2.5: SA can predict a plausible reattachment length, but SST $k - \omega$ follows the local velocity recovery more closely. The thesis-specific Ansys Fluent backstep results support the same caution without being treated as a separate experimental validation case.

2.8 Conclusion

This chapter answers the first research question regarding the predictive capabilities and physical limitations of the Spalart-Allmaras model for internal turbulent flows relevant to topology optimization. The SA model is attractive because it is a one-equation, wall-resolved closure that avoids empirical wall functions and remains substantially easier to stabilize than two-equation alternatives. The channel-like and mildly curved evidence supports its use as a practical forward model for

wall-resolved internal-flow topology optimization. This is especially relevant for optimization problems where the objective function or design goal is aligned with the simplified SA response: even if SA is not closure-independent truth, it can still guide useful hydraulic improvements that are later checked by independent validation. This statement is not only an assumption of the present thesis. Yoon [61] used the SA model directly inside turbulent-flow topology optimization, and Dilgen et al. [22] likewise showed that RANS-based turbulent topology optimization can produce improved flow layouts. These papers are best read as evidence that SA/RANS closures can be productive optimization models, not as proof that their predicted final performance is independent of the turbulence closure.

The limitation is not simply high Reynolds number. The main issue is curvature- and separation-driven Reynolds-stress anisotropy under a scalar eddy-viscosity closure. Literature cases at $De \approx 9,000$ and $De \approx 30,000$ provide useful supporting evidence, while the thesis-specific Ansys comparisons give a rough operating scale for the present bend cases: the U-bend case at $De \approx 15,000$ remains credible, whereas the tighter 90° bend at $De \approx 18,000$ already shows clear SA deterioration for the selected profile comparison. The separated-flow evidence points in the same direction. In the Driver-Seegmiller validation data, SST $k - \omega$ follows the measured backward-facing-step velocity profiles more closely than SA, and the thesis-specific backward-facing-step case confirms that the predicted recirculation and recovery are visibly model dependent. These diagnostics are therefore treated as case-specific evidence of model sensitivity, not as universal thresholds.

Consequently, using SA as the turbulence model inside the FEniCS topology optimization loop is defensible for this thesis because of its numerical simplicity and compatibility with wall-resolved optimization. The resulting layouts should be interpreted as SA-optimized design candidates, not as closure-independent predictions by themselves. For attached or mildly curved cases, independent sharp-geometry validation can show whether the SA-optimized layout remains robust. For designs containing strong curvature or local separation, quantitative performance claims must be conditioned on an independent closure check, such as SST $k - \omega$, because the separated-flow losses may be strongly model dependent.

Chapter 3

Implementation of the Spalart-Allmaras Model in FEniCS

3.1 Introduction

With the theoretical foundation of the Spalart-Allmaras (SA) model established in Chapter 2, this chapter details its numerical translation into a working **FEniCS** [27] solver. The focus is methodological: the finite-element weak forms, Picard coupling strategy, pressure-correction flow solver, SA transport equation, and relaxed Eikonal wall-distance calculation are described before the solver is verified separately in Chapter 4.

The numerical simulations and eventual topology optimization in this thesis are built upon FEniCS, an open-source computing platform for solving partial differential equations (PDEs). Unlike commercial computational fluid dynamics (CFD) software such as Ansys, which typically relies on the Finite Volume Method (FVM), FEniCS uses the Finite Element Method (FEM) and requires the user to explicitly define the mathematical formulation of the problem [36]. FEniCS is useful in this thesis because the same weak-form language used to assemble the forward equations can also be used to assemble residual derivatives for the topology-optimization adjoint in Chapter 5.

To solve a PDE in FEniCS, the governing equations (the “strong form”) must be mathematically transformed into a weak formulation, or variational form. FEniCS utilizes the Unified Form Language (UFL) to express these weak forms in Python code that closely mirrors the mathematical notation [34]. This high-level mathematical scripting is particularly advantageous for topology optimization, as it makes the connection between the implemented residuals and the sensitivity equations explicit.

For reproducibility, two project folders accompany this thesis. **Turbulence Models** contains the standalone SA solver and the verification setup files. **FluidTO** contains the topology-optimization code and the frozen-adjoint benchmark setup files. The appendices refer to these folders when tabulating the numerical setups.

3.1.1 Derivation of a Weak Formulation

Before diving into the complex turbulence equations, it is instructive to demonstrate how a continuous PDE is translated into a discrete FEniCS formulation. To illustrate this methodology, consider the Poisson equation for pressure, which is a fundamental component of incompressible fluid solvers. The strong form of the Poisson equation is given by:

$$-\nabla^2 p = f \quad \text{in } \Omega \quad (3.1)$$

where p is the pressure field, f is a source term derived from the velocity divergence, and Ω is the computational domain. To obtain the weak form, the equation is multiplied by a sufficiently smooth test function v and integrated over the entire domain Ω , following standard Galerkin finite element procedures [36, 34]:

$$-\int_{\Omega} (\nabla^2 p) v \, dx = \int_{\Omega} f v \, dx. \quad (3.2)$$

Applying integration by parts to the left-hand side yields:

$$\int_{\Omega} \nabla p \cdot \nabla v \, dx - \int_{\partial\Omega} (\nabla p \cdot \mathbf{n}) v \, ds = \int_{\Omega} f v \, dx, \quad (3.3)$$

where $\partial\Omega$ is the boundary of the domain and $\mathbf{n}$ is the outward-pointing normal vector. This integral formulation lowers the continuity requirements of the solution, requiring only first-order derivatives to be square-integrable. Neumann boundary conditions can also be substituted directly into the boundary integral. This variational approach forms the foundation of all fluid and turbulence equations implemented in this thesis.

3.2 Numerical Solution Strategy for Fluid Dynamics

The RANS-SA system is solved in segregated form rather than as one monolithic nonlinear block. This choice is motivated by the nonlinear convection terms, the velocity-pressure saddle-point structure, and the feedback between the velocity field and the turbulent eddy viscosity. The verification solver therefore uses Picard coupling between the velocity-pressure equations and the SA transport equation.

3.2.1 Decoupling the Physics: The Picard Iteration Scheme

The coupling is circular: the momentum equation needs the turbulent eddy viscosity ν_t , while the SA transport equation is convected by the velocity field $\mathbf{u}$. A fully simultaneous solve would keep all of this coupling inside one nonlinear algebraic system, which was not the most robust option for the verification code developed here.

The solver instead applies **Picard iteration**, or successive substitution, a standard segregated strategy for RANS calculations [28, 57]. At iteration k , the velocity-pressure solve uses the eddy viscosity assembled from the previous SA field. The updated velocity is then used in the next SA solve, after which the eddy viscosity is recomputed. Repeating this sequence forms the outer fixed-point loop for the coupled RANS-SA state.

Jury question: Why not solve the complete RANS-SA system monolithically?

Answer: A monolithic solve is cleaner mathematically but harder to make robust while the topology changes and intermediate designs create near-solid regions. Picard decoupling gives a stable forward state for the design loop. Where the final flow residual is differentiated, the code can use a steady SNES solve so the adjoint is assembled from a consistent final velocity-pressure residual.

3.2.2 Incremental Pressure Correction Scheme (IPCS)

Inside each Picard iteration, the incompressible velocity-pressure problem is advanced with an Incremental Pressure Correction Scheme (IPCS). IPCS separates the velocity and pressure updates instead of solving them in one coupled block, following the fractional-step projection ideas of Chorin [19] and Temam [56]. The weak forms below use the Galerkin finite-element notation used in FEniCS [36, 34]. A pseudo-time step Δt is retained as a steady-solver device for the nonlinear convective terms. Each pseudo-time step consists of the following three subproblems:

1. **Tentative velocity step:** The momentum equation is first solved with the old pressure p^n , producing an intermediate velocity $\mathbf{u}^*$ that is not yet divergence-free. The RANS diffusion term uses the effective kinematic viscosity $\nu + \nu_t$. With explicit convection and implicit diffusion, the semi-discrete equation is:

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} + (\mathbf{u}^n \cdot \nabla) \mathbf{u}^* - \nabla \cdot (2(\nu + \nu_t^n) \boldsymbol{\varepsilon}(\mathbf{u}^*)) = -\nabla p^n + \mathbf{f}, \quad (3.4)$$

where $\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^T)$ is the symmetric rate-of-strain tensor.

2. **Pressure correction step:** The pressure is updated through a Poisson problem chosen so that the final velocity satisfies $\nabla \cdot \mathbf{u}^{n+1} = 0$:

$$\nabla^2 p^{n+1} = \nabla^2 p^n + \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^*. \quad (3.5)$$

3. **Velocity update step:** The pressure increment is then used to project the tentative velocity onto the divergence-free space:

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla (p^{n+1} - p^n). \quad (3.6)$$

In finite-element form, the three IPCS subproblems are written on a velocity space V_h and pressure space Q_h . For all test functions $\mathbf{v} \in V_h$ and $q \in Q_h$, the tentative

velocity step solves

$$\begin{aligned} \int_{\Omega} \frac{1}{\Delta t} \mathbf{u}^* \cdot \mathbf{v} \, dx + \int_{\Omega} ((\mathbf{u}^n \cdot \nabla) \mathbf{u}^*) \cdot \mathbf{v} \, dx + \int_{\Omega} 2(\nu + \nu_t^n) \boldsymbol{\varepsilon}(\mathbf{u}^*) : \boldsymbol{\varepsilon}(\mathbf{v}) \, dx \\ = \int_{\Omega} \frac{1}{\Delta t} \mathbf{u}^n \cdot \mathbf{v} \, dx - \int_{\Omega} \nabla p^n \cdot \mathbf{v} \, dx + \int_{\Omega} \mathbf{f} \cdot \mathbf{v} \, dx. \end{aligned} \quad (3.7)$$

The pressure correction is then obtained from

$$\int_{\Omega} \nabla p^{n+1} \cdot \nabla q \, dx = \int_{\Omega} \nabla p^n \cdot \nabla q \, dx - \int_{\Omega} \frac{1}{\Delta t} (\nabla \cdot \mathbf{u}^*) q \, dx, \quad (3.8)$$

and the corrected velocity is recovered through

$$\int_{\Omega} \mathbf{u}^{n+1} \cdot \mathbf{v} \, dx = \int_{\Omega} \mathbf{u}^* \cdot \mathbf{v} \, dx - \int_{\Omega} \Delta t \nabla (p^{n+1} - p^n) \cdot \mathbf{v} \, dx. \quad (3.9)$$

In the forward-flow verification scripts, pressure is treated as a kinematic pressure and the viscous operator is written in symmetric strain-rate form. The momentum equation does not include a Streamline-Upwind/Petrov–Galerkin (SUPG) term. SUPG stabilization [17] is reserved for the SA working-variable transport equation, where it damps element-scale oscillations in convection-dominated cases. Brinkman and density-penalty terms are not part of the standalone verification solver; they are introduced in Chapter 5.

The Picard loop is stopped when the L_2 norm of the update residual falls below the prescribed tolerance ϵ_{tol} for both the fluid and turbulence fields.

3.3 Detailed Formulation of the Spalart-Allmaras Model

Chapter 2 selected the SA model because it offers a relatively compact RANS closure for the flow cases considered here. In the implementation, the SA equation is solved inside the outer Picard loop as a steady transport problem for the working variable $\tilde{\nu}$. Some convection-dominated verification runs add the SUPG and pseudo-time terms described below. In the topology-optimization runs, the SA pseudo-time term is disabled so the frozen adjoint is assembled from the same steady residual used in the final forward state. The continuous steady SA equation is

$$\mathbf{u} \cdot \nabla \tilde{\nu} = c_{b1} \tilde{S} \tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{y} \right)^2 + \frac{1}{\sigma} [\nabla \cdot ((\nu + \tilde{\nu}) \nabla \tilde{\nu}) + c_{b2} (\nabla \tilde{\nu})^2] \quad (3.10)$$

Here ν is the molecular kinematic viscosity, $\sigma = 2/3$ is the turbulent Prandtl number, c_{b1} and c_{w1} are calibrated SA constants, $\tilde{S}$ is the modified vorticity magnitude, f_w is the wall-destruction damping function, and y is the distance to the nearest solid wall [50].

3.3.1 Picard-Linearized Weak SA Transport Form

The FEniCS implementation does not solve Eq. 3.10 as a fully nonlinear scalar problem. Instead, it uses the outer Picard state to freeze the nonlinear SA closure terms and then solves a linearized weak problem for the next working-variable field. The closure functions used in this weak form are defined in the following subsection.

Let $\tilde{\nu}^k$ denote the previous Picard value of the SA working variable, while $\tilde{\nu}^{k+1}$ is the unknown being solved. The modified vorticity $\tilde{S}^k$, wall function f_w^k , diffusion coefficient, and explicit source terms are evaluated from $\tilde{\nu}^k$ and the current velocity field. The scalar finite-element space for the transported turbulence variable is denoted by T_h , analogous to the velocity space V_h and pressure space Q_h used in the IPCS weak forms. The Picard-linearized SA step is: find $\tilde{\nu}^{k+1} \in T_h$ such that, for all scalar test functions $\xi \in T_h$,

$$\begin{aligned} & \int_{\Omega} (\mathbf{u} \cdot \nabla \tilde{\nu}^{k+1}) \xi \, dx + \frac{1}{\sigma} \int_{\Omega} (\nu + \tilde{\nu}^k) \nabla \tilde{\nu}^{k+1} \cdot \nabla \xi \, dx \\ & + \int_{\Omega} c_{w1} f_w^k \frac{\tilde{\nu}^k}{y^2} \tilde{\nu}^{k+1} \xi \, dx \\ & = \int_{\Omega} \left[c_{b1} \tilde{S}^k \tilde{\nu}^k + \frac{c_{b2}}{\sigma} \nabla \tilde{\nu}^k \cdot \nabla \tilde{\nu}^k \right] \xi \, dx. \end{aligned} \quad (3.11)$$

The left-hand side contains the advection term $(\mathbf{u} \cdot \nabla \tilde{\nu}^{k+1})$, implicit diffusion, and implicit wall destruction. Production and cross-diffusion remain on the right-hand side as explicit Picard sources. After solving Eq. 3.11, the working variable is under-relaxed and bounded before the turbulent eddy viscosity ν_t is recomputed from the standard SA algebraic relation.

When pseudo-time stabilization is enabled for the SA solve, as in the steady U-bend verification configuration, the left-hand side of Eq. 3.11 is augmented by

$$\int_{\Omega} \frac{\tilde{\nu}^{k+1} - \tilde{\nu}^k}{\Delta t_{\tilde{\nu}}} \xi \, dx. \quad (3.12)$$

This is a numerical damping term for the Picard update of the transported working variable; it does not alter the steady SA closure being targeted after convergence. It is therefore treated as optional forward-solver stabilization, not as a defining part of the topology-optimization formulation.

For convection-dominated internal-flow cases, the implementation may augment the SA transport solve with SUPG stabilization [17]. This stabilization is applied to the transported working variable $\tilde{\nu}$ only; it is not a modification of the RANS momentum equation and it is not an additional turbulence-model source term.

Jury question: Does SUPG change the turbulence model and bias your optimized shapes?

Answer: It changes the numerical transport discretization of `nu_tilde`, not the physical SA closure. The purpose is to suppress element-scale oscillations in convection-dominated SA transport. It is still a numerical modeling choice, so it belongs in the solver setup and should remain fixed when comparing designs.

With element size h and local speed $|\mathbf{u}|$, the added streamline contribution has the form

$$F_{\text{SUPG},\tilde{\nu}} = \int_{\Omega} \tau_{\tilde{\nu}}(\mathbf{u} \cdot \nabla \xi) \mathcal{R}_{\tilde{\nu}} dx, \quad \tau_{\tilde{\nu}} = C_{\text{SUPG}} \frac{h}{2|\mathbf{u}|}, \quad (3.13)$$

where $\mathcal{R}_{\tilde{\nu}}$ is the strong residual corresponding to Eq. 3.11: the pointwise advection, diffusion, wall-destruction reaction, and explicit production and cross-diffusion sources are collected on one side before being projected along the streamline test direction. When Eq. 3.12 is active, its pseudo-time contribution is included in the same residual. The coefficient C_{SUPG} is a user-selected stabilization factor. In the verification benchmarks and in the turbulent topology-optimization runs, Eq. 3.13 is used to suppress element-scale oscillations in $\tilde{\nu}$ while preserving the same underlying SA closure.

3.3.2 Closure Terms and Calibration Constants

The FEniCS implementation uses the standard vorticity-based SA closure functions and calibration constants from the reference model [50, 46]. These functions define the modified vorticity $\tilde{S}$, the near-wall damping functions, the wall-destruction term, and the empirical constants used in Eq. 3.11. The full list of closure expressions is moved to Appendix B.1, because it is standard model-definition material rather than a thesis-specific derivation. The only thesis-specific modeling choice in this part is that the geometric trip functions are omitted, following the topology-optimization convention of Yoon [61], because the solid-fluid boundary is not known a priori during optimization.

3.3.3 Wall-Distance Calculation via the Relaxed Eikonal Equation

A major computational challenge in the SA model is the calculation of the wall distance y . In a standard static CFD simulation, y is purely geometric and can be pre-computed.

However, to ensure this verification solver translates into the topology optimization code, where solid boundaries evolve continuously, the implementation uses a PDE-based approach.

Jury question: Why solve a PDE for wall distance instead of computing exact geometric distance?

Answer: Exact distance is simple for a fixed wall but becomes awkward and non-smooth when the wall is created by a density field. The relaxed reciprocal-distance PDE gives a differentiable field that can be penalized in topology-created solids, which is more compatible with FEniCS/UFL and the optimization loop.

The true geometric wall distance satisfies the standard Eikonal equation, $|\nabla y| = 1$. However, this formulation contains singularities at the medial axes of the domain. To guarantee continuous differentiability, a relaxed reciprocal-distance equation proposed by Yoon [61] is solved:

$$\nabla G \cdot \nabla G + \sigma_w G (\nabla \cdot \nabla G) = (1 + 2\sigma_w) G^4. \quad (3.14)$$

Here, G is an intermediate reciprocal-distance field variable. The parameter σ_w is a relaxation constant, typically set to 0.1, that controls the smoothness of the field. The PDE is subjected to a Dirichlet boundary condition at the physical walls, $G = G_0$, where G_0 is an arbitrarily large constant. Once G is solved over the domain, the physical wall distance y is recovered algebraically using the inverse relationship:

$$y = \frac{1}{G} - \frac{1}{G_0}. \quad (3.15)$$

The code applies small numerical floors and non-negativity safeguards during this recovery to prevent division-by-zero and roundoff artifacts, but the mathematical equation used in the thesis remains the Yoon formulation above.

This system is implemented as a nonlinear weak residual for G . Following the Galerkin statement of Yoon’s relaxed wall-distance equation [61], the PDE is multiplied by a scalar test function and the $\sigma_w G \nabla^2 G$ term is integrated by parts, so that the residual contains only first derivatives of G :

$$(1 - \sigma_w) \int_{\Omega} |\nabla G|^2 z \, dx - \sigma_w \int_{\Omega} G \nabla G \cdot \nabla z \, dx - (1 + 2\sigma_w) \int_{\Omega} G^4 z \, dx = 0. \quad (3.16)$$

3.4 Picard-IPCS Coupling Algorithm

The coupling sequence is the same for the fixed-wall verification benchmarks, so it is described here as part of the solver rather than repeated for each geometry. The relaxed Eikonal problem in Eq. 3.16 is solved first to obtain the wall-distance field y required by the SA closure. The RANS-SA fields are then advanced with the outer Picard loop shown in Figure 3.1.

During Picard iteration k , the previous SA field $\tilde{\nu}^k$ is converted to ν_t^k and inserted into the tentative-velocity problem through $\nu + \nu_t^k$. The velocity-pressure state is advanced by Eqs. 3.7–3.9. The resulting velocity $\mathbf{u}^{k+1}$ then convects the SA weak problem in Eq. 3.11, with the optional pseudo-time and SUPG additions of Eqs. 3.12 and 3.13. After this scalar solve, the new working variable is under-relaxed and bounded, the eddy viscosity is updated, and the Picard residual is evaluated.

The pressure-drop and profile-error definitions are intentionally not part of this algorithmic section. They depend on the benchmark geometry and boundary conditions, so they are introduced with the channel and U-bend verification cases in Chapter 4.

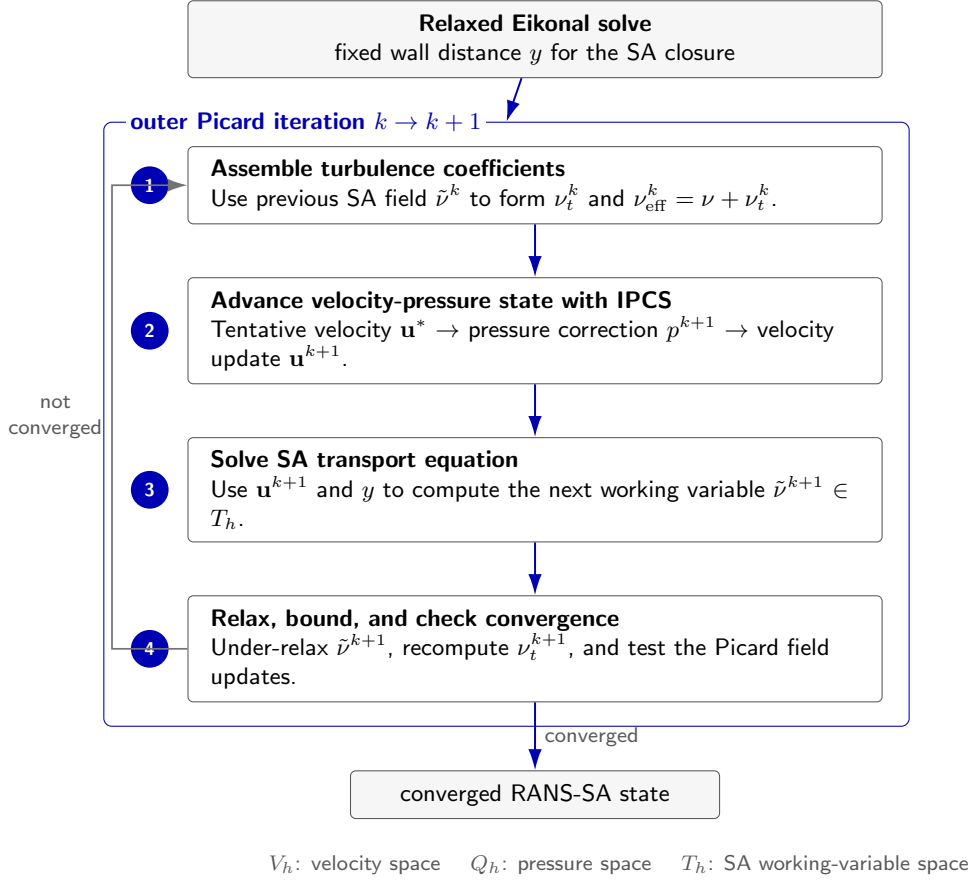


FIGURE 3.1: Nested Picard-IPCS coupling workflow (ChatGPT, 2026).

3.5 Conclusion

This chapter has established the numerical formulation of the FEniCS-SA forward solver used for the verification benchmarks. The key ingredients are the weak-form finite-element implementation, Picard decoupling of the velocity-pressure and turbulence fields, IPCS pseudo-time stepping for the incompressible flow solve, the steady SA transport equation, and the relaxed Eikonal wall-distance formulation.

Separating these methods from their verification keeps the thesis logic clean: this chapter explains how the forward SA solver is constructed, Chapter 4 verifies its turbulent-flow behavior against Ansys Fluent benchmark cases, and Chapter 5 extends the verified forward solver to density-based topology optimization and frozen adjoint sensitivities.

Chapter 4

Verification of the Spalart-Allmaras Solver for Turbulent Internal Flows

4.1 Introduction

Chapter 3 introduced the numerical implementation of the Spalart-Allmaras (SA) model in FEniCS. Before using this solver inside the topology optimization framework, the forward fluid solver is verified against independent Ansys Fluent calculations on two benchmark flows. The first benchmark is a fully developed turbulent channel, which isolates wall damping, turbulence production, and Picard coupling in a simple geometry. The second benchmark is a curved U-bend based on the VMFL048 operating point discussed in Chapter 2, where SA remained credible in the Ansys comparison. It therefore tests the same implementation under stronger convection, streamline curvature, and possible local separation, without turning the verification case into a validation of strongly separated flow.

This chapter keeps the essential benchmark definitions in the main text: geometry type, boundary conditions, Reynolds number, wall resolution, and the comparison metrics used to judge agreement. Detailed mesh hierarchies, solver controls, and configuration tables are collected in Appendix C. The purpose is to verify the forward-flow solver developed in Chapter 3, because this is the turbulent-flow core on which the topology optimization framework is built.

4.2 Channel Verification Benchmark

To verify the fundamental correctness of the custom fluid and turbulence solvers, the code was first tested on a standard 2D turbulent channel flow. To cleanly isolate turbulence production, wall damping, and Picard coupling without the artifacts of a developing boundary layer, this benchmark was formulated as a pressure-driven periodic channel rather than a conventional inlet-outlet duct.

Because the SA formulation is used here as a low-Reynolds wall-resolved model, the mesh must resolve the viscous sublayer instead of relying on empirical wall functions. The channel and U-bend verification meshes are therefore designed for approximately $y^+ = 1$ at the no-slip walls. Appendix C documents the channel mesh-independence checks, the selected FEniCS U-bend mesh, and the Ansys U-bend grid-independence baseline.

The same finite-element implementation choices are used for both verification benchmarks: the momentum equation is assembled with the symmetric effective-viscosity stress form, while SUPG stabilization is restricted to the SA working-variable transport equation.

4.2.1 Geometry and Boundary Conditions

The velocity field ($\mathbf{u}$) and Spalart-Allmaras working variable ($\tilde{\nu}$) were constrained periodically between the streamwise boundaries, while the top and bottom boundaries were treated as standard no-slip walls ($\mathbf{u} = 0, \tilde{\nu} = 0$). Flow was driven entirely by a fixed pressure drop from the upstream plane ($p = 2$ Pa) to the downstream plane ($p = 0$ Pa). Based on a reference bulk velocity of $U_{\text{ref}} = 18.5$ m/s and using the full channel height ($H = 2.0$ m) as the characteristic length scale, the solver was targeted to resolve a fully turbulent flow regime at a channel Reynolds number of $Re_H \approx 20,300$. The detailed physical and numerical settings are listed in Appendix C.

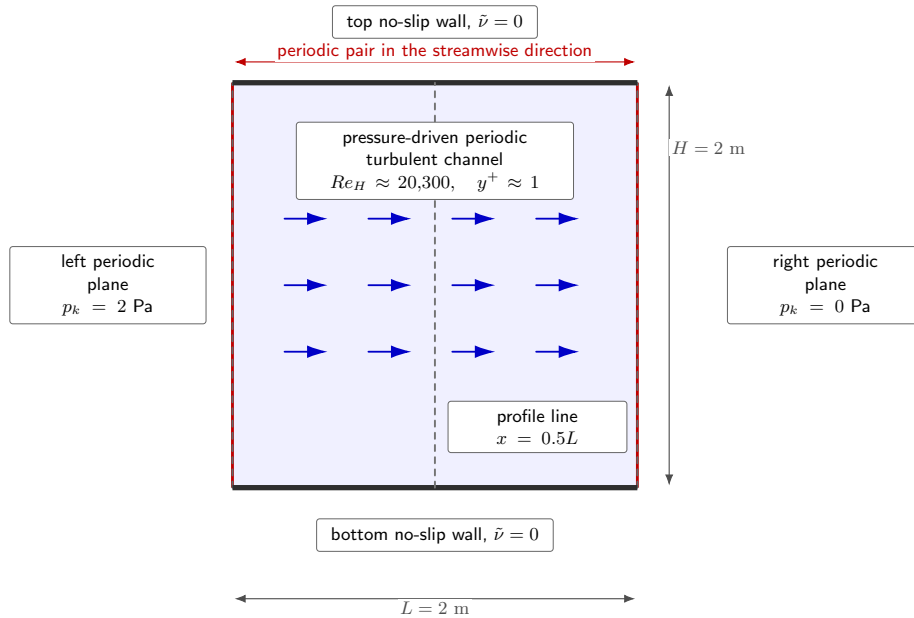


FIGURE 4.1: Pressure-driven periodic channel setup (ChatGPT, 2026).

4. VERIFICATION OF THE SPALART-ALLMARAS SOLVER FOR TURBULENT INTERNAL FLOWS

4.2.2 Validation Against Ansys Fluent

The channel flow was simulated in FEniCS using the decoupled Picard iteration method and compared against an Ansys Fluent SA baseline. Both solvers were run on wall-resolved meshes, and the grid-independence checks are reported in Appendix C. The vertical dashed line in Figure 4.1 marks the mid-plane profile extraction location used for the velocity-profile comparison.

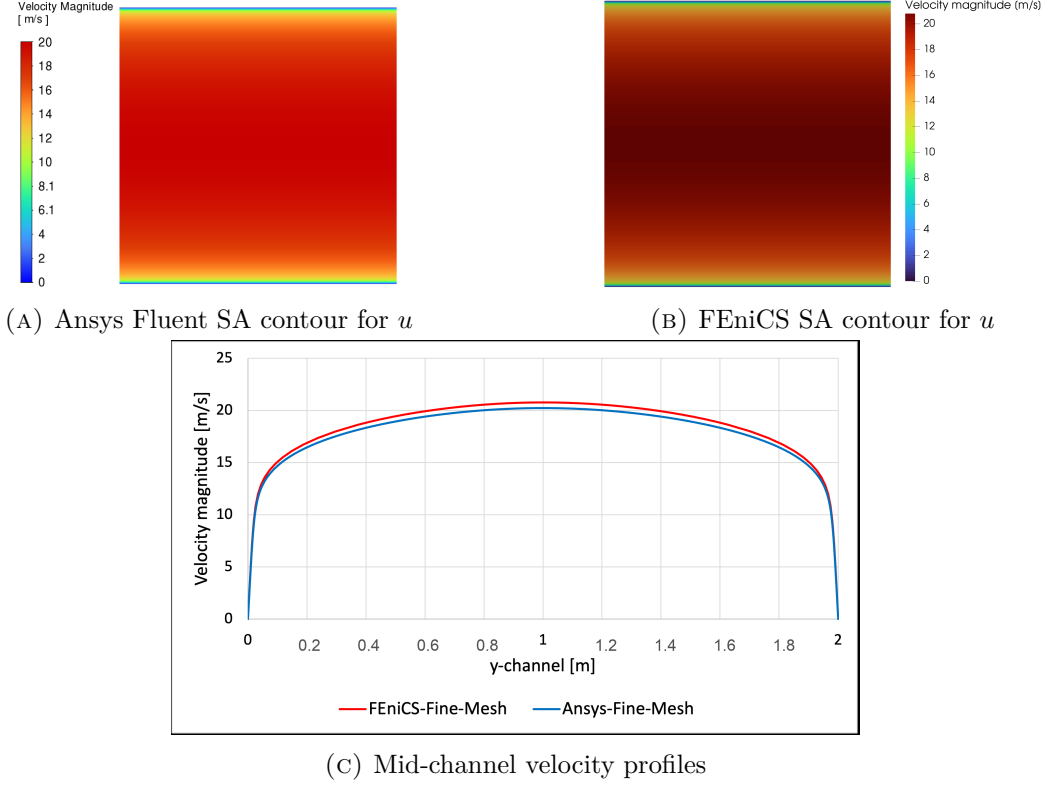


FIGURE 4.2: Channel-flow validation against Ansys Fluent.

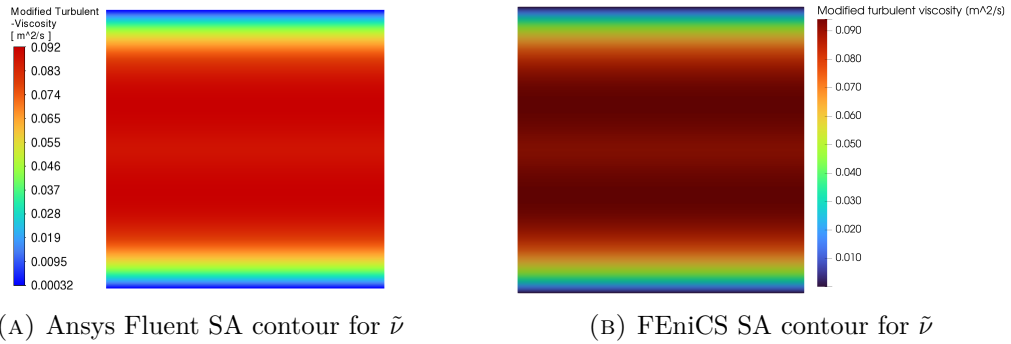


FIGURE 4.3: Modified turbulent-viscosity comparison for the channel benchmark.

As shown in Figure 4.2, the qualitative velocity contours of the custom FEniCS implementation show good agreement with the Ansys Fluent baseline. Both solvers produce a fully developed, streamwise-uniform channel field with zero velocity at the walls and a broad high-speed core around the centerline. The remaining visual differences are confined mainly to the thickness of the near-wall color bands and the exact width of the central high-velocity region, which are expected to be sensitive to mesh resolution, plotting scale, and the different finite-element and finite-volume discretizations. For the quantitative comparison, velocity profiles are extracted from both solvers across the middle of the channel, i.e. at $x = 0.5L$, with y varied from the lower wall to the upper wall. The custom solver resolves the logarithmic region of the turbulent boundary layer and predicts the turbulent diffusion toward the channel center.

Figure 4.3 shows the corresponding modified turbulent viscosity field. The two contours have the same physical structure: $\tilde{\nu}$ is suppressed at the no-slip walls by the SA boundary condition and wall-destruction term, then rises rapidly through the buffer/log region toward an almost uniform core value. The FEniCS field is slightly smoother in the wall-normal transition, but it preserves the same symmetry and wall-to-core trend as the Ansys reference. This agreement is important because $\tilde{\nu}$, rather than velocity alone, controls the eddy-viscosity field used by the momentum equation.

Because the channel is driven by a prescribed pressure difference, $\Delta p = 2.0$ Pa is identical by construction and is not an independent comparison metric.

Jury question: Why do you report bulk velocity instead of pressure drop for the channel verification?

Answer: The pressure drop is imposed, so matching it would not verify the solver. The meaningful response is the flow rate or bulk velocity produced by that pressure gradient, plus the velocity and nu_tilde profiles.

The retained global metric is therefore the bulk velocity,

$$U_{\text{bulk}} = \frac{1}{H} \int_0^H u_x(0.5L, y) dy, \quad (4.1)$$

where u_x is the streamwise velocity component and H is the channel height.

The two error metrics are cross-code discrepancies, not separate errors for each solver. The exported fine-mesh profiles provide velocity magnitude along the wall-normal channel line; for this fully developed channel this is equivalent to the streamwise velocity within the exported precision. After interpolating the FEniCS profile onto the Ansys line coordinates y_i , the pointwise difference is

$$e_i = u_{x,\text{FEniCS}}(y_i) - u_{x,\text{Ansys}}(y_i). \quad (4.2)$$

The maximum pointwise velocity discrepancy is normalized by the maximum Ansys profile velocity to avoid singular relative errors at the no-slip walls,

$$E_{\infty} = \frac{\max_i |e_i|}{\max_i |u_{x,\text{Ansys}}(y_i)|} \times 100\%, \quad (4.3)$$

and the relative profile error is evaluated as a line-integrated norm,

$$E_{L_2} = \left(\frac{\int_0^H |u_{x,\text{FEniCS}} - u_{x,\text{Ansys}}|^2 dy}{\int_0^H |u_{x,\text{Ansys}}|^2 dy} \right)^{1/2} \times 100\%. \quad (4.4)$$

These profile metrics are computed from exported solver data after interpolating the FEniCS profile onto the Ansys profile coordinates.

TABLE 4.1: Macroscopic flow metric and fine-mesh profile discrepancies for the 2D channel benchmark.

Metric	Ansys	FEniCS	Discrepancy
Bulk velocity (U_{bulk}) [m/s]	17.977	18.458	2.67%
Maximum profile discrepancy (E_∞)	fine-mesh velocity profile		2.66%
Relative profile norm (E_{L_2})	fine-mesh velocity profile		2.64%

This agreement supports the verification that the core Navier-Stokes solver, relaxed Eikonal wall-distance field, and SA production/destruction terms behave correctly in a canonical wall-bounded turbulent flow. The main-text parameters are sufficient to interpret the result, while Appendix C provides the detailed mesh and solver settings needed to reproduce it.

4.3 U-Bend Verification Benchmark

Following the channel flow verification, the solver was tested on the highly convective U-bend operating point evaluated in Chapter 2. This case is closer to the curved internal-flow topologies studied later in the thesis because the flow experiences strong streamline curvature, pressure-gradient-driven momentum redistribution, and may include local near-wall separation. It was selected from the VMFL048 comparison because SA performed reasonably there; the more severe VMFL030 bend and the backward-facing-step evidence remain the cautionary cases for stronger curvature- and separation-sensitive predictions.

4.3.1 Geometry and Boundary Conditions

The verification used a 2D geometry corresponding to the symmetry mid-plane of the full 3D VMFL048 case.

Since the FEniCS domain is mathematically planar, it cannot resolve true 3D Dean secondary vortices. The Dean number is therefore retained only as an inherited descriptor of the corresponding 3D curved-pipe operating point, not as evidence that the 2D model contains Dean-vortex physics.

Jury question: Does the 2D U-bend verification prove your solver captures Dean-vortex physics?

Answer: No. It verifies the 2D wall-resolved SA implementation against a matched 2D Ansys baseline. Dean number is retained to connect the setup to the original physical case, but secondary vortices and 3D separation are explicitly outside the present model scope.

The straight inlet and outlet legs are shorter than in the original VMFL048 Ansys mesh to reduce meshing and computational effort; both the FEniCS and Ansys verification runs use this same short-leg U-bend geometry.

To drive the flow, a uniform prescribed velocity ($\mathbf{u}_{in} = 1.42 \text{ m/s}$) was imposed at the inlet alongside a fully turbulent boundary value ($\tilde{\nu}_{in} = 9.34 \times 10^{-5} \text{ m}^2/\text{s}$), while a zero relative static pressure ($p = 0 \text{ Pa}$) was assigned to the outlet. While the computational domain is 2D, the 3D physical pipe diameter ($D = 0.028 \text{ m}$) and radius of curvature ($R_c = 0.125 \text{ m}$) were retained as the characteristic length scales. This gives the same nominal Reynolds number ($Re_D \approx 44,700$) and reference Dean number ($De \approx 15,000$) as the 3D VMFL048 operating point. The detailed setup is listed in Appendix C.

As in the channel benchmark, the U-bend mesh must be wall resolved because the SA model is integrated to the wall rather than coupled to wall functions. The selected FEniCS verification mesh and the Ansys grid-independence information are kept in Appendix C.

The U-bend is significantly more convection dominated than the periodic channel benchmark, but it uses the same SA-transport SUPG stabilization described in Chapter 3. This stabilization is applied only to the transported working variable $\tilde{\nu}$; no additional SUPG term is introduced in the momentum equation.

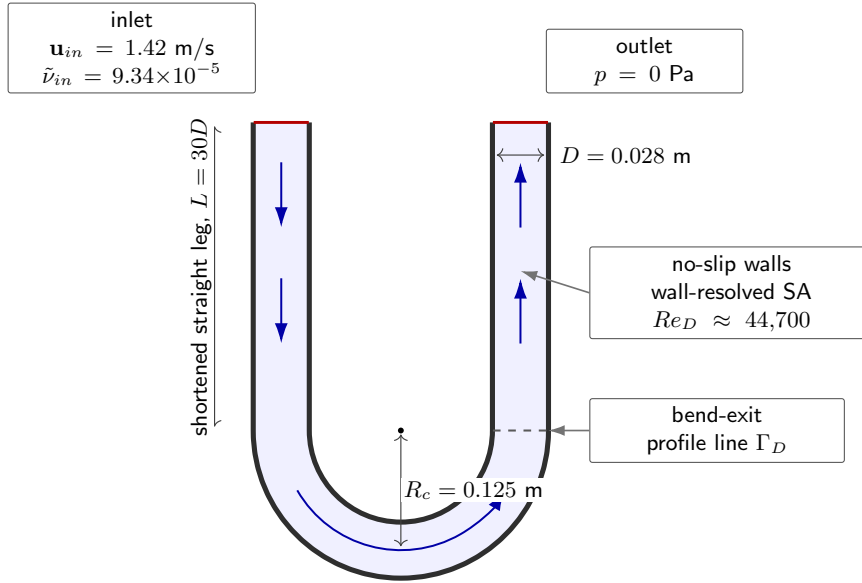


FIGURE 4.4: Shortened 2D U-bend verification setup (ChatGPT, 2026).

4.3.2 Validation Against Ansys Fluent

The U-bend was simulated on the shortened 2D domain in both FEniCS and Ansys Fluent. The setup is based on the VMFL048 operating point from the Ansys verification manual [9], and the profile line Γ_D in Figure 4.4 is used for the bend-exit

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velocity comparison. The Ansys baseline was checked for grid independence, and the corresponding mesh information is documented together with the selected FEniCS medium mesh in Appendix C. The side-by-side velocity contour panels below are cropped and scaled to the same geometry window while retaining the original color scales, so the visual comparison is not driven by unequal image canvases.

The FEniCS–Ansys comparison is therefore made on the selected `Medium_FEniCS` mesh. The Ansys baseline is supported by a separate grid-independence check, but no full coarse–medium–fine FEniCS mesh-convergence study is claimed for this U-bend case. The reported agreement should be interpreted as cross-code verification on the selected wall-resolved FEniCS mesh, not as a FEniCS mesh-independent result.

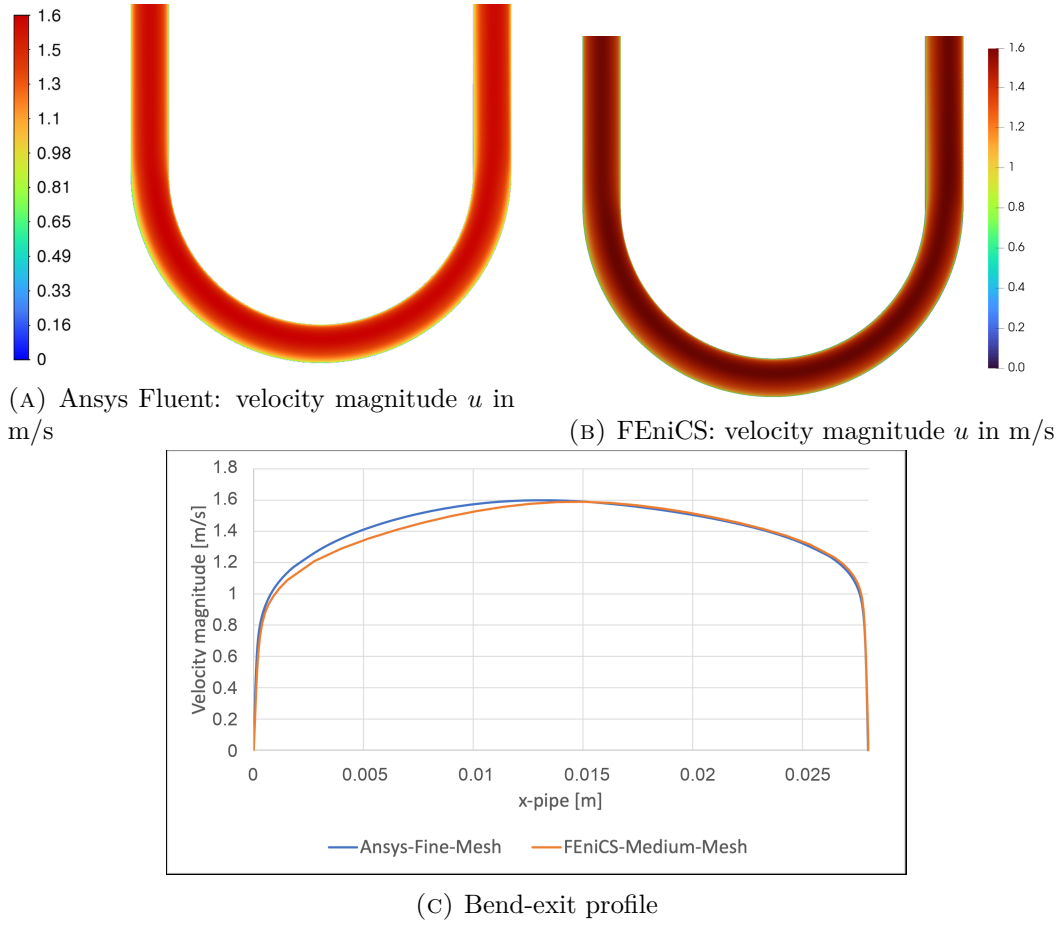


FIGURE 4.5: U-bend velocity validation.

As shown in Figure 4.5, the velocity-magnitude contours do not show a large recirculation bubble or extensive detached region in either solver. This does not rule out a small local separation region near the bend wall, because separation is more directly identified from wall shear, streamlines, or velocity-vector reversal than from velocity magnitude alone. The plotted fields mainly show that both solvers recover

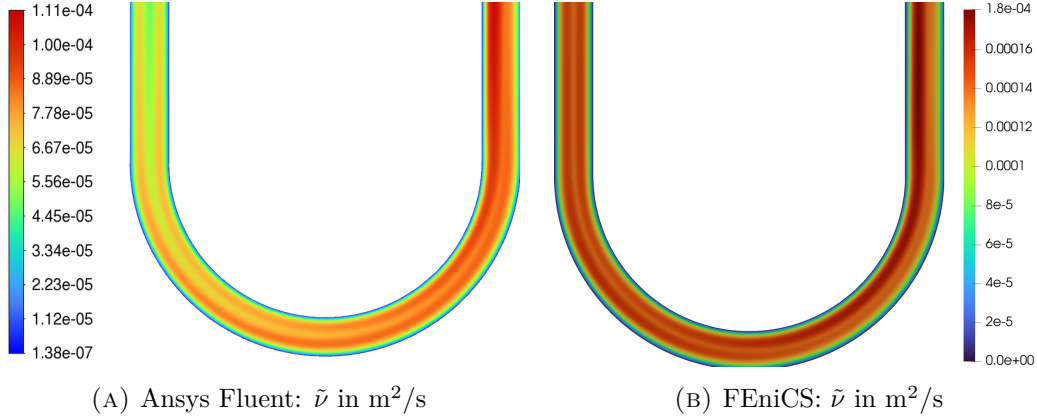


FIGURE 4.6: U-bend modified turbulent viscosity.

the same curvature-dominated velocity structure: low-speed wall layers and a broad high-speed core convected through the bend. This is consistent with the VMFL048 evidence in Chapter 2, where SA behaved reasonably for this moderate-curvature U-bend despite its known limitations in separated flows. The most relevant differences are local: the bend-exit profile shows Ansys predicting slightly higher velocity over part of the inner-side half of the diameter, while FEniCS is slightly higher over part of the outer-side half. The two curves nevertheless overlap closely over most of the diameter and reach nearly the same peak velocity.

Figure 4.6 provides the companion check on the SA transport variable. The two fields share the same organization as in the channel case: $\tilde{\nu}$ is suppressed at the walls and rises in the interior where the modeled turbulent diffusion is active. The important difference is the magnitude. The FEniCS contour peaks around $1.8 \times 10^{-4} \text{ m}^2/\text{s}$, while the Ansys contour peaks around $1.1 \times 10^{-4} \text{ m}^2/\text{s}$. This should not be read as a different turbulence model equation, but as a local cross-code discrepancy in the transported working variable caused by the combined effects of mesh, finite-element versus finite-volume discretization, pressure-velocity coupling, stabilization, and boundary treatment. Since the velocity profile and pressure drop remain close, the discrepancy is acceptable for forward-solver verification, but it prevents a stronger claim of pointwise turbulent-viscosity agreement.

For the U-bend, the inlet velocity is prescribed rather than produced by a fixed pressure gradient, so pressure drop remains a meaningful global comparison metric. In FEniCS, the pressure-drop diagnostic is evaluated from mean pressures on the inlet and outlet boundaries,

$$\Delta p_{\text{FEniCS}} = \rho \left[\frac{1}{|\Gamma_{\text{in}}|} \int_{\Gamma_{\text{in}}} p_k \, d\Gamma - \frac{1}{|\Gamma_{\text{out}}|} \int_{\Gamma_{\text{out}}} p_k \, d\Gamma \right], \quad (4.5)$$

where p_k is the kinematic pressure solved by the IPCS formulation, $\rho = 997 \text{ kg}/\text{m}^3$, and $|\Gamma| = \int_{\Gamma} 1 \, d\Gamma$. The density factor converts the kinematic pressure difference to pascals. The same local comparison metrics used for the channel are computed from

the exported bend-exit profiles as

$$E_\infty = \frac{\max_i ||\mathbf{u}_{\text{FEniCS}}(s_i)| - |\mathbf{u}_{\text{Ansys}}(s_i)||}{\max_i |\mathbf{u}_{\text{Ansys}}(s_i)|} \times 100\%, \quad (4.6)$$

and

$$E_{L_2} = \left(\frac{\int_{\Gamma_D} (|\mathbf{u}_{\text{FEniCS}}| - |\mathbf{u}_{\text{Ansys}}|)^2 ds}{\int_{\Gamma_D} |\mathbf{u}_{\text{Ansys}}|^2 ds} \right)^{1/2} \times 100\%, \quad (4.7)$$

where Γ_D denotes the line segment across the bend-exit diameter and s is the corresponding arclength coordinate. As in the channel case, the profile norms are evaluated from exported solver data after interpolating the FEniCS profile onto the Ansys profile coordinates.

TABLE 4.2: Pressure-drop comparison and bend-exit profile discrepancies for the 2D U-bend benchmark.

Metric	Ansys	FEniCS	Discrepancy
Pressure drop (Δp) [Pa]	792.995	815.130	2.79%
Maximum profile discrepancy (E_∞)	bend-exit velocity profile		8.50%
Relative profile norm (E_{L_2})	bend-exit velocity profile		2.75%

4.4 Conclusion

The two verification cases answer the second research question by establishing the practical operating range of the custom FEniCS-SA solver before it is embedded in the topology optimization loop. The periodic channel confirms that the solver reproduces canonical wall-bounded turbulent behavior, including the near-wall damping and production balance that are central to the SA model. On the fine mesh, the channel bulk velocity differs from the Ansys reference by 2.67%, while the exported diameter profile gives $E_\infty = 2.66\%$ and $E_{L_2} = 2.64\%$. The U-bend then checks the same implementation in a curved, convection-dominated internal flow. Its pressure drop differs from Ansys by 2.79%, and the bend-exit profile gives $E_\infty = 8.50\%$ and $E_{L_2} = 2.75\%$.

These results should be interpreted as a targeted verification of the forward solver, not as a claim that the custom implementation perfectly aligns with Ansys Fluent or that SA is generally reliable in separated flow. The known limitations of a 2D SA model remain: secondary Dean vortices, anisotropic Reynolds stresses, and full three-dimensional separation physics are outside the present solver scope. The turbulent-viscosity fields also do not show a universal FEniCS over- or under-prediction: the channel fields are close in magnitude, while the U-bend FEniCS field has a higher local peak than Ansys. Within this scope, however, the implemented SA equations, wall-distance calculation, and segregated coupling strategy produce stable and physically plausible turbulent internal-flow fields that are sufficiently accurate for use as the forward-flow solver in the topology optimization studies.

Chapter 5

Turbulent Topology Optimization Methodology

5.1 Introduction

Building upon the Spalart-Allmaras (SA) fluid dynamics solver [49, 50] developed in Chapter 3 and verified in Chapter 4, this chapter introduces the density-based topology optimization (TO) formulation used for the turbulent benchmark cases. The forward model remains the same wall-resolved RANS-SA framework, but it is embedded in a fixed-grid porous-medium design loop with Brinkman momentum penalization, topology-aware SA and wall-distance penalties, density filtering, projection, and gradient-based updates.

The purpose of the chapter is to define the mathematical ingredients that are needed to interpret Chapter 6. More detailed implementation support, including the RAMP and Heaviside curves, the design-bound sketch, the DG0 filter form, and inlet-turbulence conversion formulas, is collected in Appendix D.

5.2 Brinkman Penalization for Density-Based Flow TO

In density-based topology optimization, the computational domain Ω remains fixed while a density-like topology field identifies fluid and solid regions. In the convention used here, the physical topology value $\bar{\gamma} = 1$ represents fluid, while $\bar{\gamma} = 0$ represents solid material. The solid-fluid boundary is not fitted by the mesh during the optimization. Instead, low-density regions are treated through a porous-medium relaxation that strongly damps the local velocity and converges toward a sharp solid boundary after projection and extraction.

Following standard density-based flow topology optimization formulations [15, 29], the momentum resistance is introduced through an inverse permeability, or Brinkman penalty, $\alpha(\bar{\gamma})$. The implementation uses the RAMP interpolation [52]

$$\alpha(\bar{\gamma}) = \alpha_{solid} + (\alpha_{fluid} - \alpha_{solid}) \frac{\bar{\gamma}(1+q)}{\bar{\gamma}+q}, \quad (5.1)$$

where α_{solid} is the solid resistance, α_{fluid} is the fluid resistance, and q controls the interpolation curve. Other interpolation choices, including SIMP-type laws, are also used in the fluid-TO literature; Eq. 5.1 is the convention used in the present code and results.

5.2.1 Density Chain and Domain Bounds

The optimizer updates a raw DG0 design field γ , but the governing equations are assembled with the bounded physical field $\bar{\gamma}$. The density chain is

$$\gamma \longrightarrow \tilde{\gamma} \longrightarrow \bar{\gamma}, \quad (5.2)$$

where $\tilde{\gamma}$ is the filtered density and $\bar{\gamma}$ is the filtered, projected, and bounded density. In the implementation this final mapping is

$$\bar{\gamma} = \gamma_{lower} + (\gamma_{upper} - \gamma_{lower})\Pi_{\beta}(\tilde{\gamma}). \quad (5.3)$$

The fields γ_{lower} and γ_{upper} are not additional optimizer variables. They are cell-wise lower and upper bounds used to impose prescribed non-design regions: freely optimized cells use bounds 0 and 1, fixed-fluid cells use bounds 1 and 1, and fixed-solid cells use bounds 0 and 0. This keeps inlet ducts, outlet ducts, separator bars, and other case-specific regions fixed during the optimization. The same masking convention is used for objective and volume integrations; the relevant regions are shown with each benchmark setup in Chapter 6.

Jury question: Could the optimizer cheat by changing inlet, outlet, or separator regions?

Answer: No. Those cells are removed from the active design through lower and upper density bounds and through the objective/volume masks. The optimizer only updates active DG0 design variables; fixed fluid or fixed solid regions are restored after filtering/projection.

5.3 Penalization of the Governing Equations

Topology enters the governing equations through penalty terms. The Brinkman momentum term and the default SA working-variable damping use the bounded projected density $\bar{\gamma}$. The reciprocal wall-distance penalty also follows the current design density in the benchmark configurations; implementation variants are summarized in Appendix D.

5.3.1 Modified Reynolds-Averaged Navier-Stokes Equations

The steady incompressible RANS momentum equation is modified by adding the Brinkman sink term:

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nabla \cdot [(\mu + \mu_t)(\nabla\mathbf{u} + (\nabla\mathbf{u})^T)] - \alpha(\bar{\gamma})\mathbf{u}. \quad (5.4)$$

As $\bar{\gamma} \rightarrow 0$, the Brinkman resistance approaches α_{solid} and drives the local velocity toward zero.

The final velocity-pressure state used for the frozen adjoint is represented by a steady weak residual. With $\mu_{\text{eff}} = \mu + \mu_t^{fr}$ and test functions $(\mathbf{v}, q)$, the residual is

$$\begin{aligned} R_{\text{NS}}((\mathbf{u}, p), (\mathbf{v}, q)) = & \int_{\Omega} \rho(\mathbf{u} \cdot \nabla) \mathbf{u} \cdot \mathbf{v} \, dx + \frac{1}{2} \int_{\Omega} \mu_{\text{eff}} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) : (\nabla \mathbf{v} + \nabla \mathbf{v}^T) \, dx \\ & + \int_{\Omega} \nabla p \cdot \mathbf{v} \, dx + \int_{\Omega} q \nabla \cdot \mathbf{u} \, dx + \int_{\Omega} \alpha(\bar{\gamma}) \mathbf{u} \cdot \mathbf{v} \, dx = 0. \end{aligned} \quad (5.5)$$

When the final forward solve is performed with SNES, SNES solves Eq. 5.5 for the accepted final state. The adjoint is then assembled by linearizing the same residual at that state.

5.3.2 Modified Spalart-Allmaras Transport Equation

Topology-created solid regions should not retain turbulent working-variable content. A sink term is therefore added to the SA transport equation [61, 22]:

Jury question: Why add a topology penalty to nu_tilde instead of relying only on Brinkman damping?

Answer: Brinkman damping suppresses velocity, but it does not automatically make the turbulence working variable behave like it is adjacent to a wall. Without a nu_tilde sink, solid-like regions could retain artificial eddy viscosity. The penalty makes topology-created solids more wall-like for the SA model.

$$\begin{aligned} \mathbf{u} \cdot \nabla \tilde{\nu} = & c_{b1} \tilde{S} \tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{y} \right)^2 + \frac{1}{\sigma} [\nabla \cdot ((\nu + \tilde{\nu}) \nabla \tilde{\nu}) + c_{b2} (\nabla \tilde{\nu})^2] \\ & - \alpha_{\tilde{\nu}} (1 - \bar{\gamma})^{N_{\tilde{\nu}}} \tilde{\nu}. \end{aligned} \quad (5.6)$$

The factor $(1 - \bar{\gamma})$ is the local solid indicator. The coefficient $\alpha_{\tilde{\nu}}$ sets the damping strength and the exponent $N_{\tilde{\nu}}$ controls how sharply the damping grows as material becomes solid. In the reported cases, $\alpha_{\tilde{\nu}}$ can be either fixed or advanced by a continuation schedule, while $N_{\tilde{\nu}}$ remains fixed.

In the Picard-linearized finite-element solve, the topology penalty enters as an additional reaction term. Using the previous Picard field $\tilde{\nu}^k$ to evaluate the nonlinear coefficients, the SA update solves

$$\begin{aligned} & \int_{\Omega} (\mathbf{u} \cdot \nabla \tilde{\nu}^{k+1}) \xi \, dx + \frac{1}{\sigma} \int_{\Omega} (\nu + \tilde{\nu}^k) \nabla \tilde{\nu}^{k+1} \cdot \nabla \xi \, dx \\ & + \int_{\Omega} c_{w1} f_w^k \frac{\tilde{\nu}^k}{y^2} \tilde{\nu}^{k+1} \xi \, dx + \int_{\Omega} \alpha_{\tilde{\nu}} (1 - \bar{\gamma})^{N_{\tilde{\nu}}} \tilde{\nu}^{k+1} \xi \, dx \\ & = \int_{\Omega} \left[c_{b1} \tilde{S} \tilde{\nu}^k + \frac{c_{b2}}{\sigma} \nabla \tilde{\nu}^k \cdot \nabla \tilde{\nu}^k \right] \xi \, dx. \end{aligned} \quad (5.7)$$

5.3.3 Reciprocal Penalized Wall-Distance Equation

The SA model requires a wall distance y . In topology optimization, the relevant wall is not only the fixed mesh boundary; it also includes topology-created solid regions.

The reciprocal relaxed Eikonal formulation therefore penalizes the reciprocal field G toward a large value inside solid material [61]:

$$\nabla G \cdot \nabla G + \sigma_w G (\nabla \cdot \nabla G) - (1 + 2\sigma_w) G^4 - \alpha_G (1 - \bar{\gamma})^{N_G} (G - G_0) = 0. \quad (5.8)$$

The recovered wall distance is

$$y = \frac{1}{G} - \frac{1}{G_0}. \quad (5.9)$$

Thus, forcing G toward G_0 collapses the recovered distance toward zero in topology-created solid. As with the SA working-variable penalty, α_G can be continued by schedule, while N_G is kept fixed in the reported cases.

Jury question: What is the physical meaning of forcing the reciprocal wall-distance variable in solid regions?

Answer: It is not a new physical wall law; it is a topology-consistent way to tell the SA model that a density-created obstacle should act like a nearby wall. Driving G to G_0 makes the recovered wall distance approach zero, increasing wall-destruction behavior around emerging solids.

The weak residual used in the nonlinear wall-distance solve is

$$\begin{aligned} R_G(G, z) = & (1 - \sigma_w) \int_{\Omega} |\nabla G|^2 z \, dx - \sigma_w \int_{\Omega} G \nabla G \cdot \nabla z \, dx - (1 + 2\sigma_w) \int_{\Omega} G^4 z \, dx \\ & - \int_{\Omega} \alpha_G (1 - \bar{\gamma})^{N_G} (G - G_0) z \, dx = 0. \end{aligned} \quad (5.10)$$

5.4 Optimization Problem Formulation

The optimization problem combines the penalized state equations with a performance objective and a volume constraint. The two objectives used in the framework are defined below.

5.4.1 Objective Formulation

The dissipation objective is a volume power-loss functional consisting of physical viscous dissipation and Brinkman drag over the prescribed objective mask Ω_A [61, 22]:

$$J_D(\mathbf{u}, \bar{\gamma}) = \int_{\Omega_A} \left[\frac{1}{2} \mu_{\text{eff}} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) : (\nabla \mathbf{u} + \nabla \mathbf{u}^T) + \alpha(\bar{\gamma}) \mathbf{u} \cdot \mathbf{u} \right] d\Omega. \quad (5.11)$$

The framework also supports an average inlet-pressure objective,

$$J_p(p) = \frac{1}{|\Gamma_{\text{in}}|} \int_{\Gamma_{\text{in}}} p \, d\Gamma. \quad (5.12)$$

Here Γ_{in} is the physical inlet boundary of the fixed non-design extension.

The two objectives target the same physical idea—reducing hydraulic losses—but they expose it differently to the optimizer. The dissipation objective J_D measures losses as a volume integral of viscous dissipation and Brinkman drag in the objective region. The pressure objective J_p measures the mean inlet pressure directly; when the inlet flow rate and outlet reference pressure are fixed, minimizing J_p is equivalent to minimizing the mean pressure drop, and the corresponding hydraulic power loss is proportional to $\Delta p Q$. In the final turbulent benchmark set, the pipe-bend and U-bend use J_p , while the diffuser uses J_D .

Jury question: Is it inconsistent to use pressure objective for bends and dissipation for the diffuser?

Answer: The objectives are both hydraulic-loss objectives, but they match different reference benchmarks and boundary-value problems. The thesis avoids hiding this by reporting common sharp-geometry validation metrics, pressure drop and viscous dissipation, for all final designs.

5.4.2 Volume Constraint

The volume constraint is evaluated over the prescribed volume mask, written here as Ω_A :

$$g(\bar{\gamma}) = \frac{\int_{\Omega_A} \bar{\gamma} d\Omega}{\int_{\Omega_A} 1 d\Omega} - V_{\max} \leq 0. \quad (5.13)$$

This constraint depends only on the density field, not on the flow state:

$$\frac{\partial g}{\partial \mathbf{u}} = 0, \quad \frac{\partial g}{\partial p} = 0. \quad (5.14)$$

5.4.3 Continuous Optimization Statement

For a selected objective J_i , with $i = D$ for Eq. 5.11 or $i = p$ for Eq. 5.12, the continuous problem is

$$\begin{aligned} & \underset{\gamma(\mathbf{x}) \in [0,1]}{\text{minimize}} && J_i \\ & \text{subject to} && R_{\text{NS}} = 0, \\ & && R_{\bar{\nu}} = 0, \\ & && R_G = 0, \\ & && g(\bar{\gamma}) \leq 0. \end{aligned} \quad (5.15)$$

After finite-element discretization, this problem is solved over the active DG0 design variables. The Method of Moving Asymptotes (MMA) [54] is used as the outer optimizer. MMA does not solve the flow equations itself; it receives the current objective value, constraint values, and their gradients with respect to the active raw design variables. At each design iteration it builds a local separable approximation of the constrained problem, solves this cheaper approximate subproblem subject to bounds and move limits, and returns the next raw density vector. The moving asymptotes adapt between iterations to damp overly aggressive changes while still

allowing consistent descent when the gradient direction remains stable. In this work, prescribed non-design regions are excluded from the MMA variable vector and are restored through the bound mapping in Eq. 5.3.

5.5 Sensitivity Analysis and the Frozen Turbulence Assumption

After the objective and volume constraint are defined, the remaining task is to compute gradients with respect to the design field. Direct finite differences over all design cells are infeasible because the number of design variables is the number of active DG0 cells. A central finite-difference gradient would require two perturbed forward solves per active cell, in addition to the baseline state, and each perturbation would require the wall-distance solve, SA/Picard updates, and final velocity-pressure solve. An adjoint method avoids this scaling: for one scalar objective, one adjoint solve gives the sensitivity with respect to all design variables [43, 44].

Several sensitivity models are possible. The most complete option would be a coupled turbulent adjoint containing adjoint equations for the velocity, pressure, SA working variable, and wall-distance field, or a fully discrete adjoint of the complete IPCS/Picard/SNES update sequence. Such an approach would include the design dependence of $\tilde{\nu}$, ν_t , and G , but it would also require differentiating the nonlinear SA transport equation, the reciprocal wall-distance equation, eddy-viscosity clipping and relaxation, and the solver controls used to recover difficult intermediate designs. This is substantially more complex and less robust than the laminar adjoint used in many density-based flow-TO formulations.

The implementation therefore uses a frozen-turbulence adjoint of the final steady velocity-pressure residual. The forward loop still solves the topology-dependent wall-distance and SA equations, but their infinitesimal design derivatives are omitted during gradient assembly.

Jury question: What exactly is frozen in the frozen-turbulence adjoint?

Answer: The converged SA working variable, eddy viscosity, and wall-distance fields are treated as fixed coefficients while differentiating the final velocity-pressure residual with respect to density. The velocity and pressure response to Brinkman changes is still included through the adjoint.

The final SA working variable, eddy viscosity, and reciprocal wall-distance fields from the accepted forward solve are treated as fixed coefficients:

$$\frac{\partial \tilde{\nu}^{fr}}{\partial \bar{\gamma}} = 0, \quad \frac{\partial \nu_t^{fr}}{\partial \bar{\gamma}} = 0, \quad \frac{\partial G^{fr}}{\partial \bar{\gamma}} = 0. \quad (5.16)$$

This assumption is a practical compromise. It keeps the gradient computation close to the implemented final flow residual, captures the direct Brinkman design effect and the resulting velocity-pressure response, and avoids an unstable or incomplete turbulence-adjoint implementation. The trade-off is that terms such as $\partial \nu_t / \partial \bar{\gamma}$ and $\partial G / \partial \bar{\gamma}$ are missing. The resulting gradient is therefore the derivative of an approximate frozen-turbulence model, not the derivative of a fully coupled RANS-SA

topology-optimization system. This limitation matters most where topology changes strongly alter turbulent production, wall distance, or separated-flow behavior, and it motivates the frozen finite-difference checks and independent CFD validation used later in the thesis.

Let $\mathbf{s} = (\mathbf{u}, p)$ denote the final velocity-pressure state. The differentiated residual is

$$R(\mathbf{s}, \bar{\gamma}; \nu_t^{fr}, G^{fr}) = 0, \quad (5.17)$$

which corresponds to Eq. 5.5. For the selected objective J_i , the adjoint state $\mathbf{s}^*$ is found from

$$\left(\frac{\partial R}{\partial \mathbf{s}}\right)^T \mathbf{s}^* = -\left(\frac{\partial J_i}{\partial \mathbf{s}}\right)^T. \quad (5.18)$$

Equation 5.18 is the adjoint equation. After finite-element assembly, R is a residual vector and $\partial R/\partial \mathbf{s}$ is the Jacobian matrix of that residual with respect to the velocity-pressure degrees of freedom. The adjoint state $\mathbf{s}^*$ is therefore a vector of adjoint degrees of freedom. Products such as $(\mathbf{s}^*)^T \partial R/\partial \bar{\gamma}$ are ordinary algebraic dot products over assembled residual entries, equivalently the finite-dimensional pairing between a residual functional and adjoint test coefficients; they are not pointwise products of continuous fields.

The design derivative with respect to the projected density is then

$$\frac{dJ_i}{d\bar{\gamma}} = \frac{\partial J_i}{\partial \bar{\gamma}} + (\mathbf{s}^*)^T \frac{\partial R}{\partial \bar{\gamma}}. \quad (5.19)$$

The direct objective term depends on the selected objective. For the pressure objective, $\partial J_p/\partial \bar{\gamma} = 0$ because J_p contains only the boundary pressure; density dependence enters through the adjoint residual term. For the dissipation objective, $\partial J_D/\partial \bar{\gamma}$ contains the explicit Brinkman-drag contribution $\alpha'(\bar{\gamma})\mathbf{u} \cdot \mathbf{u}$, while the frozen-turbulence approximation omits any derivative through μ_t^{fr} . The resulting derivative is chained through the projection and density filter before being passed to MMA.

Because Eq. 5.19 is built under the frozen-turbulence assumption, the finite-difference checks in Appendix E verify the implemented frozen derivative rather than a fully coupled turbulence derivative.

Jury question: What do your finite-difference and Taylor checks actually prove?

Answer: They prove that the derivative supplied to MMA is consistent with the implemented frozen model and the selected solver path. They do not prove that the omitted coupled turbulence terms are negligible, and the thesis says that explicitly.

5.6 Density Filtering, Projection, and Algorithmic Workflow

The raw DG0 density is filtered to impose a minimum length scale before it is used in the physical penalties. Since a DG0 field is constant inside each element, the

Helmholtz-type density filter [16, 35] is written with an interior-facet jump term:

$$\int_{\Gamma_{int}} r^2 \frac{\alpha_{dg}}{h_{avg}} \llbracket \tilde{\gamma} \rrbracket \cdot \llbracket v \rrbracket dS + \int_{\Omega} \tilde{\gamma} v dx = \int_{\Omega} \gamma v dx. \quad (5.20)$$

Here v is the DG0 test function, Γ_{int} denotes internal element facets, and $\llbracket \cdot \rrbracket$ is the discontinuous-Galerkin jump operator; this facet coupling follows the standard interior-penalty DG treatment of elliptic operators [11]. In the optimization loop, Eq. 5.20 smooths the raw cell densities and suppresses checkerboard-scale changes.

The filtered density is then sharpened by a continuous Heaviside projection [30, 58],

$$\Pi_{\beta}(\tilde{\gamma}) = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\gamma} - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}, \quad (5.21)$$

where η is the projection threshold and β controls the steepness. This step pushes filtered intermediate densities toward the solid or fluid limits before the bound mapping in Eq. 5.3 restores prescribed regions and produces the physical density $\bar{\gamma}$. The mask normalization and projection curves are shown in Appendix D.

Figure 5.1 summarizes the optimization workflow. Each design iteration filters and projects the design density, restores prescribed bounds, updates the wall-distance and SA turbulence fields in the forward loop, solves the final frozen-viscosity velocity-pressure residual, computes the adjoint gradient for the selected objective, maps objective and volume sensitivities back to the free raw variables, and calls MMA for the next design. The “Forward SA” block denotes the accepted fields $(\mathbf{u}, p, \tilde{\nu}, G)$; these fields are recomputed for the current design before the frozen adjoint is assembled.

Continuation is used to improve robustness. The projection steepness β , RAMP parameter q , move limit, and selected penalty amplitudes can be advanced stage by stage. The exact schedules are benchmark-specific and are reported with the numerical settings in Appendix E.

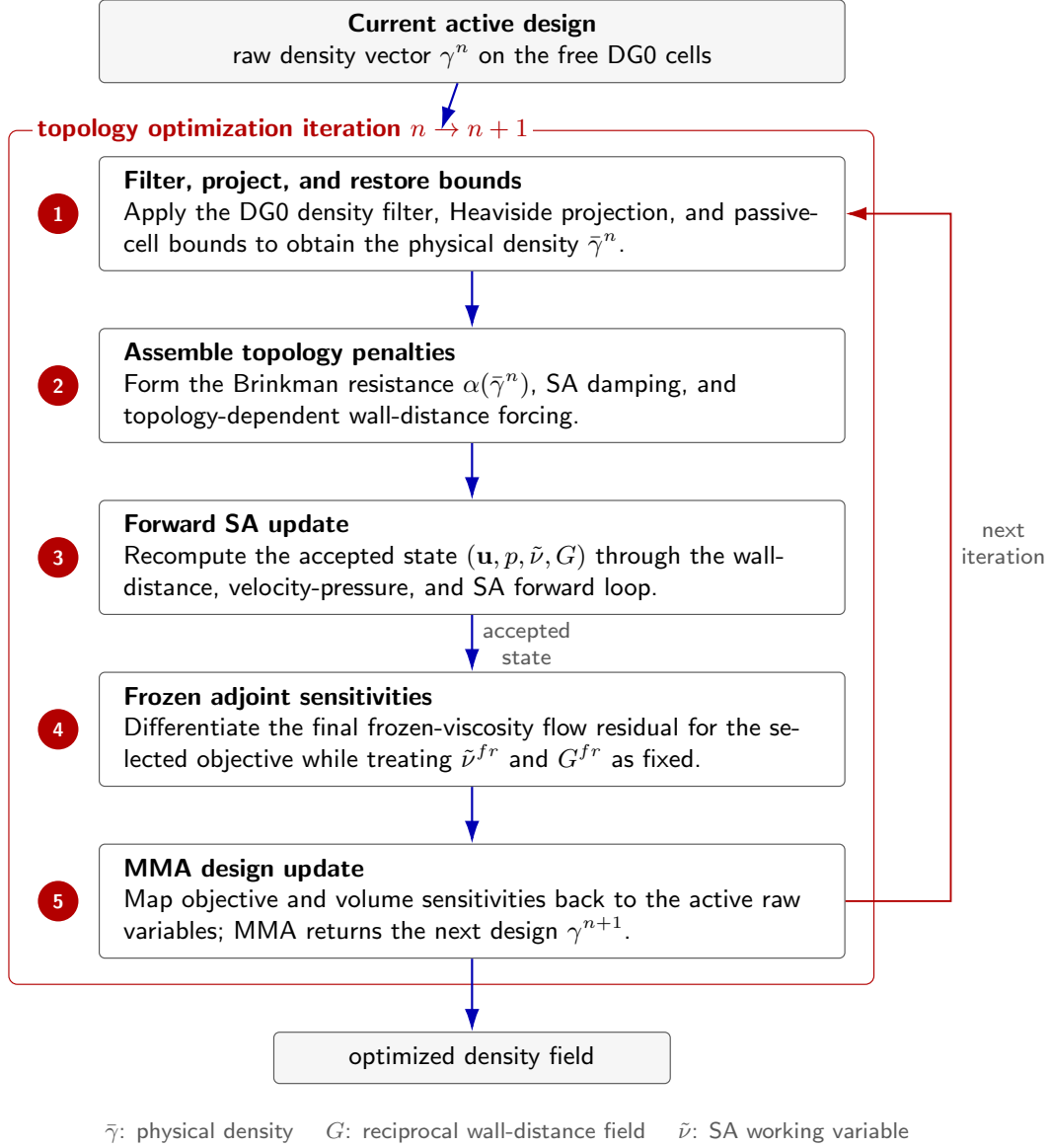


FIGURE 5.1: Algorithmic path of the density-based frozen-turbulence topology optimization loop (ChatGPT, 2026).

5.7 Conclusion

This chapter has defined the methodology used for the turbulent topology optimization results. The fixed-grid design is represented by a bounded density field, Brinkman resistance suppresses velocity in solid material, and additional topology penalties suppress the SA working variable and create topology-dependent wall distance. The optimization uses either an inlet-pressure objective or a dissipation objective, a density-volume constraint, and frozen-turbulence adjoint sensitivities

for MMA updates.

The most important limitation is the frozen derivative: the turbulence fields are updated in the forward solve, but their design derivatives are neglected. The results in Chapter 6 should therefore be read together with the frozen-sensitivity checks and independent CFD validation reported in the appendices.

Chapter 6

Turbulent Topology Optimization Results

6.1 Introduction

Building upon the FEniCS-SA solver verified in Chapter 4 and the topology optimization methodology developed in Chapter 5, this chapter evaluates the turbulent topology optimization solver on the pipe-bend and U-bend internal-flow benchmarks from Bayat et al. [13] and the diffuser benchmark from Yoon [61]. Table 6.1 gives the operating points and validation analyses. Each benchmark section then introduces only the geometry, objective, and interpretation needed for that case.

The optimized layouts are interpreted through projected density fields, velocity fields, and independent sharp-geometry CFD validation in Ansys Fluent. Appendix E retains the supporting frozen-sensitivity verification, FEniCS convergence histories, numerical settings, and Ansys grid-independence studies, grouped by benchmark case.

6.2 Benchmark Cases and Evaluation Metrics

The three benchmarks cover curved internal flows and a diffuser with a prescribed outlet contraction. The Bayat cases use reciprocal wall-distance penalization, topology-dependent SA and wall-distance reaction terms, an average inlet-pressure objective, and a frozen-turbulence adjoint. The diffuser uses the same frozen-SA optimization machinery, but it minimizes the dissipation objective J_D .

The pipe-bend and U-bend cases minimize the average inlet-pressure objective J_p defined in Eq. 5.12, which keeps their setup comparable with the Bayat benchmark objective. The diffuser case minimizes the dissipation objective J_D defined in Eq. 5.11, matching the formulation used by Yoon [61].

All extracted geometries are then evaluated as sharp-wall Ansys Fluent models, where the Brinkman penalty is removed and the validation metrics are recomputed on the exported flow domain.

6. TURBULENT TOPOLOGY OPTIMIZATION RESULTS

TABLE 6.1: Benchmark cases evaluated in Chapter 6. Detailed physical and numerical parameters are listed in Appendix E.

Case	V_f	Re	Analyses reported
Pipe bend [13]	0.25	5,000	FEniCS optimization; Ansys-SA and Ansys SST $k - \omega$ CFD validation; laminar-design comparison
U-bend [13]	0.27	5,000	FEniCS optimization; Ansys-SA and Ansys SST $k - \omega$ CFD validation; laminar-design comparison
Diffuser [61]	0.30	3,000	Velocity-outlet baseline; pressure-outlet FEniCS optimization with J_D ; Ansys-SA and Ansys SST $k - \omega$ CFD validation; laminar-design comparison for the pressure-outlet case

Jury question: Why validate sharp geometries if the optimizer solved a diffuse Brinkman problem?

Answer: Because a manufactured design would be a sharp-wall flow path, not a porous density field. The diffuse field is an optimization device. Sharp-geometry validation tests whether the design transfers after thresholding, smoothing, and removing the artificial Brinkman resistance.

The primary validation metric is the physical viscous dissipation integrated over the full exported CFD domain, including the non-design inlet, outlet, and fixed regions that are retained in the final geometry,

$$\Phi_D = \int_{\Omega} \frac{1}{2} \mu_{\text{eff}} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) : (\nabla \mathbf{u} + \nabla \mathbf{u}^T) d\Omega, \quad (6.1)$$

where Ω denotes the complete sharp-geometry flow domain exported to Ansys and $\mathbf{S} = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^T)$ is the symmetric strain-rate tensor. The integrand in Eq. 6.1 can therefore also be written as $2\mu_{\text{eff}} \mathbf{S} : \mathbf{S}$. In the Ansys Fluent post-processing, the scalar strain-rate magnitude is $\dot{\gamma} = \sqrt{2\mathbf{S} : \mathbf{S}}$, so the custom field function $\mu_{\text{eff}} \dot{\gamma}^2$ is equivalent to the integrand in Eq. 6.1. Because the reported Ansys validations all re-solve the extracted geometries as turbulent flows, the effective viscosity used in these post-processed metrics is $\mu_{\text{eff}} = \mu + \mu_t$.

The second global validation metric is the static pressure drop between the physical inlet and outlet sections,

$$\Delta p = \bar{p}_{\text{in}} - \bar{p}_{\text{out}}, \quad \bar{p}_{\Gamma} = \frac{1}{|\Gamma|} \int_{\Gamma} p_s d\Gamma, \quad (6.2)$$

where p_s is the static pressure and $|\Gamma| = \int_{\Gamma} 1 d\Gamma$ is the boundary length in the two-dimensional model. In Ansys Fluent, p_s is taken directly from the area-weighted average static pressure on the inlet and outlet surfaces. For FEniCS results that use kinematic pressure, the corresponding static pressure is obtained as $p_s = \rho p_k$ before evaluating Eq. 6.2. Pressure drop is retained throughout Chapter 6 because it is the most direct engineering interpretation of the same mechanical losses measured by Φ_D . All two-dimensional dissipation quantities are evaluated per unit depth.

6.3 Pipe Bend Benchmark

The pipe-bend benchmark introduces a direct comparison with the turbulent topology optimization results of Bayat et al. [13]. While the reference study employs a standard k - ε RANS model with wall functions, the present framework tackles the same problem using a wall-resolved SA model under the frozen-turbulence assumption. This comparison tests whether the fundamental topological solution, a streamlined and separation-avoiding bend, remains consistent across different turbulence closures.

6.3.1 Geometry and Boundary Conditions

The boundary conditions and geometric constraints replicate the setup shown in Fig. 10 of Bayat et al. [13], specifically targeting the $V_f = 0.25$ volume-fraction case. The inlet is placed on the left side of the non-design extension, the outlet is placed along the lower boundary, and the outlet pressure is fixed to zero. The optimization minimizes the average pressure on this prescribed inlet boundary required to drive the flow through the evolving bend. The local benchmark geometry used for the present implementation is shown in Figure 6.1.

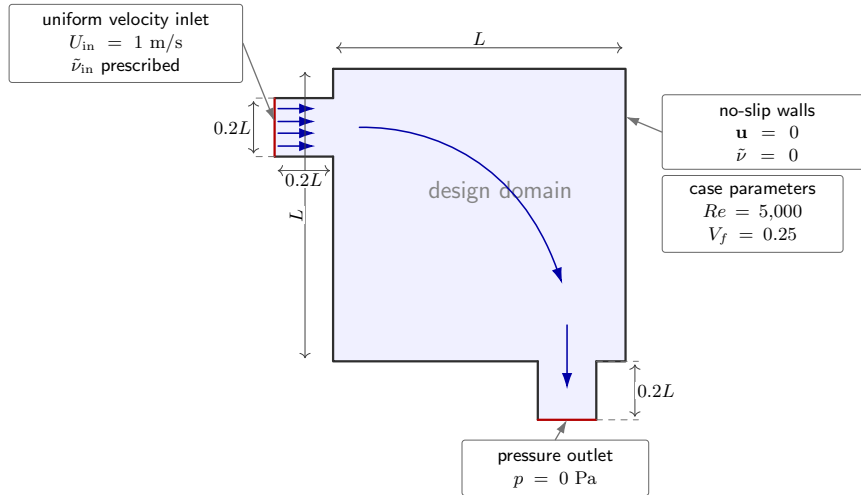


FIGURE 6.1: Pipe-bend benchmark geometry and boundary conditions, adapted from Fig. 10 of Bayat et al. [13] (ChatGPT, 2026).

6.3.2 Optimization Results

Figure 6.2 illustrates the projected density field $\bar{\gamma}$ and the corresponding velocity magnitude. The supporting coordinate finite-difference and Taylor checks for the frozen adjoint are reported in Appendix E. The FEniCS-SA result shows the same broad routing strategy as the k - ε reference from Bayat et al. [13]: the main high-speed path bypasses the sharpest corners to reduce momentum loss. A direct overlay

is provided in Figure E.1. The exact gray-scale density distribution near the walls still differs because the turbulence closure and near-wall treatment are not the same.

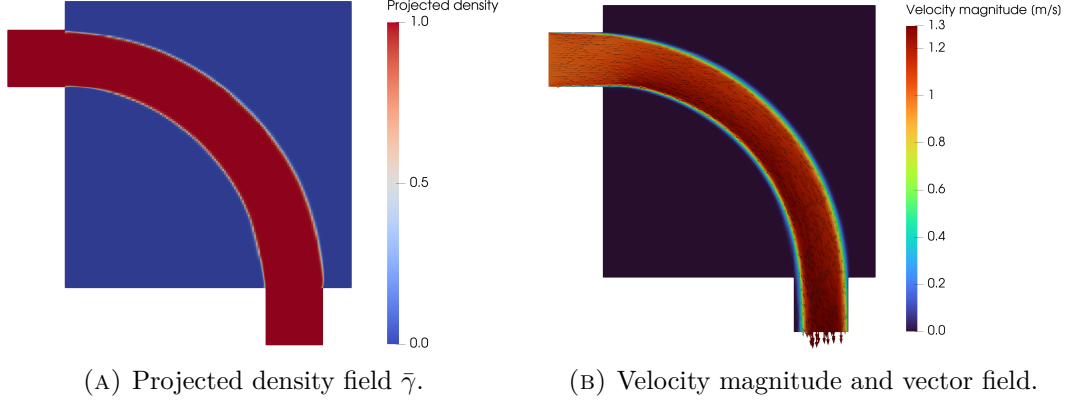


FIGURE 6.2: FEniCS-SA optimization result for the pipe-bend benchmark at $V_f = 0.25$. Final objective value $J_p = 0.2130$ Pa.

For the same $V_f = 0.25$, $Re = 5,000$ operating point, Bayat et al. [13] report $J_p = 0.4023$ Pa for their conventional $k-\varepsilon$ topology optimization. This value is used only as a reference scale, not as a direct validation target, because the turbulence closure and near-wall treatment differ from the present wall-resolved SA formulation.

The reported FEniCS value is the endpoint of a short 70-iteration run without continuation. It is used here as a provisional design state for comparing topology and validation metrics, not as a converged optimum. The convergence history in Figure E.2 shows that the normalized pressure objective is still monotonically decreasing, so the run is promising but should be extended with more MMA iterations and continuation.

Jury question: Can you claim the pipe-bend result is successful if the optimization was not converged?

Answer: The success claim is about transfer and robustness of the extracted design candidate, not final optimality. The thesis clearly labels it as provisional. The important evidence is that the candidate remains strong under Ansys-SA and SST $k-\omega$ and strongly beats the laminar reference under identical turbulent conditions.

6.3.3 CFD Validation

The extracted 70-iteration pipe-bend topology is evaluated in Table 6.2. Because the SST $k-\omega$ model is notably more sensitive to adverse pressure gradients than SA, evaluating the extracted geometry under both closures serves as a stress test for the provisional shape.

TABLE 6.2: CFD validation metrics for the extracted pipe-bend topology.

Solver	Model	Φ_D [W/m]	Δp [Pa]	Comment
FEniCS diffuse design	SA	0.0138	0.2130	70-iteration non-continuation state
Ansys sharp geometry	SA	0.0153	0.1457	Cross-code comparison
Ansys sharp geometry	SST $k - \omega$	0.0166	0.1522	Turbulence-model sensitivity

Using the Ansys-SA result as the reference, the SST $k - \omega$ evaluation increases the pressure drop by approximately 4.5% and the viscous dissipation by approximately 8.5%. The SST $k - \omega$ model therefore predicts slightly higher losses, consistent with a closure that is more sensitive to adverse pressure gradients, but the changes remain modest. This suggests that the extracted pipe-bend geometry is not narrowly tuned to the SA model. It does not, however, establish that the underlying FEniCS optimization has converged.

Jury question: Why is the pipe bend your strongest validation case?

Answer: The closure shift is modest and the laminar-vs-turbulent ranking is decisive under SST $k - \omega$. That combination suggests the geometry is not merely exploiting an SA-specific separation prediction, even though the optimization endpoint itself is not claimed to be the final optimum.

6.3.4 Comparison with Laminar Design

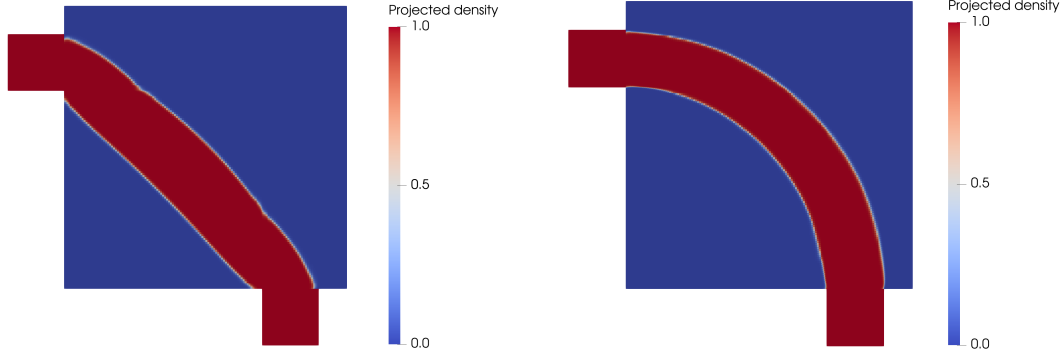
The laminar reference uses the same pipe-bend mesh, design-domain restrictions, inlet and outlet treatment, objective type, and volume fraction, but replaces the SA turbulent state equations by the laminar flow formulation and is optimized at $Re = 1$. This laminar optimization favors a sharper, shorter turn that is appropriate for the creeping-flow reference problem but less suitable once the extracted geometry is operated at the target turbulent Reynolds number. The comparison is therefore interpreted under identical turbulent operating conditions, rather than by comparing the laminar and turbulent optimization objectives directly.

Figure 6.3 compares the laminar and turbulent optimized topologies before sharp-geometry extraction. The laminar-design mesh-independence study supporting the SST $k - \omega$ result is reported in Table E.8 of Appendix E.

Figure 6.4 shows the corresponding SST $k - \omega$ velocity contours through the extracted laminar and turbulent pipe-bend designs. These contours provide the flow-field evidence behind the scalar metrics: sharper turns and corners in the laminar design promote stronger local separation and momentum loss when the geometry is operated at the turbulent Reynolds number.

The final SST $k - \omega$ comparison in Table 6.3 evaluates the extracted laminar and turbulent pipe-bend designs under identical turbulent operating conditions. The turbulent design reduces the viscous dissipation from 0.0313 to 0.0166 W/m and the pressure drop from 0.5444 to 0.1522 Pa. These reductions, 47.0% and 72.0% respectively, show that the turbulent topology optimization framework outperforms the laminar surrogate for the target turbulent operating point.

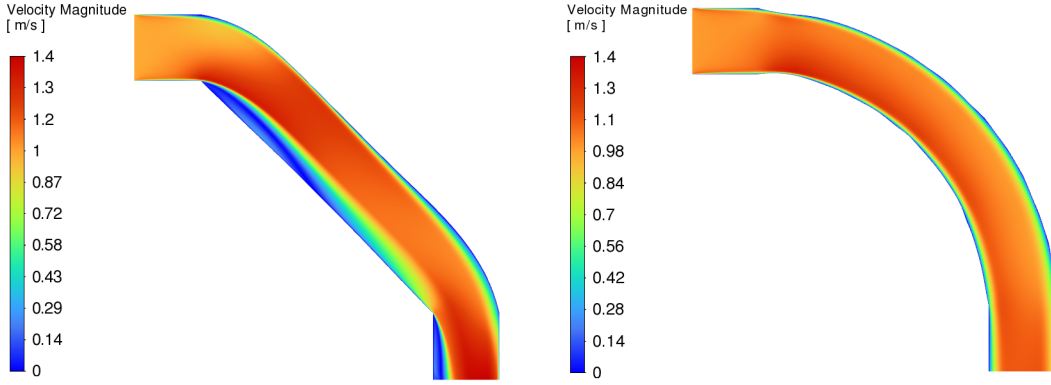
6. TURBULENT TOPOLOGY OPTIMIZATION RESULTS



(A) Projected density field $\bar{\gamma}$: laminar-optimized design ($Re = 1$).

(B) Projected density field $\bar{\gamma}$: turbulent-optimized design ($Re = 5,000$).

FIGURE 6.3: Laminar and turbulent topology comparison for the pipe-bend benchmark.



(A) Velocity magnitude field u : laminar-optimized design.

(B) Velocity magnitude field u : turbulent-optimized design.

FIGURE 6.4: Velocity-contour comparison for the extracted pipe-bend designs under identical turbulent operating conditions (SST $k - \omega$ model).

TABLE 6.3: Performance comparison of the extracted pipe-bend topologies under identical turbulent operating conditions.

Design origin	Model	Φ_D [W/m]	Δp [Pa]	Relative performance
Laminar TO, $Re = 1$	SST $k - \omega$	0.0313	0.5444	Baseline
Turbulent TO, SA	SST $k - \omega$	0.0166	0.1522	47.0% lower Φ_D ; 72.0% lower Δp

6.4 U-Bend Benchmark

The U-bend is the companion benchmark to the pipe bend. It uses the same frozen-SA topology optimization framework, but the geometry forces a 180° return path between two ports on the same side of the domain. This makes the case useful for

checking whether the optimizer can form a smooth turbulent return duct without relying on the shorter 90° routing available in the pipe-bend case.

6.4.1 Geometry and Boundary Conditions

The geometry and boundary conditions follow Fig. 11 and Table 4 of Bayat et al. [13]. The square design region has side length L , with a non-design lead duct of length $0.2L$ on the left. The inlet and outlet ports both have height $0.2L$; the top port acts as the inlet and the lower port acts as the pressure outlet. A fixed internal separator bar of thickness $0.10L$ extends from the lead duct into the design domain, forcing the flow to turn around its rounded tip. The target volume fraction is $V_f = 0.27$, and the Reynolds number is $Re = 5,000$ based on the port height and maximum inlet velocity. Figure 6.5 shows the corresponding local geometry and boundary-condition layout used for this benchmark.

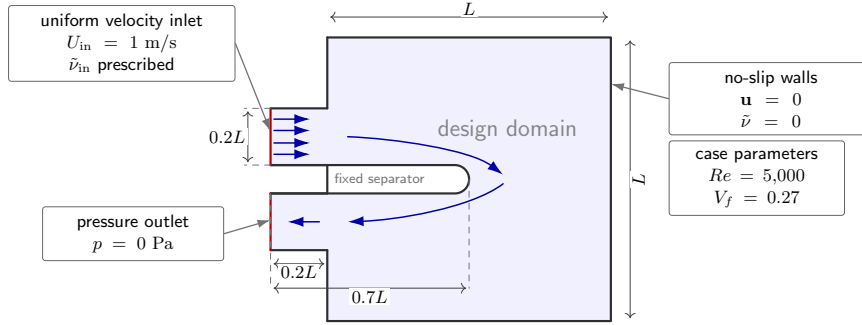


FIGURE 6.5: Geometry and boundary conditions for the U-bend benchmark, adapted from Fig. 11 and Table 4 of Bayat et al. [13] (ChatGPT, 2026).

6.4.2 Optimization Results

Figure 6.6 reports the final projected density and velocity-magnitude field for the U-bend. The main feature to assess is whether the design forms a rounded return passage around the separator tip, rather than a short path with tight local curvature. Because the objective is average inlet pressure, a successful layout reduces the pressure required to maintain the imposed inlet flow while respecting the prescribed volume fraction.

For the matching $V_f = 0.27$, $Re = 5,000$ U-bend case, the conventional $k-\varepsilon$ result reported by Bayat et al. [13] is $J_p = 0.8327$. As for the pipe bend, this value provides a reference scale only; the comparison remains qualitative because the present solver uses a different turbulence model and near-wall treatment.

The turbulent U-bend topology should also be read in the same way as the provisional pipe-bend topology: it is an extracted optimization endpoint from the available run, not a fully converged final optimum. The available U-bend conver-

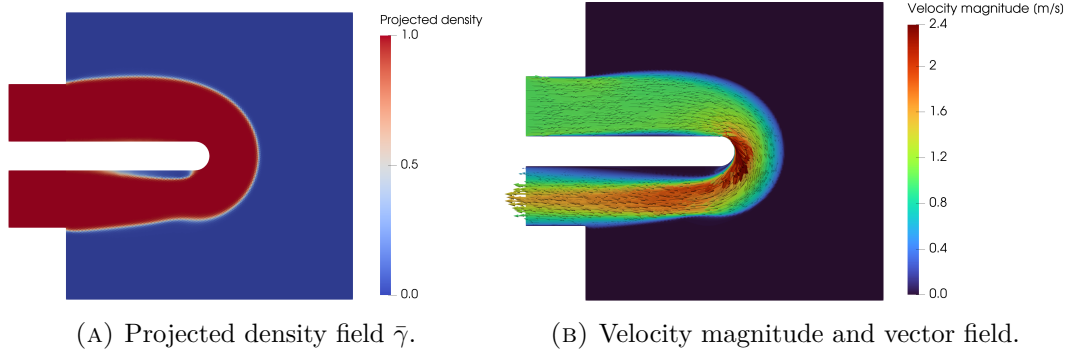


FIGURE 6.6: FEniCS-SA optimization result for the U-bend benchmark at $V_f = 0.27$. Final objective value $J_p = 0.8250$ Pa.

gence history is reported in Figure E.4 of Appendix E, using the same combined objective and volume-residual format as the pipe-bend diagnostic.

6.4.3 CFD Validation

The extracted U-bend topology is validated with the same sharp-geometry workflow as the pipe bend. The Ansys-SA solve checks the design under the same closure family used in the optimization loop, while the Ansys SST $k - \omega$ solve tests whether the extracted turbulent design remains robust under a two-equation model that is more sensitive to separation and adverse pressure gradients. Table 6.4 summarizes the FEniCS-SA optimization diagnostic and the two sharp-geometry Ansys validation solves for the turbulent U-bend design; the supporting validation mesh hierarchy is documented in Table E.11 of Appendix E.

TABLE 6.4: CFD validation metrics for the extracted turbulent U-bend topology.

Solver	Model	Φ_D [W/m]	Δp [Pa]	Comment
FEniCS diffuse design	SA	0.0936	0.8250	Optimization-state diagnostic
Ansys sharp geometry	SA	0.0785	0.8071	Cross-code comparison with the optimization closure family
Ansys sharp geometry	SST $k - \omega$	0.1016	1.1973	Independent turbulence-closure check

Using the Ansys-SA result as the reference, the SST $k - \omega$ validation predicts larger losses for the same extracted turbulent geometry: Φ_D increases by approximately 29.4% and Δp by approximately 48.3%. This trend is physically plausible because the fixed separator forces a compact return flow, where SST $k - \omega$ is more responsive to adverse pressure gradients and local recirculation downstream of the separator. The U-bend validation is therefore interpreted as a closure-sensitivity result for the forced 180° return flow rather than as closure-independent agreement.

Jury question: Does the U-bend result undermine the method?

Answer: It narrows the method’s validated envelope. Under SA the turbulent topology improves on the laminar reference, but SST k - ω shows strong closure sensitivity. That is exactly why the thesis frames the U-bend as a cautionary case rather than a clean success.

6.4.4 Comparison with Laminar Design

The laminar reference uses the same U-bend mesh, inlet and outlet ports, separator geometry, objective type, and volume fraction, but it is optimized at $Re = 1$ without the SA turbulence model. The extracted laminar and turbulent layouts are then evaluated under the same target turbulent operating conditions. This comparison tests whether the SA-optimized turbulent return path remains advantageous when the sharp geometry is re-solved with an independent turbulence closure.

Figure 6.7 compares the laminar and turbulent optimized U-bend topologies. The laminar-design mesh-independence study supporting the SST $k - \omega$ result is reported in Table E.13 of Appendix E.

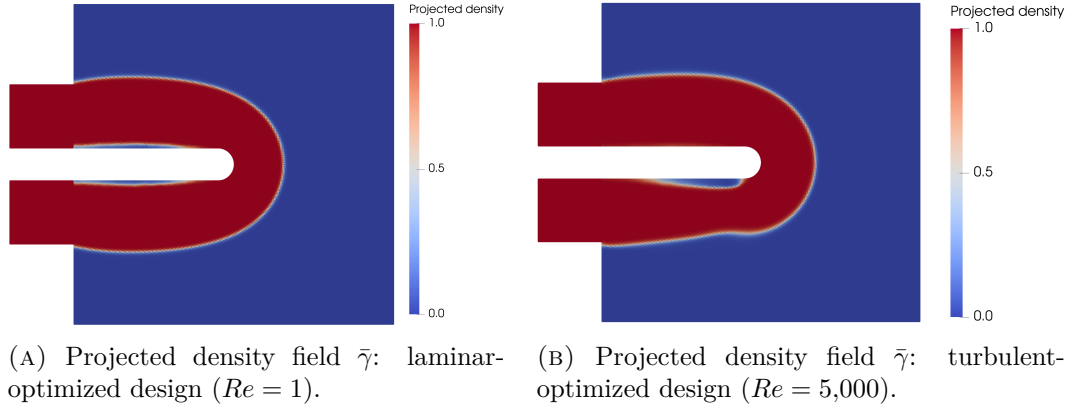


FIGURE 6.7: Laminar and turbulent topology comparison for the U-bend benchmark.

Figure 6.8 shows SST $k - \omega$ velocity contours through the extracted U-bend designs, plotted with the same color scale. The fields show broadly similar return-flow organization, with a high-speed outer path around the bend and a low-momentum band along the inner side downstream of the separator. The close-up vector diagnostic in Figure E.6 of Appendix E shows local recirculation in this low-momentum region, so the velocity fields are interpreted as qualitative evidence of curvature-induced separation and recovery rather than as the primary performance metric.

Table 6.5 gives the corresponding laminar-design comparison under identical turbulent operating conditions.

Under Ansys-SA, the turbulent U-bend design gives the expected improvement over the laminar reference. Under SST $k - \omega$, however, the two designs are much closer: the turbulent design has a slightly lower viscous dissipation but a slightly higher pressure drop. The U-bend therefore supports the same cautious message as the velocity fields: the SA-optimized return path is beneficial under the design

6. TURBULENT TOPOLOGY OPTIMIZATION RESULTS

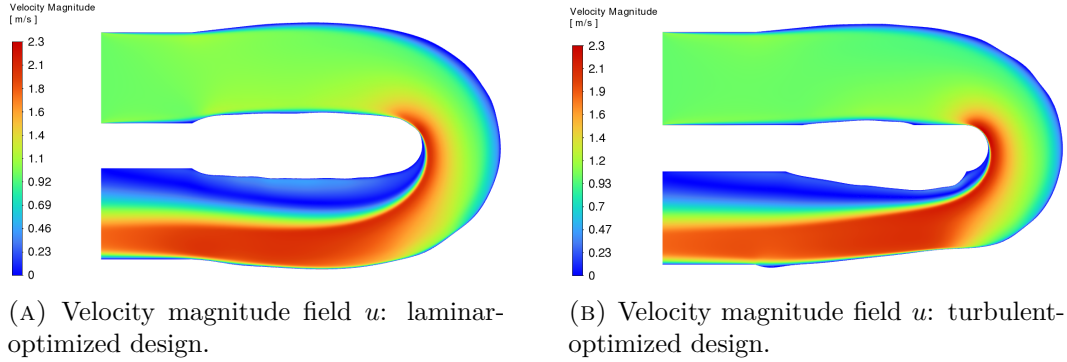


FIGURE 6.8: SST $k - \omega$ velocity-contour comparison for the extracted U-bend designs under identical turbulent operating conditions.

TABLE 6.5: Performance comparison of the extracted U-bend topologies under identical turbulent operating conditions.

Design origin	Validation model	Φ_D [W/m]	Δp [Pa]	Relative performance
Laminar TO, $Re = 1$	SA	0.1137	0.9646	Baseline under SA
Turbulent TO, SA	SA	0.0785	0.8071	31.0% lower Φ_D ; 16.3% lower Δp
Laminar TO, $Re = 1$	SST $k - \omega$	0.1069	1.1496	Baseline under SST $k - \omega$
Turbulent TO, SA	SST $k - \omega$	0.1016	1.1973	5.0% lower Φ_D ; 4.1% higher Δp

closure, but the 180° turn remains sensitive to the turbulence model used for sharp-geometry validation.

6.5 Diffuser Benchmark

The diffuser benchmark follows the third numerical example of Yoon [61]. It adds a downstream-boundary-condition sensitivity to the closure-sensitivity checks already used for the bend cases. A velocity-outlet baseline follows the published setup, while the primary result replaces the prescribed outlet velocity profile with a pressure outlet so that the outlet velocity distribution is solved rather than imposed.

6.5.1 Velocity-Outlet Boundary-Condition Sensitivity

The velocity-outlet baseline prescribes parabolic velocity profiles at both the full-height inlet and the center-third outlet. It produces a more strongly diverging channel near the outlet, which is interpreted as partly boundary-condition-driven. Its geometry, FEniCS fields, and Ansys diagnostic contour are reported in Appendix E; the pressure-outlet variant is used as the primary diffuser result below.

6.5.2 Pressure-Outlet Setup and Optimization Results

The pressure-outlet variant changes only the downstream boundary treatment relative to the velocity-outlet baseline documented in Appendix E. Its local geometry and boundary-condition layout are shown in Figure 6.9. The inlet remains the shifted parabolic velocity profile with $U_{\max, \text{in}} = 3 \text{ m/s}$ and the same $Re = 3,000$ definition, while the center outlet is assigned a fixed static pressure $p = 0 \text{ Pa}$. No outlet velocity profile is prescribed. This makes the outlet mass flux and velocity distribution part of the solution and removes the direct Dirichlet forcing that affected the velocity-outlet topology.

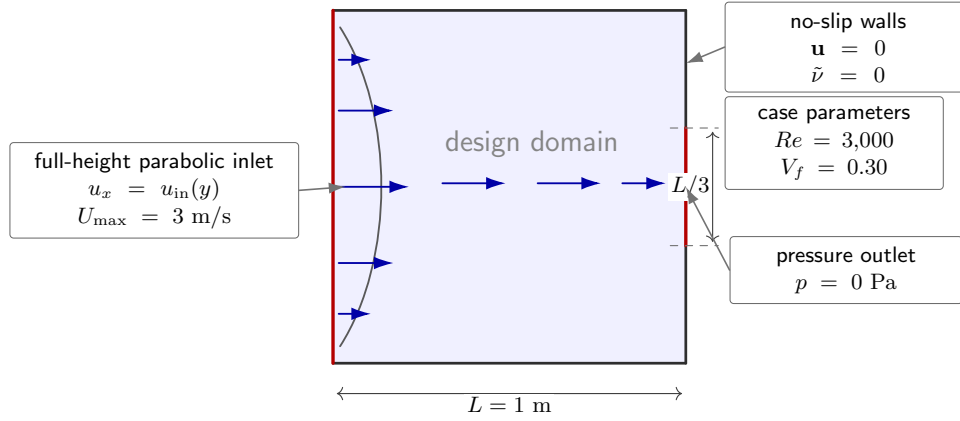


FIGURE 6.9: Pressure-outlet diffuser geometry and boundary conditions (ChatGPT, 2026).

Figure 6.10 reports the pressure-outlet FEniCS-SA topology and velocity field. Compared with the velocity-outlet result in Appendix E, the outlet region is less strongly widened. The optimized pressure-outlet objective is also lower, decreasing from $J_D = 3.1164 \times 10^5 \text{ W/m}$ to $J_D = 2.7664 \times 10^5 \text{ W/m}$, or by about 11.2%. This is not a direct performance ranking between equivalent designs, because the boundary-value problem has changed. The conclusion is narrower: removing the outlet Dirichlet velocity profile reduces the amount of geometry change required near the outlet and gives a less forced diffuser design.

The corresponding pressure-outlet convergence history through this common comparison point is reported in Figure E.10 of Appendix E.

6.5.3 Pressure-Outlet CFD Validation

The extracted pressure-outlet topology is validated with the same sharp-geometry workflow as the pipe bend and U-bend. The Ansys-SA solve checks the transfer from the diffuse FEniCS design to a sharp-wall geometry under the same closure family, while the Ansys SST $k - \omega$ solve tests whether the ranking is robust under an independent two-equation model. Because the optimization objective is J_D , the validation emphasizes both physical viscous dissipation and static pressure drop.

6. TURBULENT TOPOLOGY OPTIMIZATION RESULTS

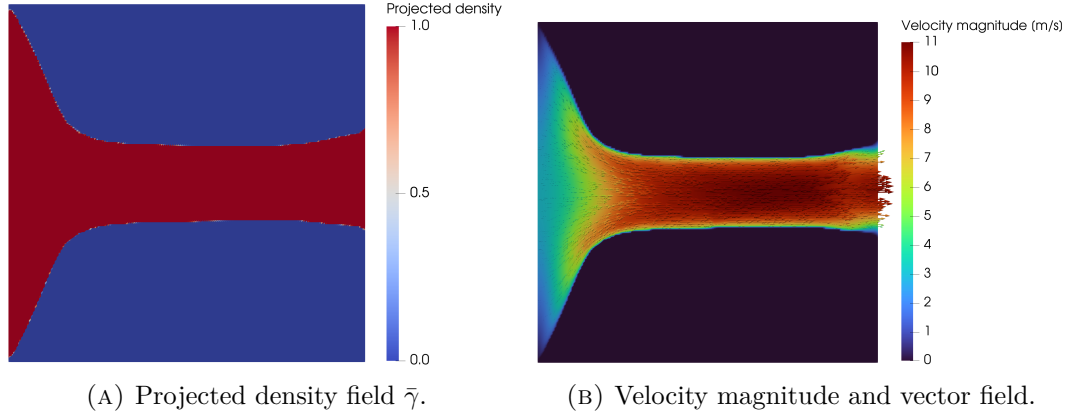


FIGURE 6.10: FEniCS-SA optimization result for the pressure-outlet diffuser at $V_f = 0.30$. Final objective value $J_D = 2.7664 \times 10^5$ W/m.

TABLE 6.6: CFD validation metrics for the extracted pressure-outlet diffuser topology.

Evaluation	Φ_D [W/m]	Δp_s [Pa]	Interpretation
FEniCS-SA diffuse topology	2.6307×10^5	2.6022×10^5	Diffuse-domain diagnostic; total objective $J_D = 2.7664 \times 10^5$ W/m
Ansys-SA sharp geometry	1.1885×10^5	1.1541×10^5	Sharp-wall validation with the optimization closure family
Ansys SST $k - \omega$ sharp geometry	1.2834×10^4	5.0006×10^4	Independent closure check; much lower modeled dissipation

The FEniCS-SA values in Table 6.6 are approximately 2.21 times larger in Φ_D and 2.25 times larger in Δp_s than the Ansys-SA sharp-geometry values, so this case cannot be claimed as quantitative cross-code agreement. The result is interpreted as a transfer limitation between the diffuse Brinkman-domain optimization diagnostic and the sharp-wall extracted geometry.

Jury question: Is the diffuser validation a failure?

Answer: It is a failed quantitative transfer case, but still useful evidence. It shows where diffuse-to-sharp extraction, adverse pressure gradients, and closure model form dominate the interpretation. The thesis does not claim quantitative agreement for the diffuser.

The Ansys validation rows are also strongly closure-sensitive: for the turbulent pressure-outlet design, the SA pressure drop is 2.31 times the SST $k - \omega$ value, and the SA dissipation is 9.26 times the SST $k - \omega$ value. This is the opposite trend from the U-bend: here SST $k - \omega$ predicts lower modeled losses, while SA appears more loss-producing in the adverse-pressure-gradient diffuser flow. Because the split is large, it is treated as model-form uncertainty rather than evidence that either closure gives a closure-independent physical truth. Additional velocity-magnitude, turbulent-viscosity-ratio, and local dissipation-density fields were inspected but are

not retained as main figures because they do not change the interpretation. The appendix wall-shear diagnostic and SST $k - \omega$ vector close-up in Figures E.11 and E.12 identify local reverse-flow regions in the pressure-outlet geometry; the integrated pressure-drop and dissipation metrics remain the primary evidence for this diffuser.

6.5.4 Pressure-Outlet Laminar-Design Comparison

The laminar reference for the final diffuser comparison uses the same pressure-outlet boundary treatment, objective type, and volume fraction, but it is optimized at $Re = 1$ without the SA turbulence model. The extracted laminar and turbulent pressure-outlet layouts are then evaluated under the same target turbulent operating conditions. This comparison isolates whether the SA-optimized pressure-outlet design remains beneficial when the final sharp geometry is re-solved with the same closure and with SST $k - \omega$.

Table 6.7 compares the extracted laminar and turbulent pressure-outlet diffuser topologies under identical turbulent operating conditions. The topology and velocity fields were inspected, but they do not by themselves reveal the performance ranking clearly enough to justify a separate figure. Because the diffuser is optimized with J_D , lower Φ_D is the primary performance indicator, while the static pressure drop remains the most direct hydraulic interpretation of the extracted sharp geometry. Both SA and SST $k - \omega$ validation solves are included to quantify whether the final ranking changes with the validation model.

TABLE 6.7: Performance comparison of the extracted pressure-outlet diffuser topologies under identical turbulent operating conditions.

Evaluation	Φ_D [W/m]	Δp_s [Pa]	Relative performance
Laminar TO, Ansys-SA	1.3495×10^5	1.3979×10^5	Baseline under SA
Turbulent TO, Ansys-SA	1.1885×10^5	1.1541×10^5	11.9% lower Φ_D ; 17.4% lower Δp_s
Laminar TO, Ansys SST $k - \omega$	1.1289×10^4	3.8332×10^4	Baseline under SST $k - \omega$
Turbulent TO, Ansys SST $k - \omega$	1.2834×10^4	5.0006×10^4	13.7% higher Φ_D ; 30.5% higher Δp_s

The laminar-design comparison reinforces this closure-sensitive interpretation. Under Ansys-SA, the turbulent topology reduces the pressure drop by 17.4% and the viscous dissipation by 11.9% relative to the pressure-outlet laminar topology. Under SST $k - \omega$, the ranking reverses: the turbulent topology gives a 30.5% higher pressure drop and a 13.7% higher dissipation.

Jury question: How should you defend a ranking reversal in the diffuser?

Answer: Do not soften it. The reversal is evidence that the SA-optimized diffuser is model-form sensitive in adverse-pressure-gradient flow. The honest conclusion is that this case needs a stronger turbulence model, better extraction checks, or experimental validation before making design claims.

The appendix wall-shear and vector diagnostics identify where local reverse flow appears in the pressure-outlet geometry, but the case-specific conclusion comes from

the scalar ranking reversal. The diffuser is therefore retained as a closure-sensitivity case for the present SA-based framework.

6.6 Conclusion

This chapter answers the third research question by evaluating whether the turbulent topology optimization framework produces designs that remain credible after independent CFD validation. The pipe bend provides the strongest validation evidence among the reported cases: the Appendix E frozen-sensitivity checks confirm the derivative supplied to MMA for J_p , and the extracted 70-iteration topology remains robust when validated in Ansys Fluent with both SA and SST $k - \omega$. The FEniCS optimization endpoint itself is provisional rather than converged, but the extracted geometry is not narrowly tuned to the SA model.

The U-bend extends the assessment to a forced 180° return path and gives a more cautious result. Like the pipe-bend topology, the turbulent U-bend topology is an optimization endpoint rather than a fully converged final optimum, so the result is interpreted as validation of the extracted geometry and not as a convergence claim. For the turbulent topology alone, replacing Ansys-SA by SST $k - \omega$ increases Φ_D by 29.4% and Δp by 48.3%. Compared with the laminar reference, the turbulent design improves both metrics under SA, but under SST $k - \omega$ it gives only a 5.0% lower dissipation and a 4.1% higher pressure drop. The appendix close-up vector diagnostic explains this case-specific sensitivity through local recirculation in the low-momentum inner return region.

The diffuser adds a separate caution: outlet treatment and validation closure both affect the final ranking. The velocity-outlet baseline shows that a prescribed outlet profile can influence the optimized outlet geometry, so the pressure-outlet variant is used as the primary diffuser result. For this pressure-outlet diffuser, the turbulent SA design improves on the laminar design under Ansys-SA, reducing Δp_s by 17.4% and Φ_D by 11.9%. The same two geometries evaluated with SST $k - \omega$ give the opposite ranking, with the turbulent design increasing Δp_s by 30.5% and Φ_D by 13.7%. Overall, the pipe bend provides the cleanest transfer case, while the U-bend and diffuser show why extracted turbulent designs must still be checked under an independent closure before making closure-independent claims.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

The objective of this thesis was to build and assess a flow-only turbulent topology optimization framework for internal configurations relevant to energy and thermal-management systems. The framework does not solve the energy equation and does not yet optimize conjugate heat transfer. Instead, it addresses the hydraulic prerequisite for those applications: generating low-loss internal flow paths under turbulent operating conditions. This narrower scope is deliberate, because turbulent momentum transport, wall-distance treatment, Brinkman penalization, geometry extraction, and adjoint sensitivity assembly already form a difficult optimization problem.

The work was organized as a sequence of checks. Chapter 2 assessed the useful range of the Spalart-Allmaras (SA) closure. Chapters 3 and 4 implemented and verified the FEniCS-SA solver. Chapters 5 and 6 embedded that solver in a density-based topology optimization loop with Brinkman penalization, filtering, projection, MMA, topology-dependent SA and wall-distance penalties, a frozen-turbulence adjoint, sensitivity checks, and independent sharp-geometry CFD validation.

Research Question 1

How accurately does the Spalart-Allmaras turbulence model predict curved and separated internal turbulent flows, and where do its physical limitations appear?

The answer is conditional. SA is defensible as a practical design-model closure for attached wall-bounded and mildly curved internal flows, including the moderate-curvature VMFL048 U-bend evidence. It is not closure-independent truth for strongly curved or separated flows.

The Ansys comparisons in this thesis make that boundary concrete. The VMFL048 U-bend at $De \approx 15,000$ remains credible, while the tighter VMFL030 pipe bend at $De \approx 18,000$ already shows clear SA deterioration in the selected profile comparison. The backward-facing-step evidence gives the same warning for separated recovery:

SA can predict plausible integral behavior, but SST $k - \omega$ follows the local velocity recovery more closely. The limiting issue is therefore curvature- and separation-driven Reynolds-stress anisotropy under a scalar eddy-viscosity closure, not a single Reynolds-number threshold.

Research Question 2

Does the custom FEniCS implementation of the Spalart-Allmaras model reproduce Ansys Fluent predictions for the selected turbulent internal-flow benchmarks?

Within the two-dimensional, wall-resolved scope of the thesis, the answer is yes. The solver reproduces the governing ingredients needed by the optimizer: incompressible flow coupling, SA transport, eddy-viscosity reconstruction, wall-distance evaluation, and near-wall resolution.

In the periodic channel, the fine-mesh bulk velocity differs from the Ansys reference by 2.67%, while the exported profile errors are $E_\infty = 2.66\%$ and $E_{L_2} = 2.64\%$. In the U-bend verification case, the pressure drop differs from Ansys by 2.79%, and the bend-exit profile gives $E_\infty = 8.50\%$ and $E_{L_2} = 2.75\%$. These results verify the FEniCS implementation as a sufficiently consistent forward solver for the topology optimization studies, while leaving the model-form limitations from Chapter 2 in place.

Research Question 3

To what extent do SA-driven topology optimizations produce final designs whose predicted performance remains consistent under independent Ansys Fluent validation with SA and SST $k - \omega$, and how do they compare with laminar-optimized designs under identical turbulent operating conditions?

Chapter 6 demonstrates that the reliability of an SA-optimized design depends fundamentally on the flow physics dictated by the target geometry.

The pipe bend represents the cleanest transfer case, showing that the framework successfully produces robust, closure-independent designs in regimes dominated by attached flow and moderate curvature. Even as a provisional 70-iteration endpoint, the extracted pipe-bend topology remains highly credible: replacing Ansys-SA by SST $k - \omega$ increases pressure drop by only about 4.5% and viscous dissipation by 8.5%. Under identical SST $k - \omega$ operating conditions, the turbulent design decisively outperforms the laminar-optimized reference, reducing pressure drop by 72.0% and viscous dissipation by 47.0%. Furthermore, the derivative supplied to MMA is internally verified, with coordinate finite differences giving a maximum relative discrepancy of 5.92×10^{-5} and the directional Taylor test showing approximately second-order behavior.

Conversely, the U-bend and diffuser benchmarks expose the severe limitations of relying on a one-equation closure for topology optimization in separation-dominated flows. Because the quantitative results for these cases are highly sensitive to the validation model, the exact performance metrics lose their predictive value. For geometries forcing strong recirculation (such as the 180° return of the U-bend) or adverse pressure gradients (the diffuser), the optimizer heavily tunes the layout to the SA model’s specific, and often overly forgiving, prediction of flow separation and recovery. When these extracted shapes are cross-examined using SST $k - \omega$, which is notably more responsive to adverse pressure gradients, the predicted flow structures disagree and the performance deteriorates significantly.

Consequently, the advantage of a turbulent-optimized design over a laminar baseline is not absolute. In attached flows, the turbulent formulation is a clear necessity and delivers massive performance gains. But in heavily separated regimes, the optimized geometry becomes so deeply coupled to the SA model’s simplifying assumptions that the turbulent designs offer only inconclusive, marginal gains over a simple laminar surrogate, or can even perform worse when evaluated under a different turbulence closure.

Overall Conclusion

The central conclusion is balanced. A frozen-SA density-based topology optimization framework can produce credible low-loss turbulent internal-flow design candidates when the flow remains within the closure’s practical range and when the extracted geometry is independently validated. It is not a substitute for final CFD validation, experimental testing, or a higher-fidelity turbulence model in separated, strongly three-dimensional flows. Independent sharp-geometry CFD validation is the mandatory engineering step between an optimized density field and anything that could be manufactured.

Jury question: What is the one-sentence contribution of the thesis?

Answer: It builds and verifies an open FEniCS frozen-SA topology-optimization workflow and, more importantly, defines when its optimized turbulent internal-flow candidates should and should not be trusted.

Within its validated envelope, the framework can still be valuable as an early-stage design and research tool for internal hydraulic layouts where pressure loss or dissipated mechanical energy is the main design constraint. It can generate turbulence-aware topology candidates, compare them with laminar surrogate designs, test objective and penalization choices, and expose whether a candidate remains robust when the turbulence closure is changed. The contribution of this thesis is therefore a reproducible bridge between turbulence-model knowledge, open-source finite-element optimization, and engineering judgment about when a turbulent topology-optimized design is credible.

7.2 Limitations

The limitations below define how far the conclusions can be taken. They are separated from the future-work actions so that the same point is not repeated twice.

1. **Simplified turbulence physics and two-dimensional scope:** The framework uses a wall-resolved one-equation SA closure in 2D planar domains. This suppresses Dean vortices, secondary motion, Reynolds-stress anisotropy, and three-dimensional separation structures that can be important in physical internal-flow devices.
2. **Frozen adjoint and sensitivities:** The finite-difference and Taylor checks verify the derivative used by MMA under the same frozen-turbulence assumption. They do not verify the derivative of a fully coupled turbulence, wall-distance, and geometry-extraction pipeline. The optimizer is therefore internally consistent with its chosen approximation, but the approximation remains a modeling decision.
3. **Diffuse-to-sharp geometry transfer:** The optimized design is a continuous Brinkman-penalized density field, while validation is performed on an extracted sharp-wall geometry. The chosen threshold and smoothing procedure affect wall placement, local curvature, and separation behavior, so FEniCS-Ansys discrepancies cannot always be attributed to solver physics alone.
4. **Validation level:** Independent Ansys Fluent validation with SA and SST $k-\omega$ is a necessary engineering step, but it is still numerical validation. Experimental pressure-drop and flow-field measurements would be required before using these layouts as physically validated device designs.
5. **Optimization robustness:** The reported turbulent pipe-bend and U-bend topologies should be interpreted as optimization endpoints rather than converged final results. Although pressure drop and dissipation are physically related in steady internal flows, domain-integrated dissipation can provide a less localized optimization signal than an average inlet-pressure objective.

7.3 Future Work

The verified turbulent fluid mechanics framework developed in this thesis establishes a foundation for several targeted extensions:

7.3.1 Benchmark Completion and Optimization Robustness

The short 70-iteration, non-continuation pipe-bend run and the 275-iteration turbulent U-bend endpoint reported in Chapter 6 are useful as provisional validation states, but they should be replaced by converged or deliberately polished runs before making final quantitative claims about the FEniCS optima. Future runs should use a longer MMA budget together with continuation of the RAMP, projection,

move-limit, and topology-wall penalty parameters. In particular, the turbulent-design U-bend should be repeated as a more robust optimization campaign tuned for the forced 180° return-flow geometry. The same future campaign should rerun the frozen-sensitivity diagnostic on the exact continuation settings used for the final design.

7.3.2 Conjugate Heat Transfer (CHT) and Thermal Management

The ultimate engineering application for high-Reynolds topology optimization is not merely fluid routing, but active thermal management. Historically, the immense computational cost of resolving Conjugate Heat Transfer (CHT) has forced optimization frameworks to rely on laminar approximations (e.g., natural convection as demonstrated by Alexandersen et al. [3]) or simplified flow proxies such as Darcy models [63]. Even when high-resolution, density-based solvers are successfully deployed for microelectronics cooling, they typically circumvent the complexities of turbulent momentum transport [12].

However, high-performance heat sinks still require models that capture turbulent convection. Recent work has introduced turbulence into thermal topology optimization using extended finite element (XFEM) level-set approaches with the SA model [41], computationally reduced multi-layer thermofluid approximations [62], and standard density-based frameworks coupled to two-equation $k - \epsilon$ closures [53].

With the turbulent momentum and SA transport equations implemented and verified within the current FEniCS architecture, the framework provides a practical starting point for turbulent CHT optimization. Extending the current methodology to CHT will require implementing a turbulent Prandtl number (Pr_t) approach to model the turbulent heat flux vector, alongside deriving the corresponding thermal adjoint sensitivities to minimize peak temperatures or maximize overall heat transfer coefficients. Furthermore, the capacity for optimizing two-fluid heat exchangers, currently dominated by laminar assumptions [31], could be improved by adapting this turbulent solver.

7.3.3 Three-Dimensional Topologies

As noted in the limitations, turbulent secondary flows are inherently 3D phenomena. To fully evaluate the geometric limits of the turbulence models and design physically realizable internal-flow components, the current 2D FEniCS implementation must be extended to three dimensions. The mathematical structure of the governing equations and adjoint derivations remains similar, but the computational cost changes by orders of magnitude because wall-resolved turbulent optimization requires fine near-wall resolution over a three-dimensional design domain. Large-scale topology optimization studies have demonstrated that parallel finite-element optimization at extreme design-variable counts is possible when the formulation is paired with efficient domain decomposition and scalable solvers [1]. A 3D turbulent extension of the present work should therefore prioritize parallel mesh generation, robust algebraic multigrid preconditioning, and automated sharp-geometry extraction.

7.3.4 Adjoint Model Hierarchies

This thesis deliberately uses the frozen-turbulence adjoint as the robust baseline. Future frameworks should investigate stable ways to add selected SA transport sensitivities, especially in separated flows where the constant-eddy-viscosity assumption is least accurate. Existing turbulent topology optimization studies already span several levels of adjoint complexity, from frozen or constant-eddy-viscosity approximations [22, 65] to more complete RANS sensitivity formulations and implementation studies [38, 60]. A practical next step is therefore not necessarily to jump directly to a fully coupled turbulence adjoint, but to build an adjoint hierarchy: frozen SA sensitivities as the baseline, selected SA transport sensitivities as an intermediate model, and two-equation closures such as SST $k - \omega$ for cases where curvature, separation, and anisotropic shear dominate the design response.

Appendices

Appendix A

Spalart-Allmaras Validity Benchmark Support

This appendix supports the turbulence-modeling discussion in Chapter 2. It contains the common Ansys Fluent convergence-monitor definitions for the Chapter 2 benchmark runs and the backward-facing-step material that is useful as separated-flow model-sensitivity evidence. The backward-facing-step case is secondary to the main Chapter 2 argument based on the Driver–Seegmiller/NASA literature comparison and is not used as a FEniCS solver-verification benchmark.

Unless stated otherwise, all grid-sensitivity relative differences in this appendix are reported as positive percentage magnitudes. For a mesh-dependent metric ϕ , the coarse–medium difference is computed as $|\phi_{\text{coarse}} - \phi_{\text{medium}}|/|\phi_{\text{coarse}}| \times 100\%$, while the medium–fine difference is computed as $|\phi_{\text{medium}} - \phi_{\text{fine}}|/|\phi_{\text{fine}}| \times 100\%$.

A.1 Ansys Fluent Benchmark Convergence Monitors

For the Ansys Fluent benchmark runs in Section 2.7, convergence was assessed using Ansys Fluent report monitors. Three physical monitors were tracked in addition to the solver residuals: the inlet-to-outlet static pressure drop, the domain-integrated viscous dissipation, and the relative mass imbalance. The pressure-drop monitor was defined from area-weighted static pressures,

$$\Delta p = \bar{p}_{\text{in}} - \bar{p}_{\text{out}}, \quad (\text{A.1})$$

while the viscous-dissipation monitor used an Ansys Fluent custom field function,

$$\Phi_D = \int_{\Omega} \mu_{\text{eff}} \dot{\gamma}^2 d\Omega, \quad (\text{A.2})$$

where the custom field function **viscous-dissipation-density** was defined as $\mu_{\text{eff}} \dot{\gamma}^2$ using the Ansys Fluent **effective-viscosity** and **strain-rate-magnitude** variables. Here $\mu_{\text{eff}} = \mu + \mu_t$ for turbulent runs and $\dot{\gamma}$ is the Ansys Fluent scalar strain-rate magnitude. For a monitored scalar q , the final-window change over N

iterations was evaluated as

$$\delta_N(q) = \frac{|q^{(k)} - q^{(k-N)}|}{|q^{(k)}|}. \quad (\text{A.3})$$

The relative mass imbalance was computed from the signed inlet and outlet mass-flow-rate reports,

$$\epsilon_m = \frac{|\dot{m}_{\text{in}} + \dot{m}_{\text{out}}|}{|\dot{m}_{\text{out}}|}, \quad (\text{A.4})$$

where $\dot{m}_{\text{out}} < 0$ under the Ansys Fluent boundary-normal sign convention for these cases.

TABLE A.1: Convergence metrics used for the Ansys Fluent benchmark runs in Section 2.7. The residual criterion refers to all Ansys Fluent scaled residuals, including continuity, momentum, and turbulence-transport equations.

Benchmark	Window	Pressure-drop	Dissipation	Mass	Scaled residuals
		change $\delta_N(\Delta p)$	change $\delta_N(\Phi_D)$	imbalance ϵ_m	
VMFL030 90° bend, 3D	500 iters	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	10^{-6}
VMFL048 U-bend, 3D	500 iters	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	10^{-6}
Backward-facing step, 2D	250 iters	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	10^{-6}

A.2 Ansys SA-SST $k - \omega$ Backward-Facing-Step Benchmark

The third benchmark evaluates the backward-facing-step geometry introduced in Section 2.6. This case is included to expose turbulence-model sensitivity in separated internal flow, not to replace the experimental validation evidence from the Driver-Seegmiller/NASA comparisons discussed above. The geometry and boundary conditions are based on the backward-facing-step configuration from Marcibál's master thesis and accompanying codebase [37, 32]. They are shown in Figure A.1. The flow enters a channel of height $H_{\text{in}} = 2.0$ m with a uniform inlet velocity of $U_{\text{in}} = 25$ m/s. At $x = 0$, the lower wall drops by the step height $h = 0.25$ m, giving an outlet height of $H_{\text{out}} = 2.25$ m. The benchmark Reynolds number is defined using the step height,

$$Re_h = \frac{U_{\text{in}} h}{\nu} = 3.44 \times 10^4, \quad (\text{A.5})$$

because backward-facing-step results are conventionally compared using the step height and the reattachment length x_r/h . For reference, using the inlet-channel height would give $Re_{H_{\text{in}}} = 2.75 \times 10^5$, but this is not the primary scaling for the separated region.

The Ansys Fluent setup mirrors the wall-resolved SA assumptions used throughout this thesis. The inlet velocity is prescribed as $\mathbf{u} = (25, 0)$ m/s, the outlet pressure is fixed at $p = 0$ Pa, and the no-slip walls enforce $\tilde{\nu} = 0$. The inlet modified turbulent viscosity is set to $\tilde{\nu} = 0.18449$ m²/s, corresponding to the eddy-viscosity level

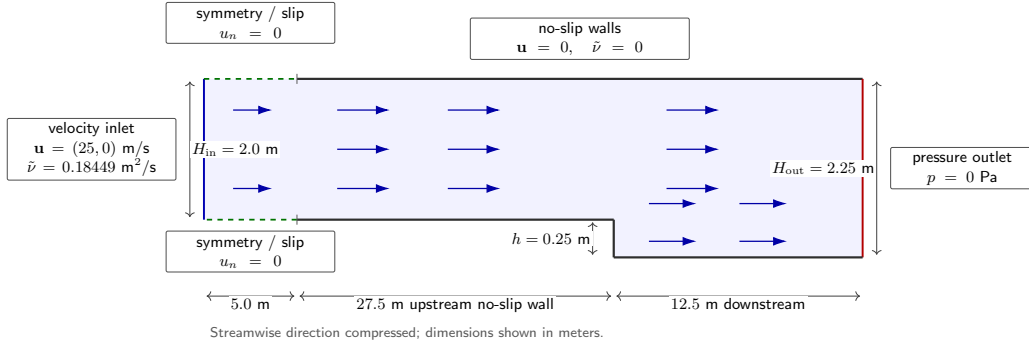


FIGURE A.1: Backward-facing-step benchmark geometry and boundary conditions, adapted from Marcibál and the associated code implementation [37, 32]. Streamwise dimension compressed for readability (ChatGPT, 2026).

used in the FEniCS backstep configuration. The important comparison quantity is the reattachment length x_r/h , obtained from the downstream lower-wall shear stress changing sign from negative to positive. Secondary diagnostics are the size of the recirculation zone, the pressure recovery, and the near-wall skin-friction distribution.

A.2.1 Wall-Resolved Ansys Grid-Sensitivity Study

The grid-sensitivity study uses the same backward-facing-step geometry and named selections as the benchmark definition above. The no-slip wall boundaries are inflated with a fixed first-layer height of $\Delta y_1 = 1.30 \times 10^{-4}$ m. This first-layer height is kept constant on all three grids because the Reynolds number and target wall resolution are unchanged. The mesh refinement is therefore applied through the number of inflation layers, inflation growth rate, and streamwise element sizing. Unlike the attached channel and bend verification cases, this separated-flow benchmark is treated as a grid-sensitivity record rather than a strict verification case, while the SA-SST $k - \omega$ comparison below uses the finest converged mesh for each closure.

The primary grid-sensitivity metric is the normalized reattachment length x_r/h , where x_r is obtained from the downstream zero-crossing of the streamwise wall shear stress on the lower wall associated with the end of the primary recirculation region. The SA reattachment length decreases from $x_r/h = 5.92$ on the coarse grid to 5.83 on the medium grid and 5.69 on the fine grid; the resulting medium-fine change remains above a strict one-percent mesh-independence criterion for this separated-flow metric. The total pressure loss Δp_t is retained as a secondary metric and is computed from the mass-flow-weighted inlet-to-outlet total-pressure difference. Velocity profiles are additionally extracted at $x/h = 1, 4, 6$, and 10, corresponding to $x = 0.25, 1.00, 1.50$, and 2.50 m downstream of the step. These stations cover the initial separated shear layer, the recirculation region, the expected reattachment region, and the downstream recovery region; the SA-SST $k - \omega$ comparison is made using the finest converged mesh available for each turbulence closure.

TABLE A.2: Wall-resolved backward-facing-step mesh hierarchy for the Ansys Fluent grid-sensitivity study.

Parameter	Coarse	Medium	Fine
<i>Mesh settings</i>			
Mesh method	Quad-dom.	Quad-dom.	Quad-dom.
Element order	Linear	Linear	Linear
Inflation option	First layer	First layer	First layer
Inflated boundaries	No-slip walls	No-slip walls	No-slip walls
Δy_1 [m]	1.30×10^{-4}	1.30×10^{-4}	1.30×10^{-4}
Inflation layers	28	36	44
Growth rate	1.18	1.15	1.12
Near-step wall size [m]	2.50×10^{-2}	1.25×10^{-2}	6.25×10^{-3}
Top-wall size [m]	5.00×10^{-2}	2.50×10^{-2}	1.25×10^{-2}
Inlet/outlet/symmetry size [m]	1.00×10^{-1}	5.00×10^{-2}	2.50×10^{-2}
Nodes	77,725	193,276	457,175
Elements	86,371	192,131	454,431
<i>Grid metrics: Spalart-Allmaras</i>			
x_r/h	5.92	5.83	5.69
$\epsilon_{\text{rel}}(x_r/h)$ [%]	–	1.52	2.46
Total pressure loss Δp_t [Pa]	46.743	46.795	46.531
$\epsilon_{\text{rel}}(\Delta p_t)$ [%]	–	0.11	0.57
<i>Grid metrics: SST $k - \omega$</i>			
x_r/h	6.09	6.13	6.08
$\epsilon_{\text{rel}}(x_r/h)$ [%]	–	0.66	0.82
Total pressure loss Δp_t [Pa]	44.741	45.139	45.142
$\epsilon_{\text{rel}}(\Delta p_t)$ [%]	–	0.89	0.01

Note. Quad-dom. = quad-dominant. The near-step wall size is applied to `upstream_lower_wall`, `step_wall`, and `downstream_lower_wall`. The inflated no-slip walls are `top_wall`, `upstream_lower_wall`, `step_wall`, and `downstream_lower_wall`; the inlet, outlet, and symmetry boundaries are not inflated. For the ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change. The SA reattachment length is reported as grid-sensitivity evidence rather than proof of strict mesh independence.

The contour plots below should therefore be read as a model-form comparison on the same local backward-facing-step geometry, not as a new validation against the NASA Driver-Seegmiller geometry. The streamwise-velocity fields identify the separated low-speed region and the downstream recovery, while the turbulent-viscosity ratio shows how each closure supplies the modeled momentum diffusion that controls this recovery. The velocity contours use the exported range of each Ansys Fluent solution, whereas the turbulent-viscosity plots use a common color scale.

Figure A.2 shows that the two closures give the same qualitative flow topology: separation at the step, a recirculating near-wall region, and redevelopment downstream. The difference is in the local recovery. The SA solution gives a smoother transition from reversed flow to positive streamwise velocity, whereas the SST $k - \omega$ solution retains a more structured low-speed region near the lower wall and a sharper separated shear layer. This is the expected sensitivity for a backward-facing step: reattachment is governed by shear-layer growth and turbulent transport, so small differences in the modeled eddy viscosity can change the local velocity field even when the global recirculation pattern remains plausible.

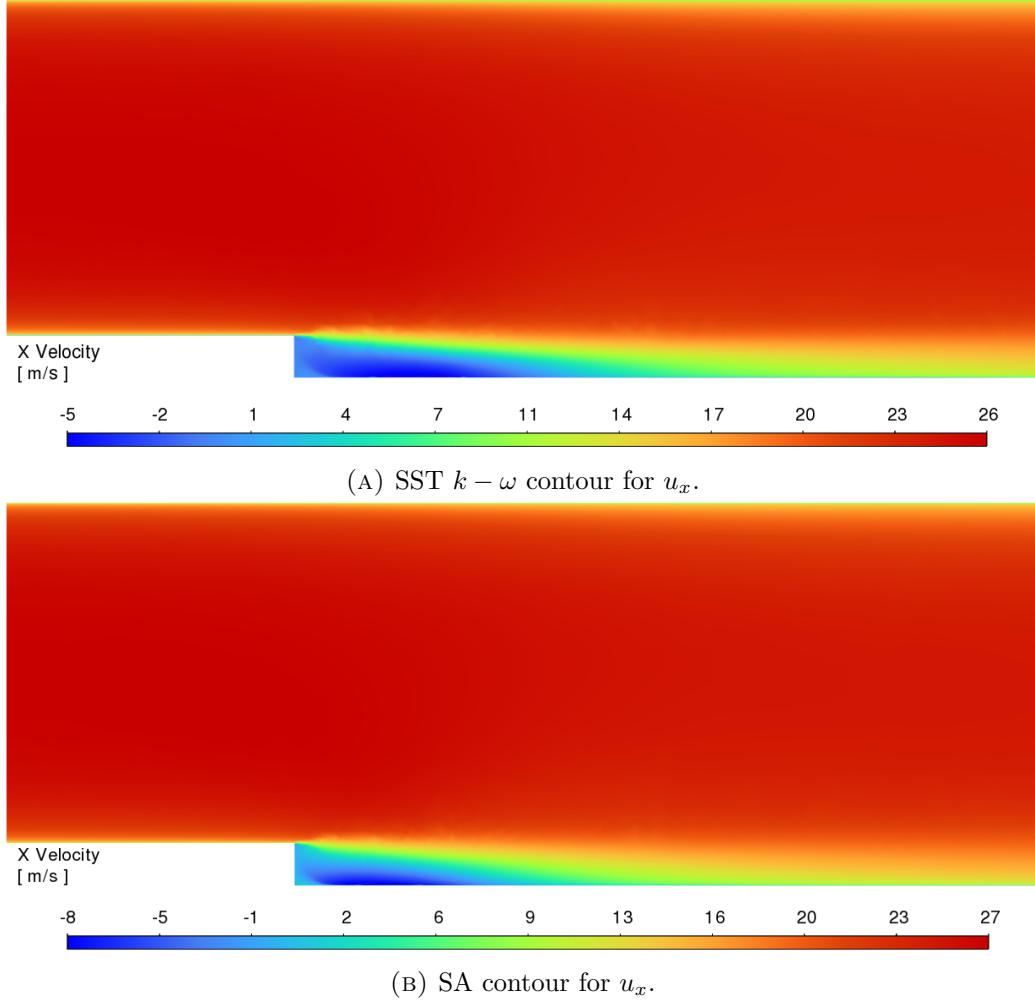


FIGURE A.2: Streamwise-velocity contours for the backward-facing-step benchmark.

The turbulent-viscosity contours in Figure A.3 explain the velocity differences. SA is a one-equation model: it transports a modified eddy-viscosity variable and reconstructs μ_t mainly from local vorticity and wall-distance effects. Its μ_t/μ field is therefore relatively smooth in this separated channel. SST $k - \omega$ solves separate equations for turbulent kinetic energy and specific dissipation rate, with near-wall $k - \omega$ behavior and outer-flow blending. In this case it preserves lower eddy viscosity in the immediate wall-bounded/reversed-flow region while carrying a stronger non-uniform eddy-viscosity distribution through the outer part of the channel. The resulting difference is not a numerical detail; it is the model-form sensitivity that makes backward-facing-step flow a useful warning case for SA-based topology optimization.

Figures A.2 and A.3 provide the full-field contour comparison, while Figure A.4

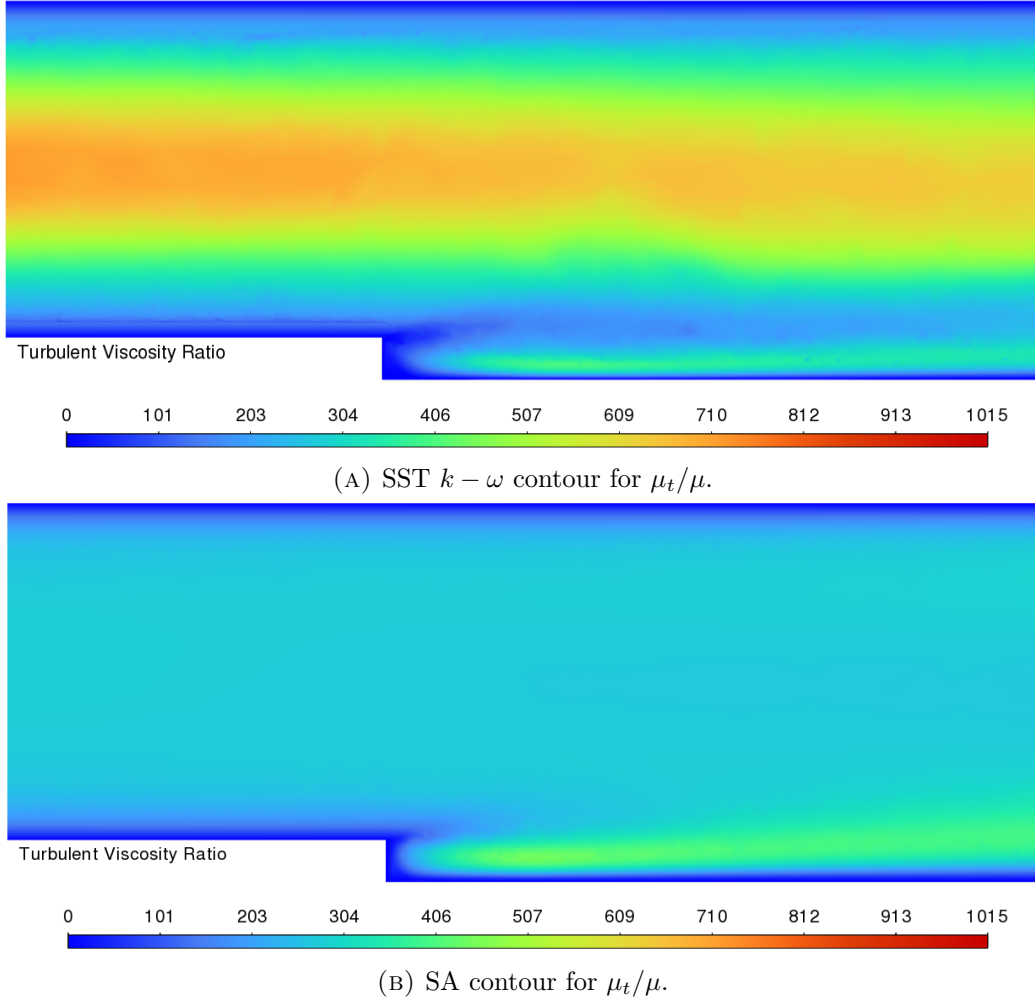


FIGURE A.3: Turbulent-viscosity-ratio contours for the backward-facing-step benchmark, plotted with a common color scale.

shows why this benchmark was included even though it is not a direct experimental validation. The purpose is to compare SA and SST $k - \omega$ on the same separated-flow geometry and determine how strongly the modeled mean recirculation and recovery depend on the selected RANS closure. Near the step, at $x/h = 1$ and $x/h = 4$, both models produce a low-speed or reversed-flow region close to the lower wall, which is consistent with a modeled recirculation bubble generated by the sudden expansion. However, the velocity profiles in this separated region are visibly model dependent, especially near the lower wall and in the developing shear layer. Farther downstream, at $x/h = 6$ and especially $x/h = 10$, the near-wall streamwise velocity becomes positive and the two profiles move closer together, consistent with modeled reattachment and redevelopment of the boundary layer.

This case is therefore interpreted as a model-sensitivity study rather than as a

A. SPALART-ALLMARAS VALIDITY BENCHMARK SUPPORT

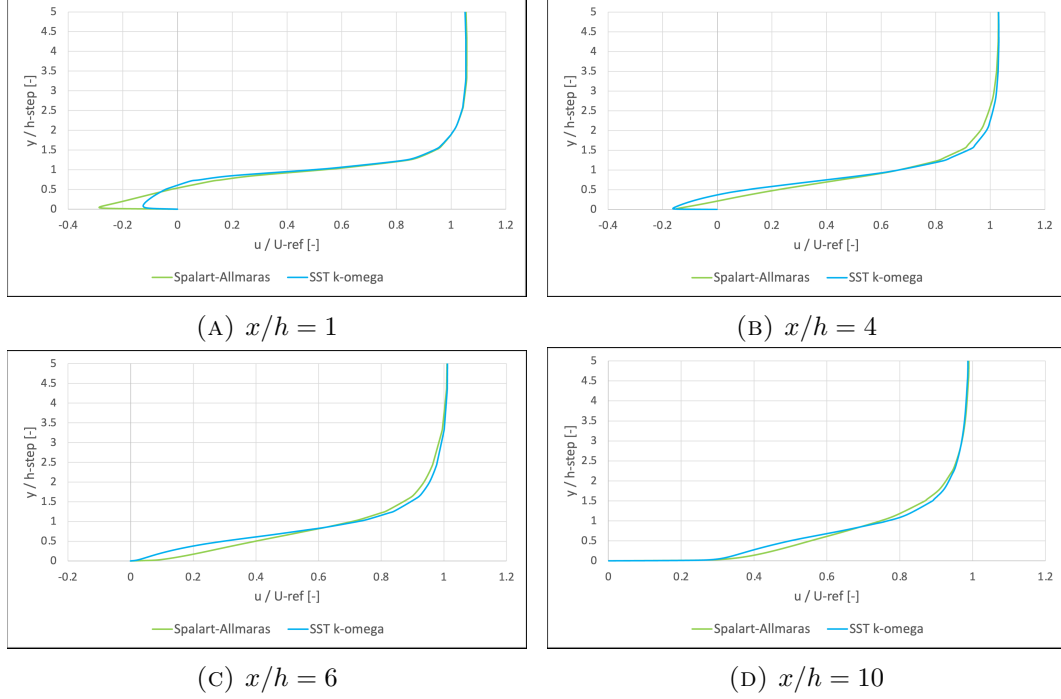


FIGURE A.4: Normalized streamwise-velocity profiles for the backward-facing-step benchmark at four downstream stations.

validation case. Without a direct comparison to the Driver-Seegmiller experimental data for the exact mesh, inlet condition, and domain definition used here, the Ansys Fluent results cannot determine which of the two closures is more accurate for this specific setup. The experimental comparison in Figure 2.5 already shows the broader trend: in separated backward-facing-step flow, SST $k - \omega$ gives better local velocity-profile agreement than SA, even though both models may predict a reasonable reattachment length. The present Ansys Fluent comparison extends that lesson to the geometry used later in the thesis. It shows that the separated region and recovery are sensitive to the turbulence model, so an SA-based optimization result containing separation cannot be interpreted as quantitatively reliable on its own. This limitation is fundamental to linear eddy-viscosity RANS closures: the Boussinesq approximation represents turbulent stresses through a scalar eddy viscosity and cannot fully represent stress anisotropy, non-local turbulence transport, or unsteady coherent structures in separated shear layers. The conclusion is therefore deliberately limited and more conservative than “SA captures separation”: SA can generate a plausible RANS mean recirculation pattern, but the separated-flow prediction is model dependent and must be checked with an independent closure such as SST $k - \omega$, and ideally with experimental or higher-fidelity validation.

Appendix B

Spalart-Allmaras FEniCS Solver Support

This appendix supports Chapter 3. It collects standard Spalart–Allmaras closure definitions used by the FEniCS implementation, while keeping the main chapter focused on the weak forms, Picard coupling, and solver structure.

B.1 SA Closure Functions and Calibration Constants

The FEniCS implementation uses the vorticity-based Spalart–Allmaras closure definition. The production term relies on a modified vorticity magnitude $\tilde{S}$ that maintains correct near-wall scaling while avoiding loss of production when the local vorticity becomes small [50, 46]. First,

$$S = \sqrt{2\Omega_{ij}\Omega_{ij}} \quad \text{and} \quad \bar{S} = \frac{\tilde{\nu}}{\kappa^2 y^2} f_{v2}. \quad (\text{B.1})$$

Here S is the scalar magnitude of the vorticity tensor, κ is the von Karman constant, and y is the distance to the nearest wall. During nonlinear iterations, the code evaluates the protected form of $\tilde{S}$ recommended in the SA reference implementation [46] so that the production term remains well defined when $S + \bar{S}$ becomes small. Small numerical floors are added in the implementation to prevent division by zero. The function f_{v2} is a near-wall modifier:

$$f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}} \quad \text{where} \quad \chi = \frac{\tilde{\nu}}{\nu} \quad \text{and} \quad f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3}. \quad (\text{B.2})$$

The wall-destruction term is modulated by the non-dimensional function f_w ,

$$f_w = g \left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6} \right)^{1/6}, \quad (\text{B.3})$$

where

$$g = r + c_{w2}(r^6 - r) \quad \text{and} \quad r = \min \left(\frac{\tilde{\nu}}{\tilde{S}\kappa^2 y^2}, 10 \right). \quad (\text{B.4})$$

The upper bound of 10 on r improves numerical stability during solver initialization when $\tilde{S}$ may momentarily approach zero.

The original formulation by Spalart and Allmaras [50] includes additional geometric trip functions (f_{t1} and f_{t2}) designed to trigger transition from laminar to turbulent flow at prescribed coordinates. Following the topology-optimization methodology of Yoon [61], these transition functions are omitted here because the topological boundaries are unknown a priori and evolve during optimization.

The empirical calibration constants are

$$\begin{array}{lll}
c_{b1} = 0.1355 & \sigma = 2/3 & c_{b2} = 0.622 \\
\kappa = 0.41 & c_{w2} = 0.3 & c_{w3} = 2.0 \\
c_{v1} = 7.1 & c_{v2} = 0.7 & c_{v3} = 0.9 \\
c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma}.
\end{array}$$

Appendix C

Spalart-Allmaras Solver Verification Details

This appendix supports Chapter 4. It contains concise FEniCS setup tables, the mesh-independence studies used for the channel benchmark, and the selected FEniCS and Ansys mesh information used for the U-bend verification case. For traceability, the active configuration file is named once near each FEniCS setup; the tables retain only the governing geometry, boundary-condition, and solver settings rather than duplicating the full Python configuration. The backward-facing-step material is placed in Appendix A, because it supports the Chapter 2 turbulence-model discussion rather than the Chapter 4 forward-solver verification.

Unless stated otherwise, all grid-independence relative differences in this appendix are reported as positive percentage magnitudes. For a mesh-dependent metric ϕ , the coarse-medium difference is computed as $|\phi_{\text{coarse}} - \phi_{\text{medium}}|/|\phi_{\text{coarse}}| \times 100\%$, while the medium-fine difference is computed as $|\phi_{\text{medium}} - \phi_{\text{fine}}|/|\phi_{\text{fine}}| \times 100\%$.

C.1 Verification Benchmark 1: Channel Flow

C.1.1 Physical and Geometric Parameters

The first verification benchmark consists of a pressure-driven periodic 2D channel. To align with fundamental turbulence research conventions, the characteristic length is defined as the full channel height (H), and the Reynolds number (Re_H) is computed strictly based on this geometric parameter rather than the hydraulic diameter. The active FEniCS setup is summarized from `ConfigChannel_SpalartAllmaras_Steady.py`, and the key physical setup is detailed in Table C.1.

C.1.2 Numerical Simulation Parameters

The channel flow was resolved using an Incremental Pressure Correction Scheme (IPCS) embedded within a Picard fixed-point iteration loop. The key numerical

TABLE C.1: Geometric, physical, and boundary-condition parameters for the periodic channel flow benchmark.

Parameter	Channel flow setting
Benchmark type	Pressure-driven, streamwise-periodic 2D channel
Domain dimensions	Length $L = 2.0$ m; full channel height $H = 2.0$ m
Characteristic length	Full channel height $H = 2.0$ m
Reference velocity	$U_{\text{ref}} = 18.5$ m/s, utilized to define the target Reynolds number
Kinematic viscosity	$\nu = 1.81818 \times 10^{-3}$ m ² /s
Reynolds number	$Re_H \approx 2.03 \times 10^4$
Body force	$\mathbf{f} = (1.0, 0)$ m/s ² in the kinematic-pressure convention
Driving condition	Streamwise body-force mode, equivalent to a pressure drop of 2 Pa over the periodic length $L = 2$ m
Velocity boundary conditions	Periodic velocity between inlet/outlet planes; no-slip on top and bottom walls
Pressure boundary conditions	Periodic pressure field with no inlet/outlet pressure Dirichlet condition; pressure mean normalized during IPCS
SA boundary conditions	Periodic $\tilde{\nu}$ between inlet/outlet planes; $\tilde{\nu} = 0$ at walls
Initial fields	$\mathbf{u}_0 = (18.5, 0)$ m/s, $p_0 = 0$ Pa, $\tilde{\nu}_0 = 5.0 \times 10^{-3}$ m ² /s
Verification mesh	Fine_FEniCS ; first layer $\Delta y_1 = 1.709 \times 10^{-3}$ m

solver settings, convergence tolerances, and relaxation factors used in FEniCS are summarized in Table C.2.

C.1.3 Spatial Verification and Grid Independence

To guarantee that the numerical results were independent of the spatial discretization, a formal grid independence study was conducted. Three distinct boundary-layer inflated meshes (Coarse, Medium, and Fine) were evaluated. The geometric properties, node counts, and resolution parameters of this mesh hierarchy are explicitly detailed in Table C.3. Crucially, the first-layer height was kept constant across all grids to strictly enforce the $y^+ \approx 1$ requirement, while the bulk and inflation-layer resolutions were systematically refined.

Visual confirmation of this spatial convergence is provided by the wall-normal velocity profiles plotted in Figure C.1. In each case, the profile is extracted at the streamwise mid-plane, $x = 0.5L$, with the wall-normal coordinate y varied across the full channel height. The **Fine_FEniCS** grid was consequently selected for the final verification against the Ansys Fluent baseline, while the coarser grids demonstrate that the wall-normal velocity profile is already asymptotically converged.

Furthermore, to ensure a rigorous cross-code validation, an equivalent grid independence study was performed for the commercial Ansys Fluent baseline. The Ansys domain was discretized using a comparable mapped mesh hierarchy, maintaining a strict $y^+ \approx 1$ resolution at the walls. In addition to the residual monitors, the bulk velocity was monitored and required to change by less than 1.0×10^{-3} over 250 iterations before accepting a converged solution. The continuity residual criterion was set to 1.0×10^{-5} , while the momentum and SA working-variable residual

TABLE C.2: Numerical simulation parameters for the FEniCS steady Spalart-Allmaras channel benchmark.

Parameter	Symbol	Setting
Velocity space	$\mathbf{u}$	Vector CG2, periodic in streamwise direction
Pressure space	p	Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1, periodic in streamwise direction
Quadrature degree	–	2
Maximum outer Picard steps	$N_{\text{Picard,max}}$	400
SA solves per Picard step	–	1
Outer velocity tolerance	$\epsilon_{\mathbf{u}}$	1.0×10^{-5}
Outer pressure tolerance	ϵ_p	1.0×10^{-6}
Outer SA tolerance	$\epsilon_{\tilde{\nu}}$	1.0×10^{-6}
Outer diagnostic convergence	–	Midline bulk velocity window of 100 Picard steps with relative tolerance 1.0×10^{-3}
IPCS pseudo-time step	Δt	5.0×10^{-3} s
Maximum IPCS steps per Picard step	–	500
IPCS velocity tolerance	–	1.0×10^{-6}
IPCS pressure tolerance	–	1.0×10^{-6} ; pressure convergence is diagnostic in body-force mode
IPCS pressure normalization	–	Enabled; the pressure correction is mean-normalized
Velocity relaxation factor	α_u	0.3
Pressure relaxation factor	α_p	0.3
SA relaxation factor	$\alpha_{\tilde{\nu}}$	0.3
SA lower bound	$\tilde{\nu}_{\min}$	1.0×10^{-12}
Wall-distance equation	–	Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$
Wall-distance Newton settings	–	Relative tolerance 1.0×10^{-8} ; absolute tolerance 1.0×10^{-10} ; maximum 80 iterations; relaxation 0.5
Linear solver strategy	–	IPCS blocks: direct MUMPS; SA transport: default solver

TABLE C.3: Wall-resolved channel mesh hierarchy for the FEniCS grid-independence study.

Parameter	Coarse	Medium	Fine
<i>Mesh settings</i>			
Streamwise cells	10	20	40
Wall-normal cells	42	51	68
Δy_1 [m]	1.709×10^{-3}	1.709×10^{-3}	1.709×10^{-3}
$\Delta y_{\min}$ [m]	1.709×10^{-3}	1.709×10^{-3}	1.709×10^{-3}
$\Delta y_{\max}$ [m]	7.707×10^{-2}	8.454×10^{-2}	7.589×10^{-2}
Inflation layers per wall	10	14	18
Growth ratio	1.45	1.35	1.25
Vertices	473	1,092	2,829
Elements	840	2,040	5,440
<i>Grid metric</i>			
U_{bulk} [m/s]	16.923	18.489	18.472
$\epsilon_{\text{rel}}(U_{\text{bulk}})$ [%]	–	9.25	0.09

Note. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

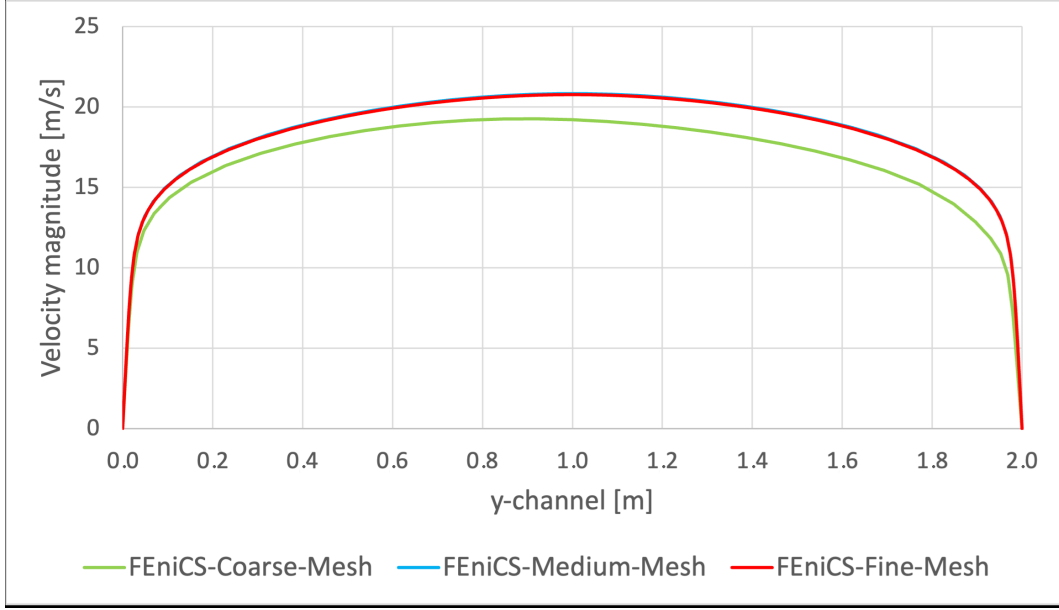


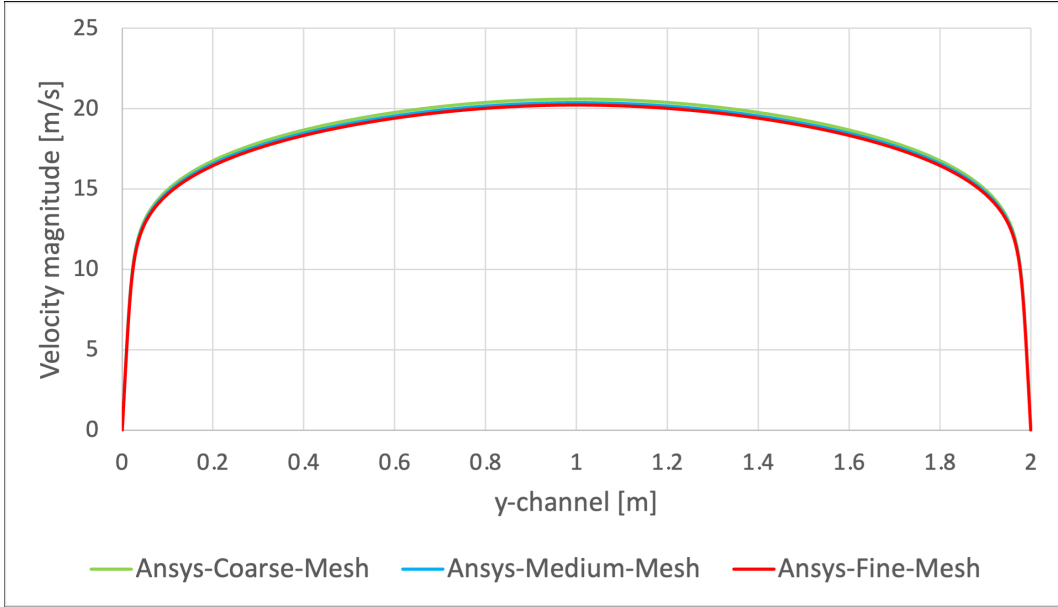
FIGURE C.1: FEniCS wall-normal velocity profiles at $x = 0.5L$ for the Coarse, Medium, and Fine grids.

criteria were set to 1.0×10^{-6} . Mass conservation was also checked from the inlet and outlet mass-flow reports. The properties of these meshes are documented in Table C.4. The wall-normal velocity profiles extracted from the Ansys solver at the same mid-channel location, illustrated in Figure C.2, exhibited identical asymptotic convergence, confirming that the baseline reference data is mathematically robust and completely independent of spatial discretization errors.

TABLE C.4: Wall-resolved channel mesh hierarchy used for the Ansys grid-independence study.

Parameter	Coarse	Medium	Fine
<i>Mesh settings</i>			
Mesh type	MultiZone	MultiZone	MultiZone
Face type	All quad	All quad	All quad
N_x	64	160	220
N_y	80	144	180
Bias type	Two-sided	Two-sided	Two-sided
Bias factor	59	28	21
Approx. Δy_1 [m]	1.700×10^{-3}	1.700×10^{-3}	1.700×10^{-3}
Nodes	7,740	26,448	43,010
Elements	7,565	26,123	42,594
<i>Grid metric</i>			
U_{bulk} [m/s]	18.305	18.114	17.977
$\epsilon_{\text{rel}}(U_{\text{bulk}})$ [%]	–	1.04	0.76

Note. N_x and N_y are the streamwise and wall-normal edge divisions. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

FIGURE C.2: Ansys wall-normal velocity profiles at $x = 0.5L$ for the Coarse, Medium, and Fine grids.

C.2 Verification Benchmark 2: U-Bend Geometry

C.2.1 Physical and Geometric Parameters

The second verification benchmark evaluates a highly convective U-bend. Because this FEniCS domain is a 2D planar approximation of the symmetry mid-plane of the 3D Ansys VMFL048 case, the 3D physical pipe diameter (D) is explicitly retained as the characteristic length scale. This ensures the Reynolds (Re_D) and Dean (De) numbers remain non-dimensionally consistent with the commercial baseline. The straight inlet and outlet legs are shorter than in the original VMFL048 Ansys mesh to reduce meshing and computational effort; both the FEniCS and Ansys runs use this same short-leg U-bend geometry. The active FEniCS setup is summarized from `ConfigUBend_SpalartAllmaras_Steady.py`, and the configuration is detailed in Table C.5.

TABLE C.5: Geometric, physical, and boundary-condition parameters for the U-bend benchmark.

Parameter	U-bend setting
Benchmark type	Short-leg 2D symmetry-plane U-bend based on Ansys VMFL048
Domain dimensions	Pipe diameter $D = 0.028$ m; centreline bend radius $R_c = 0.125$ m; straight-leg length $30D = 0.840$ m
Characteristic length	Physical pipe diameter $D = 0.028$ m
Reference / inlet velocity	Uniform prescribed inlet velocity $\mathbf{u}_{in} = (0, -1.42)$ m/s
Kinematic viscosity	$\nu = 8.9 \times 10^{-7}$ m ² /s
Reynolds number	$Re_D \approx 4.47 \times 10^4$
Dean number	$De \approx 1.50 \times 10^4$
Body force	$\mathbf{f} = (0, 0)$ m/s ²
Driving condition	Prescribed inlet velocity and zero relative static pressure at the outlet
Velocity boundary conditions	No-slip on all curved and straight pipe walls; natural velocity condition at pressure outlet
Pressure boundary conditions	$p = 0$ Pa at outlet; pressure left free at inlet and walls
SA boundary conditions	$\tilde{\nu}_{in} = 9.34 \times 10^{-5}$ m ² /s at inlet; $\tilde{\nu} = 0$ at walls; natural condition at outlet
Initial fields	$\mathbf{u}_0 = (0, -1.42)$ m/s, $p_0 = 0$ Pa, $\tilde{\nu}_0 = \tilde{\nu}_{in}$
Verification mesh	<code>Medium_WallResolved</code> ; first layer $\Delta y_1 = 1.20 \times 10^{-5}$ m

C.2.2 Numerical Simulation Parameters

Due to the adverse pressure gradients and secondary flow separation induced by the streamline curvature, the U-bend represents a significantly more demanding numerical challenge than the channel. Consequently, the FEniCS solver required a smaller pseudo-time step, tighter under-relaxation, and advanced iterative preconditioners to maintain stability, as detailed in Table C.6.

TABLE C.6: Numerical simulation parameters for the highly convective FEniCS U-bend benchmark.

Parameter	Symbol	Setting
Velocity space	$\mathbf{u}$	Vector CG2
Pressure space	p	Scalar CG1
SA working-variable space	$\tilde{v}$	Scalar CG1
Quadrature degree	–	4
Maximum outer Picard steps	$N_{\text{Picard,max}}$	1000
SA solves per Picard step	–	1
Outer velocity tolerance	$\epsilon_{\mathbf{u}}$	1.0×10^{-4}
Outer pressure tolerance	ϵ_p	1.0×10^{-4}
Outer SA tolerance	$\epsilon_{\tilde{v}}$	1.0×10^{-6}
IPCS pseudo-time step	Δt	1.0×10^{-4} s
Maximum IPCS steps per Picard step	–	200
Maximum IPCS steps for final flow	–	600
IPCS velocity tolerance	–	1.0×10^{-4}
IPCS pressure tolerance	–	2.0×10^{-3}
Pressure-drop convergence monitor	–	Final-window length 100 with relative tolerance 5.0×10^{-4} ; flow convergence required
Velocity relaxation factor	α_u	0.3
Pressure relaxation factor	α_p	0.1
SA relaxation factor	$\alpha_{\tilde{v}}$	0.01
SA pseudo-time stabilization	–	SA transport pseudo-time step 5.0×10^{-2} s
SA lower bound	$\tilde{v}_{\min}$	1.0×10^{-12}
Wall-distance equation	–	Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$
Wall-distance Newton settings	–	Relative tolerance 1.0×10^{-8} ; absolute tolerance 1.0×10^{-10} ; maximum 80 iterations; relaxation 0.5
Linear solver strategy	–	IPCS blocks: GMRES with ILU velocity/correction preconditioning and Hypre-AMG pressure preconditioning; SA transport: default solver
Restart handling	–	Restart from saved state enabled when checkpoint files are present; otherwise the configured initial fields are used

C.2.3 Selected FEniCS Mesh and Ansys Grid Independence

Managing the highly stretched, non-orthogonal inflation layers around the curved arcs of the U-bend required custom mesh generation via Gmsh. The current FEniCS comparison setup points to the wall-resolved `Medium_WallResolved` grid documented in Table C.7. Similar to the channel benchmark, the wall-adjacent element height is kept at 1.20×10^{-5} m to satisfy the low-Reynolds SA wall constraints. A separate coarse-medium-fine FEniCS mesh-convergence study is therefore not claimed for this U-bend verification case.

The reported FEniCS static pressure-drop metric is computed from boundary-measure-weighted mean pressures over the inlet and outlet sections, rather than from local point samples. Let Γ_{in} and Γ_{out} denote the inlet and outlet boundaries, and let p_k be the kinematic pressure solved in FEniCS. The physical static pressure drop reported in pascals is

$$\Delta p_s = \rho (\bar{p}_{k,\text{in}} - \bar{p}_{k,\text{out}}) = \rho \left[\frac{1}{|\Gamma_{\text{in}}|} \int_{\Gamma_{\text{in}}} p_k \, d\Gamma - \frac{1}{|\Gamma_{\text{out}}|} \int_{\Gamma_{\text{out}}} p_k \, d\Gamma \right], \quad (\text{C.1})$$

where $|\Gamma| = \int_{\Gamma} 1 \, d\Gamma$ is the FEniCS boundary measure and $\rho = 997 \text{ kg/m}^3$. In the implementation this corresponds to integrating over the marked inlet and outlet facets with `ds(2)` and `ds(3)`, respectively. The density factor is required because the IPCS formulation stores pressure in kinematic form, while the benchmark table reports the static pressure drop in Pa. Although the outlet pressure is prescribed as zero relative pressure, the outlet term is still evaluated as a boundary average so that the diagnostic follows the same inlet-minus-outlet definition on every grid.

TABLE C.7: Selected wall-resolved FEniCS U-bend mesh used for comparison with the Ansys Fluent verification baseline.

Parameter	Medium_WallResolved setting
h_{bulk} [m]	1.40×10^{-3}
Δy_1 [m]	1.20×10^{-5}
Inflation layers	14
Growth ratio	1.30
t_{infl} [m]	1.53×10^{-3}
Vertices	73,978
Elements	144,952

For the independent Ansys Fluent baseline, a separate mesh-independence study was executed in the same shortened U-bend geometry. Using a parallel unstructured meshing strategy with strict $y^+ \approx 1$ boundary layer inflation (documented in Table C.8), the Ansys velocity profiles at the bend exit across the same pipe-diameter line, shown in Figure C.3, were checked for spatial convergence. For each Ansys solve, convergence was accepted only after the scaled continuity, momentum, and Spalart–Allmaras residuals reached 1.0×10^{-6} and the monitored static pressure drop changed by less than a relative tolerance of 5.0×10^{-4} over the final 100 iterations. The inlet and outlet mass-flow rates were also compared for each grid to verify

the global mass balance before accepting the pressure-drop values. The **Fine_Ansys** grid is retained as the commercial verification baseline, while the deliberately coarser **Coarse_Ansys** level is selected to bracket the mesh-convergence trend.

TABLE C.8: Wall-resolved U-bend mesh hierarchy used for the Ansys grid-independence study.

Parameter	Coarse	Medium	Fine
<i>Mesh settings</i>			
Mesh type	Quad-dom.	Quad-dom.	Quad-dom.
h_{face} [m]	4.00×10^{-3}	2.00×10^{-3}	1.25×10^{-3}
Curvature angle	22°	12°	8°
Inflation option	First layer	First layer	First layer
Δy_1 [m]	1.20×10^{-5}	1.20×10^{-5}	1.20×10^{-5}
Inflation layers	10	16	22
Growth rate	1.35	1.22	1.16
t_{infl} [m]	6.55×10^{-4}	1.59×10^{-3}	2.17×10^{-3}
Nodes	14,525	47,707	106,460
Elements	13,984	46,640	104,763
<i>Grid metric</i>			
Static pressure drop Δp_s [Pa]	814.109	796.804	792.995
$\epsilon_{\text{rel}}(\Delta p_s)$ [%]	—	2.13	0.48

Note. Quad-dom. = quad-dominant. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

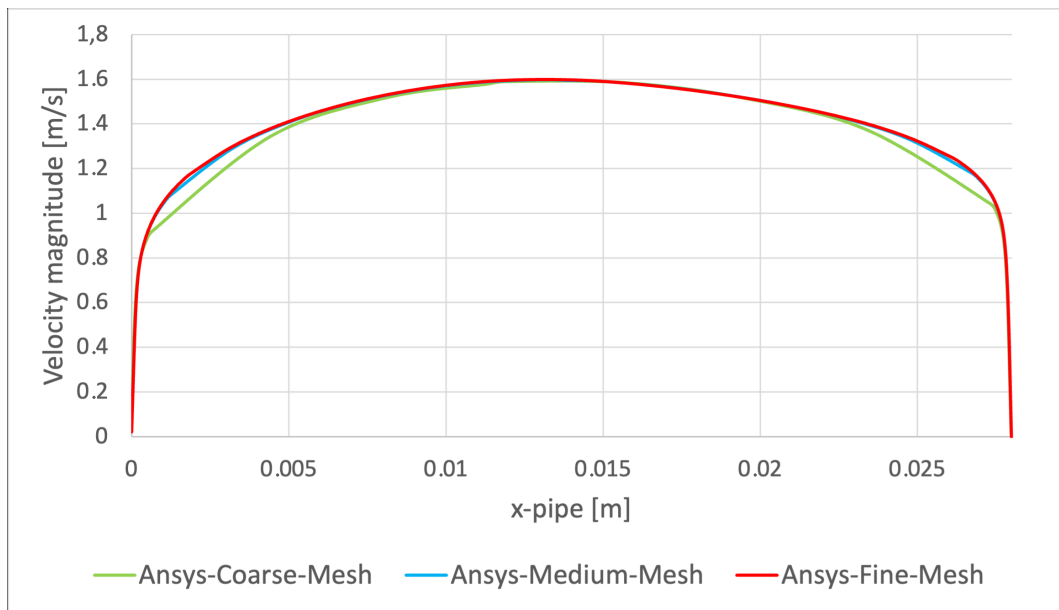


FIGURE C.3: Ansys U-bend exit velocity profiles across $D = 0.028$ m for the Coarse, Medium, and Fine grids.

Appendix D

Topology-Optimization Methodology Support

This appendix gathers supporting details for Chapter 5: interpolation curves, passive-region handling, filtering conventions, inlet turbulence conversion, and solver controls. These details are kept separate from the main methodology chapter so the central formulation remains readable.

D.1 RAMP Brinkman Interpolation

For visualization, the Brinkman resistance in Eq. 5.1 can be normalized as

$$\hat{\alpha}(\bar{\gamma}) = \frac{\alpha(\bar{\gamma}) - \alpha_{fluid}}{\alpha_{solid} - \alpha_{fluid}}. \quad (\text{D.1})$$

When $\alpha_{fluid} = 0$, this becomes $\hat{\alpha} = \alpha(\bar{\gamma})/\alpha_{solid}$. With the q convention used in this thesis, small q values make most intermediate densities comparatively fluid-like except very close to $\bar{\gamma} = 0$, while larger q values increase the resistance over a wider part of the intermediate-density range.

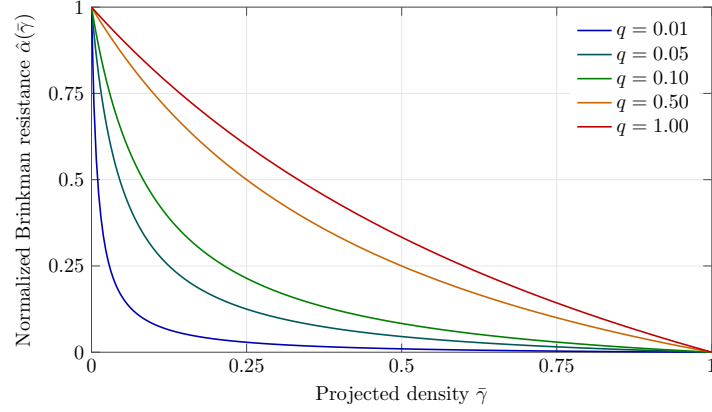


FIGURE D.1: Normalized RAMP Brinkman resistance from Eq. 5.1 for $q = \{0.01, 0.05, 0.1, 0.5, 1.0\}$ (ChatGPT, 2026).

D.2 Active and Passive Density Regions

Figure D.2 shows how the optimization code separates cells that MMA is allowed to change from cells that are prescribed by the benchmark geometry. The active cells are design variables. Passive-fluid extensions, for example inlet and outlet ducts, are held at $\bar{\gamma} = 1$ but are still included in the flow solve. Passive-solid regions, when used, are held at $\bar{\gamma} = 0$.

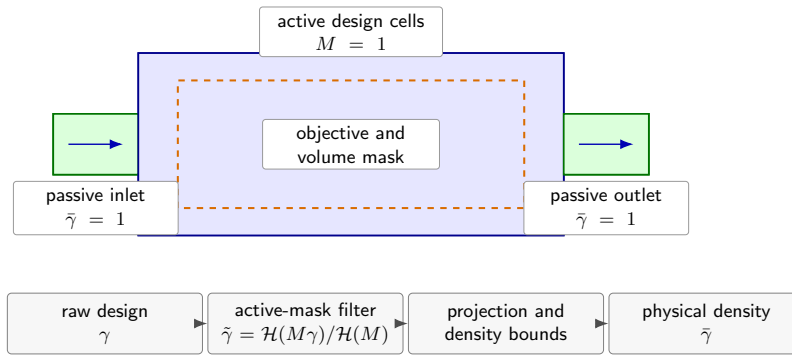


FIGURE D.2: Active design region with passive inlet and outlet extensions (ChatGPT, 2026).

D.3 DG0 Density Filtering and Active-Mask Normalization

The DG0 density filter of Eq. 5.20 is applied before the projection step. Near prescribed inlet and outlet extensions, a direct filter would blend the fixed-fluid cells into the neighboring active cells. The active-mask normalization avoids that leakage.

The implementation therefore filters the active contribution and divides by the filtered active mask:

$$\tilde{\gamma} = \frac{\mathcal{H}(M\gamma)}{\mathcal{H}(M)}. \quad (\text{D.2})$$

Here $\mathcal{H}$ denotes the filter operator and M is the active-cell indicator. After the Heaviside projection, the prescribed passive-cell bounds are restored. The sensitivity chain uses the same mapping in reverse, so MMA receives derivatives only for the active raw density variables.

D.4 Heaviside Projection Curve

Figure D.3 plots the Heaviside projection used in Eq. 5.21. Larger values of β make the projected field switch more abruptly between solid-like and fluid-like values.

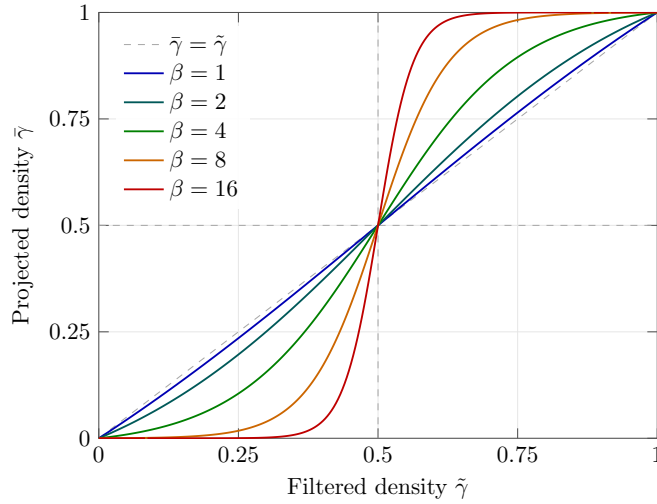


FIGURE D.3: Heaviside projection from Eq. 5.21 at $\eta = 0.5$ (ChatGPT, 2026).

D.5 Inlet Turbulence Conversion

When an inlet turbulence intensity I and turbulence length scale ℓ are provided, the code first estimates an eddy-viscosity ratio using the standard k - ϵ inlet relations,

$$k = \frac{3}{2}(UI)^2, \quad \epsilon = C_\mu^{3/4} \frac{k^{3/2}}{\ell}, \quad \frac{\nu_t}{\nu} = C_\mu^{1/4} \sqrt{\frac{3}{2}} I \frac{U\ell}{\nu}. \quad (\text{D.3})$$

If the Reynolds number and length-scale ratio are supplied instead, $U\ell/\nu$ is replaced by $Re(\ell/L)$.

The resulting ratio is converted to the SA inlet working variable by solving the SA viscosity relation

$$\nu_t = \tilde{\nu} f_{v1}, \quad f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3}, \quad \chi = \frac{\tilde{\nu}}{\nu}. \quad (\text{D.4})$$

This conversion is only an inlet initialization and boundary-condition device. The SA transport equation still determines the interior $\tilde{\nu}$ field during the forward solve.

D.6 Forward-Solver Controls

The optimization runs use IPCS and Picard updates to obtain the forward state before assembling the frozen-viscosity residual for the adjoint. Some benchmarks also activate SNES recovery schedules, convection continuation, or accepted-residual criteria when intermediate designs are difficult to solve. These controls are forward-solver devices. The frozen adjoint in Chapter 5 is formed from the final steady residual in Eq. 5.5, not by differentiating every IPCS, Picard, or recovery substep.

Appendix E

Topology-Optimization Benchmark Details

This appendix supports Chapter 6. It combines the turbulent and laminar topology-optimization support material in one place and groups it by benchmark case. Within each benchmark, the ordering is: turbulent-TO FEniCS settings and diagnostics, frozen-sensitivity checks where applicable, turbulent-design Ansys validation-mesh information, laminar-TO FEniCS settings and diagnostics, and laminar-design Ansys mesh independence where a completed hierarchy is available. For traceability, configuration file names are mentioned once in the relevant setup text; the tables summarize only the most important current settings.

Unless stated otherwise, all mesh-independence relative differences in this appendix are reported as positive percentage magnitudes. For a mesh-dependent metric ϕ , the coarse-medium difference is computed as $|\phi_{\text{coarse}} - \phi_{\text{medium}}|/|\phi_{\text{coarse}}| \times 100\%$, while the medium-fine difference is computed as $|\phi_{\text{medium}} - \phi_{\text{fine}}|/|\phi_{\text{fine}}| \times 100\%$.

E.1 Validation and Meshing Strategy

The FEniCS meshes used during topology optimization are treated as fixed optimization discretizations rather than as a separate mesh-convergence hierarchy. This distinction is important because refining the FEniCS mesh would also change the DG_0 design space, the filter stencil, the number of MMA design variables, and the diffuse Brinkman interface location. The pipe-bend optimization uses a rectangular-coordinate triangular mesh with near-wall clustering based on a smooth-wall $y^+ \approx 1$ estimate at the prescribed inlet Reynolds number. The U-bend optimization uses the corresponding wall-resolved triangular Gmsh mesh, with a near-wall size field based on the same $y^+ \approx 1$ estimate and a far-field target size of $h_{\text{max}} = 0.007L$ used to set the density-filter scale. The diffuser benchmark from Yoon [61] uses a wall-resolved rectangular-coordinate triangular mesh over the square diffuser domain, refined around the full-height inlet, the center-third outlet, and the no-slip walls. In all cases, the mesh is chosen to provide sufficient wall-resolved SA and design

resolution for the optimization problem; quantitative grid-convergence evidence is instead reported for the extracted sharp geometries in Ansys Fluent.

For each extracted turbulent topology optimization design with a completed Ansys mesh hierarchy, the validation mesh is refined through coarse, medium, and fine resolutions. Both SA and SST $k - \omega$ validation runs use the same extracted geometry, inlet condition, outlet condition, and fluid properties. The mesh-independence comparison is reported for both turbulence models where available because near-wall behavior and separated-flow predictions may respond differently to grid refinement. For the U-bend benchmark, the turbulent- and laminar-design validation tables use the same coarse–medium–fine mesh-sizing settings, while the final node and element counts differ because the extracted sharp-wall geometries are different.

Completed Ansys Fluent mesh-independence runs in this appendix use the same convergence and acceptance criteria. The static pressure drop must change by less than 5×10^{-4} over the final 100 iterations, while the viscous dissipation must change by less than 10^{-3} over the final 100 iterations. The fine mesh is accepted only when the medium–fine relative difference is below 1% for static pressure drop and below 2% for viscous dissipation for both SA and SST $k - \omega$. The reported refinement differences are computed as $|\phi_{\text{coarse}} - \phi_{\text{medium}}|/|\phi_{\text{coarse}}|$ and $|\phi_{\text{medium}} - \phi_{\text{fine}}|/|\phi_{\text{fine}}|$ for each validation metric.

E.2 Shared Turbulence-Topology Coupling Conventions

All three turbulent topology optimization benchmarks use the same basic coupling idea between the density field, the SA working variable, and the reciprocal wall-distance equation. Topology-created solids act as walls for the SA working-variable penalty and the reciprocal wall-distance solve. The wall-density source is therefore **design**, the wall-distance mode is **reciprocal_penalized**, and both the SA and wall-distance penalties use the power-law solid indicator defined in Chapter 5. The same frozen-turbulence adjoint convention is also used: the forward loop updates $\tilde{\nu}$, ν_t , and G , while the derivative freezes $\tilde{\nu}$ and G during gradient assembly.

The pipe-bend and U-bend benchmarks then share a Bayat-style pressure-loss setup, while the diffuser follows the Yoon dissipation benchmark. The common mechanisms and the benchmark-specific parameter values are summarized in Table E.1.

The pipe-bend finite-difference sensitivity verification in Section E.3.2 uses the first pressure-objective stage of the current smooth-continuation setup and stores its own finite-difference output tables. It should therefore be read as a frozen-adjoint implementation check, not as evidence of convergence for the reported optimization state.

E.2. Shared Turbulence-Topology Coupling Conventions

TABLE E.1: Shared turbulence-topology coupling conventions and benchmark-specific parameter values.

Parameter	Pipe bend and U-bend	Diffuser
SA wall-distance mode	reciprocal_penalized	reciprocal_penalized
SA wall-density source	design	design
SA penalty interpolation	Power-law solid indicator	Power-law solid indicator
Wall-distance penalty interpolation	Power-law solid indicator	Power-law solid indicator
Wall-distance recovery	Reciprocal recovery from the relaxed Eikonal variable G	Reciprocal recovery from the relaxed Eikonal variable G
Adjoint treatment	Frozen-turbulence adjoint; $\tilde{\nu}$ and G frozen during gradient assembly	Frozen-turbulence adjoint; $\tilde{\nu}$ and G frozen during gradient assembly
Solid threshold for wall source	$\bar{\gamma} = 0.50$	No separate threshold parameter in the diffuser configuration
SA working-variable penalty	Pipe bend: fixed amplitude 10^3 , exponent $n = 4$; U-bend: amplitude continuation ending at 10^5 , exponent $n = 3$	Power-law solid indicator with amplitude 10^5 and exponent $n = 3$
Wall-distance penalty	Pipe bend: fixed amplitude 10^3 , exponent $n = 4$; U-bend: amplitude continuation ending at 10^5 , exponent $n = 3$	Power-law solid indicator with amplitude 10^5 and exponent $n = 3$
Wall-distance parameters	Pipe bend: $\sigma_w = 0.01$, $G_0 = 20$; U-bend: $\sigma_w = 0.1$, $G_0 = 20$	$\sigma_w = 0.1$, $G_0 = 20$
SA inlet treatment	Turbulence-intensity estimate: $I = 7.5\%$, $\ell/H = 0.05$, eddy-viscosity ratio capped at 50	Prescribed $\tilde{\nu} = 1$ at inlet and outlet
Optimization objective	Average inlet pressure with fixed outlet pressure $p = 0$	Dissipation objective J_D ; the velocity-outlet baseline prescribes inlet and outlet profiles, while the primary pressure-outlet case prescribes the inlet profile and fixes outlet static pressure

E.3 Pipe Bend Benchmark

The pipe-bend case follows the setup of Bayat et al. [13]. The reference paper uses a standard k - ε model with wall functions; the present thesis uses the wall-resolved SA implementation and evaluates the $V_f = 0.25$ design. The active turbulent FEniCS configuration is the pressure-objective smooth-continuation path in `Config_PipeBendAlexandersen_Frozen.py`, resumed from the reconstructed iteration-70 checkpoint that completed the first $q = 1/150$, $\beta = 4$ stage. This reported state is therefore still treated as an in-progress continuation state rather than as a converged final topology-optimization result.

TABLE E.2: Geometric, physical, and boundary-condition parameters for the pipe-bend benchmark.

Parameter	Setting
Reference case	Pipe-bend benchmark from Bayat et al. [13]
Reference figure for setup	Bayat et al. Fig. 10
Reference velocity figure	Bayat et al. Fig. 15
Reference turbulence model	Standard k - ε with wall functions
Turbulence model in this thesis	Spalart-Allmaras
Design-domain dimensions	Square design region $L \times L$ with $L = 1$
Full computational domain	$x/L \in [-0.2, 1]$, $y/L \in [-0.2, 1]$
Non-design inlet duct	Length $0.2L$, height $0.2L$, inlet over $y/L \in [0.7, 0.9]$
Non-design outlet duct	Width $0.2L$, length $0.2L$, outlet over $x/L \in [0.7, 0.9]$ on the lower boundary
Volume fraction	$V_f = 0.25$
Reynolds number	$Re = 5,000$, based on $U_{in}H_{in}\rho/\mu$ with $H_{in} = 0.2L$
Density and viscosity	$\rho = 1.0$, $\mu = 4.0 \times 10^{-5}$
Inlet condition	Uniform inlet velocity $\mathbf{u} = (1, 0)$; SA inlet estimated from the shared turbulence-intensity settings in Table E.1
Outlet condition	Pressure outlet $p = 0$ on the lower outlet, with the normal/-tangential component constraint used by the FEniCS configuration
Wall conditions	No-slip velocity and $\tilde{\nu} = 0$ on external walls; topology-created solids enter through the design-density penalties
Validation metrics	Average inlet pressure, static pressure drop, Φ_D , and relative changes between FEniCS-SA, Ansys-SA, and Ansys SST k - ω

E.3.1 Reference Topology Overlay

Figure E.1 overlays the present FEniCS-SA pipe-bend topology with the corresponding Bayat et al. [13] wall-function result. The comparison is qualitative, because the reference and present designs use different turbulence closures and near-wall treatments, but the main routed bend is nearly coincident.

TABLE E.3: Numerical FEniCS solver, topology-optimization, and sensitivity-verification parameters for the reported pipe-bend state.

Parameter	Symbol / option	Setting
Flow finite elements	$(\mathbf{u}, p)$	Taylor–Hood P_2/P_1 continuous Galerkin elements
SA working-variable space	$\tilde{\nu}$	P_1 continuous Galerkin
Design-density space	γ	DG_0 cellwise constant density
FEniCS optimization mesh	–	<code>mesh_yplus1.xdmf</code> ; 39,128 vertices, 77,334 triangles, $h_{\max} = 0.007L$
Objective	J_p	Average inlet pressure
Initial design density	γ_0	$V_f = 0.25$, matched to the filtered volume
Brinkman penalty	$\alpha(\tilde{\gamma})$	RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 10^3$
RAMP continuation	q	Pressure-objective schedule $\{1/150, 1/120, 1/100, 1/75, 1/50, 1/30, 1/15\}$
Projection continuation	β	Schedule $\{4, 4.5, 5, 6, 7, 8, 9\}$ with threshold $\eta = 0.50$
MMA move limits	–	Schedule $\{0.010, 0.004, 0.0035, 0.003, 0.0025, 0.002, 0.0015\}$
Optimization iterations	–	Resume from reconstructed iteration-70 checkpoint; seven-stage budgets $\{70, 80, 90, 110, 130, 160, 200\}$ with stopping after 14 iterations below objective-change tolerance 10^{-5}
Filter radius	r_f	$4h_{\max} = 0.028L$
SA and wall-distance penalties	–	Fixed amplitudes 10^3 with power-law exponent $n = 4$; reciprocal wall-distance parameter $\sigma_w = 0.01$; no penalty continuation in the active pressure-objective path
Forward flow solver	–	IPCS flow solve for the frozen-SA Picard update, followed by a warm-started SNES Newton line-search flow solve
Frozen NS-SA coupling	–	One Picard update per design iteration with SA relaxation factor 0.15
Linear solver	–	MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves
Adjoint treatment	–	Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and G are frozen during gradient assembly
Sensitivity verification samples	–	Five active DG_0 cells, seed 13
Sensitivity verification step	Δh	10^{-6} , central finite difference
Sensitivity verification fields held fixed	–	SA working variable $\tilde{\nu}^0$ and reciprocal wall-distance field G^0

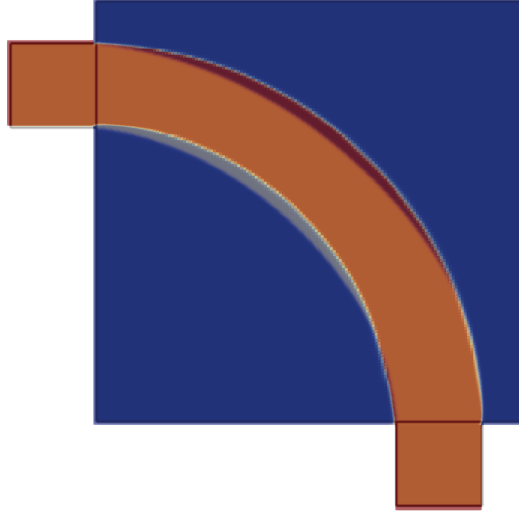


FIGURE E.1: Qualitative topology overlay for the turbulent pipe-bend benchmark; red: present FEniCS-SA fluid path, yellow: Bayat et al. [13] wall-function topology.

E.3.2 Frozen Sensitivity Verification

The frozen adjoint is checked directly on the pressure-objective pipe-bend setup inherited by `Config_PipeBendAlexandersen_Frozen_Sensitivity.py`. At the checked initial design state this corresponds to the first smooth-continuation stage in Table E.3: $\alpha_{\text{solid}} = 10^3$, $q = 1/150$, $\beta = 4$, move limit 0.010, one frozen-SA Picard update with SA relaxation 0.15, wall-distance parameter $\sigma_w = 0.01$, and power-law SA/wall-distance penalty exponent $n = 4$. No full turbulence-adjoint or wall-distance-adjoint comparison is made; the purpose is to compare the frozen derivative supplied to MMA with objective changes obtained by re-solving the perturbed velocity-pressure state while using the same derivative path as the current continuation configuration.

The checked-in coordinate finite-difference result is performed at the initial pipe-bend design state, with base objective $J_0 = 1.8343209502$. In every perturbation, the SA working variable $\tilde{\nu}^0$ and reciprocal wall-distance field G^0 are held fixed. This matches the frozen-turbulence assumption used by the adjoint: the flow state is re-solved, but the turbulence and wall-distance updates are not differentiated.

The sensitivity configuration evaluates five active DG_0 samples with seed 13. For sampled cell i , the raw density vector γ is perturbed by $\pm\Delta h \mathbf{e}_i$, where $\mathbf{e}_i$ is the unit vector for that active design variable and $\Delta h = 10^{-6}$. The central finite-difference derivative is

$$S_i^{\text{FD}} = \frac{J_p(\gamma + \Delta h \mathbf{e}_i; \mathbf{u}^+, p^+, \tilde{\nu}^0, G^0) - J_p(\gamma - \Delta h \mathbf{e}_i; \mathbf{u}^-, p^-, \tilde{\nu}^0, G^0)}{2\Delta h}. \quad (\text{E.1})$$

The frozen-adjoint value for the same cell is denoted by S_i^{fr} ; the superscripts FD and fr denote finite-difference and frozen-adjoint quantities, respectively. The absolute

and relative discrepancies are computed as

$$\Delta_i^{\text{abs}} = |S_i^{\text{fr}} - S_i^{\text{FD}}|, \quad \varepsilon_i^{\text{fr}} = \frac{\Delta_i^{\text{abs}}}{\max(|S_i^{\text{fr}}|, |S_i^{\text{FD}}|, 10^{-30})}. \quad (\text{E.2})$$

TABLE E.4: Coordinate sensitivity check for the frozen pipe-bend configuration. Values are taken from `SensitivityVerificationTable.tsv`.

Cell	Finite difference S_i^{FD}	Frozen adjoint S_i^{fr}	Absolute error Δ_i^{abs}	Relative error $\varepsilon_i^{\text{fr}}$
CV_1	-2.7011×10^{-5}	-2.7010×10^{-5}	1.5875×10^{-10}	5.8774×10^{-6}
CV_2	-2.1146×10^{-6}	-2.1148×10^{-6}	1.2513×10^{-10}	5.9171×10^{-5}
CV_3	-1.6664×10^{-4}	-1.6664×10^{-4}	4.2394×10^{-10}	2.5440×10^{-6}
CV_4	-1.7209×10^{-5}	-1.7209×10^{-5}	4.3775×10^{-10}	2.5437×10^{-5}
CV_5	2.3641×10^{-5}	2.3641×10^{-5}	2.0497×10^{-10}	8.6700×10^{-6}

The coordinate checks show close agreement between the frozen-adjoint derivative and the re-solved central finite differences across all five samples. The maximum absolute discrepancy is 4.38×10^{-10} , and the maximum relative discrepancy is 5.92×10^{-5} . The sign of the derivative also agrees in all five samples.

The same sampled active cells are also used for a directional Taylor check with steps $(10^{-3}, 3 \times 10^{-4}, 10^{-4}, 3 \times 10^{-5})$ and seed 29. The directional derivative is $9.6668564775 \times 10^{-5}$, and the central finite differences remain within 8.66×10^{-8} relative error over the tested step range. The symmetric Taylor remainder decays with approximately second-order behavior, as reported in Table E.5.

TABLE E.5: Directional Taylor check for the frozen pipe-bend sensitivity configuration. Values are taken from `TaylorCheckLog.tsv`.

Step ϵ	Frozen directional derivative	Central finite difference	Directional relative error	Symmetric Taylor order
1.0×10^{-3}	$9.6668564775 \times 10^{-5}$	$9.6668565153 \times 10^{-5}$	3.9055×10^{-9}	–
3.0×10^{-4}	$9.6668564775 \times 10^{-5}$	$9.6668563524 \times 10^{-5}$	1.2939×10^{-8}	2.0006
1.0×10^{-4}	$9.6668564775 \times 10^{-5}$	$9.6668562044 \times 10^{-5}$	2.8252×10^{-8}	2.0077
3.0×10^{-5}	$9.6668564775 \times 10^{-5}$	$9.6668573146 \times 10^{-5}$	8.6596×10^{-8}	2.1339

This check verifies the implemented frozen-turbulence derivative for the first-stage pipe-bend sensitivity setup that generated the stored TSV file. The conclusion is deliberately limited to that derivative: the SA working variable and reciprocal wall-distance field are held fixed during the checks, matching the optimization adjoint; consequently, the result should not be interpreted as validation of a fully coupled turbulence sensitivity or as convergence evidence for the resumed pipe-bend continuation state.

E.3.3 Turbulent FEniCS Optimization Diagnostics

Figure E.2 records the convergence history for the first turbulent pipe-bend pressure-objective stage. The reported endpoint is the reconstructed iteration-70 checkpoint

that seeds the current smooth-continuation configuration. The pressure objective is normalized by its initial value $J_{p,0}$, while the volume constraint is defined as the relative residual

$$r_V^{(k)} = 100 \frac{v_f^{(k)} - V_f}{V_f}, \quad (\text{E.3})$$

where $v_f^{(k)}$ is the realized fluid volume fraction at MMA iteration k . With this sign convention, the dashed line at $r_V = 0$ represents the active volume constraint. Values below this line are feasible and correspond to a design using slightly less fluid volume than the prescribed limit.

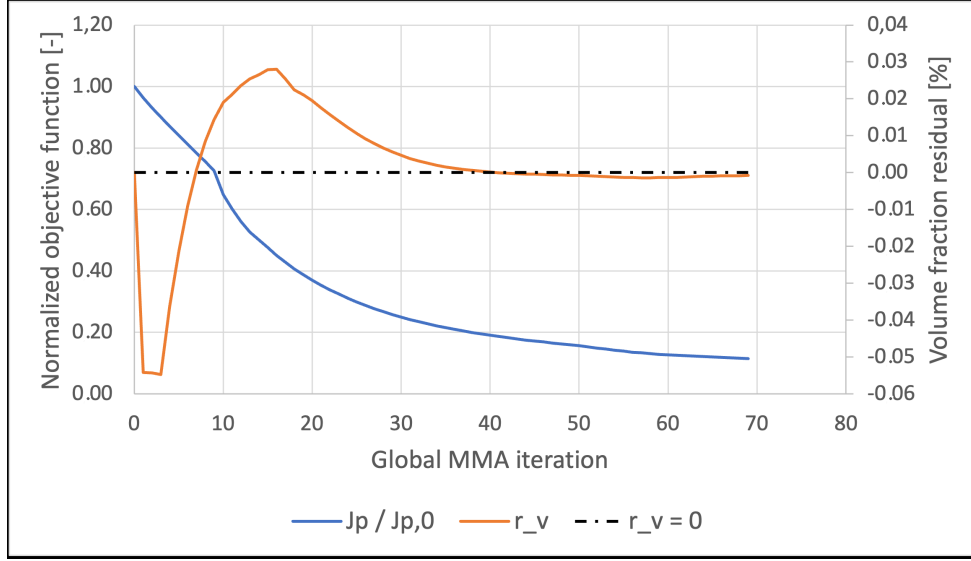


FIGURE E.2: Convergence history for the first 70-iteration turbulent pipe-bend J_p stage, showing $J_p/J_{p,0}$ and the relative volume-fraction residual r_V from Eq. (E.3).

The monotonic objective reduction supports using the checkpoint scalar values for the provisional turbulent pipe-bend discussion in Chapter 6. Since the current checked-in setup resumes from this stage into additional continuation, no final convergence claim is made from the first-stage history alone.

The extracted turbulent pipe-bend geometry is meshed in Ansys Fluent with a patch-conforming triangular bulk mesh and wall-resolved inflation on all no-slip walls. The first-layer height is kept fixed from the smooth-wall $y^+ \approx 1$ estimate, while the mesh hierarchy refines the bulk face size, curvature angle, and inflation distribution.

TABLE E.6: Ansys Fluent mesh-independence table for the extracted turbulent pipe-bend benchmark geometry.

Parameter	Coarse	Medium	Fine
<i>Mesh settings</i>			
Mesh type	Tri.	Tri.	Tri.
h_{bulk} [m]	1.10×10^{-2}	5.00×10^{-3}	3.25×10^{-3}
Curvature angle	15°	7°	4.5°
Near-wall	WR infl.	WR infl.	WR infl.
Δy_1 [m]	6.00×10^{-4}	6.00×10^{-4}	6.00×10^{-4}
Inflation layers	8	14	18
Growth rate	1.15	1.10	1.08
t_{infl} [m]	8.24×10^{-3}	1.68×10^{-2}	2.25×10^{-2}
Nodes	5,147	21,343	44,042
Elements	7,551	32,780	68,743
<i>Validation metrics</i>			
Φ_D^{SA} [W/m]	0.0143	0.0151	0.0153
$\epsilon_{\text{rel}}(\Phi_D^{\text{SA}})$ [%]	–	5.38	1.32
$\Phi_D^{\text{SST } k-\omega}$ [W/m]	0.0150	0.0164	0.0166
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST } k-\omega})$ [%]	–	9.04	1.22
Static pressure drop Δp_s^{SA} [Pa]	0.1423	0.1449	0.1457
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SA}})$ [%]	–	1.84	0.55
Static pressure drop $\Delta p_s^{\text{SST } k-\omega}$ [Pa]	0.1481	0.1516	0.1522
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SST } k-\omega})$ [%]	–	2.36	0.40
Selected validation mesh	Fine_Ansys		

Note. WR = wall-resolved; Φ_D = viscous dissipation; Δp_s = static pressure drop. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

E.3.4 Laminar FEniCS Configuration Differences

The laminar pipe-bend reference is a completed FEniCS topology-optimization run from `Config_PipeBendAlexandersen_Laminar.py`. It reuses the turbulent pipe-bend setup in Tables E.2 and E.3 wherever not stated otherwise: the geometry, wall-resolved mesh, non-design inlet and outlet ducts, uniform inlet velocity, pressure-outlet treatment, $V_f = 0.25$ volume constraint, average inlet-pressure objective J_p , filter radius, and projection threshold are unchanged. The Brinkman resistance and continuation controls are laminar-specific and are listed in Table E.7.

E.3.5 Laminar FEniCS Optimization Diagnostics

Figure E.3 records the completed laminar pipe-bend optimization history. The normalized inlet-pressure objective drops rapidly in the first continuation stage and then remains nearly flat, while the volume-fraction residual repeatedly returns to the active constraint after continuation updates.

E. TOPOLOGY-OPTIMIZATION BENCHMARK DETAILS

TABLE E.7: Laminar-specific differences from the turbulent pipe-bend FEniCS setup above.

Parameter	Laminar setting
Flow model	Laminar incompressible Navier–Stokes; the SA working variable, reciprocal wall-distance equation, topology-dependent SA penalties, and frozen-turbulence coupling are omitted
Reynolds number	$Re = 1$, based on $U_{\text{in}}H_{\text{in}}\rho/\mu$ instead of the turbulent $Re = 5,000$ setup
Density and viscosity	$\rho = 1.0$ and $\mu = 0.2$, with the same $U_{\text{in}} = 1$ and $H_{\text{in}} = 0.2L$ as the turbulent case
Brinkman penalty	RAMP form with $\alpha_{\text{fluid}} = 0$ and $\alpha_{\text{solid}} = 10^5$
Flow finite elements	Taylor–Hood P_2/P_1 for $(\mathbf{u}, p)$; no separate P_1 SA working-variable space is used
Forward and adjoint solves	SNES/MUMPS laminar Newton forward solves with relative tolerance 10^{-6} , absolute tolerance 10^{-8} , and maximum forward iteration count 220; the adjoint is solved as a linear system with MUMPS
Continuation and optimizer controls	Seven-stage schedule $q = \{0.01, 0.02, 0.04, 0.08, 0.08, 0.12, 0.20\}$, $\beta = \{1, 2, 4, 8, 8, 12, 16\}$, move limits $\{0.010, 0.008, 0.006, 0.004, 0.003, 0.002, 0.0015\}$, and inner-iteration budgets $\{12, 25, 45, 80, 120, 160, 220\}$; stop if objective change is below 10^{-6} for ten iterations

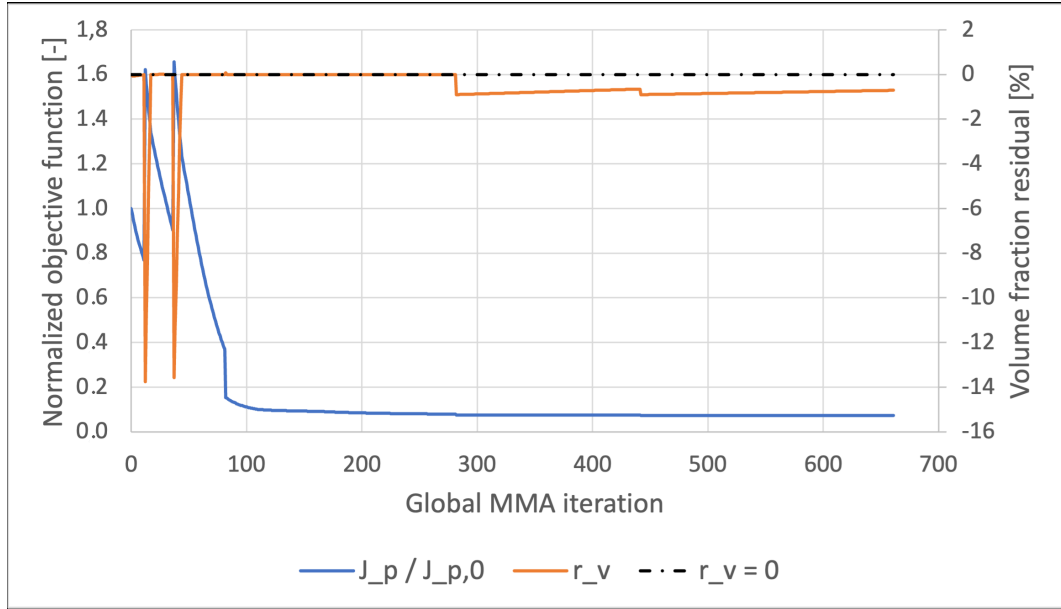


FIGURE E.3: FEniCS convergence history for the laminar pipe-bend topology-optimization run, showing $J_p/J_{p,0}$ and the relative volume-fraction residual.

E.3.6 Laminar-Design Mesh Independence

Table E.8 reports the Ansys mesh-independence study for the extracted laminar pipe-bend design. This table is separate from the turbulent-design mesh study in Table E.6 of Section E.3 because the laminar and turbulent topologies generally produce different wall shapes and near-wall mesh requirements.

TABLE E.8: Ansys mesh-independence table for the extracted laminar pipe-bend design.

Parameter	Coarse	Medium	Fine
<i>Mesh settings</i>			
Mesh type	Tri.	Tri.	Tri.
h_{bulk} [m]	4.50×10^{-3}	2.60×10^{-3}	2.00×10^{-3}
Curvature angle	8°	4°	3°
Δy_1 [m]	1.20×10^{-3}	1.20×10^{-3}	1.20×10^{-3}
Inflation layers	12	15	16
Growth rate	1.06	1.045	1.04
t_{infl} [m]	2.02×10^{-2}	2.49×10^{-2}	2.62×10^{-2}
Nodes	22,494	56,468	89,804
Elements	35,948	93,828	153,163
<i>Validation metrics</i>			
$\Phi_D^{\text{SST } k-\omega}$ [W/m]	0.0293	0.0310	0.0313
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST } k-\omega})$ [%]	–	5.82	0.95
Static pressure drop $\Delta p_s^{\text{SST } k-\omega}$ [Pa]	0.5229	0.5420	0.5444
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SST } k-\omega})$ [%]	–	3.65	0.45
Selected comparison mesh		Fine_Ansys	

Note. Φ_D = viscous dissipation; Δp_s = static pressure drop.

E.4 U-Bend Benchmark

The U-bend case follows Fig. 11 and Table 4 of Bayat et al. [13]. It uses the same turbulent topology optimization framework as the pipe bend, but forces a 180° return path around a fixed internal separator. The active turbulent FEniCS setup is summarized from `Config_UBendAlexandersen_Frozen.py`.

No separate U-bend sensitivity-verification table is reported. The U-bend uses the same average inlet-pressure objective J_p and the same frozen-turbulence $(\mathbf{u}, p)$ adjoint implementation as the pipe-bend pressure-loss benchmark, for which the direct finite-difference and Taylor checks are reported above. The omitted U-bend check would therefore mainly repeat the same derivative path, while adding a final-flow-solver robustness issue specific to the tighter U-bend continuation rather than providing independent adjoint evidence.

TABLE E.9: Geometric, physical, and boundary-condition parameters for the U-bend benchmark.

Parameter	Setting
Reference case	U-bend benchmark from Bayat et al. [13]
Reference figure/table for setup	Bayat et al. Fig. 11 and Table 4
Reference turbulence model	Standard k - ε with wall functions
Turbulence model in this thesis	Spalart-Allmaras
Design-domain dimensions	Square design region $L \times L$ with $L = 1$
Full computational domain	$x/L \in [-0.2, 1]$, $y/L \in [0, 1]$
Port geometry	Two left-side ports of height $0.2L$; inlet over $y/L \in [0.55, 0.75]$, outlet over $y/L \in [0.25, 0.45]$
Fixed separator	Bar thickness $0.10L$, rounded-tip radius $0.05L$, extending from $x/L = -0.2$ to tip at $x/L = 0.5$
Volume fraction	$V_f = 0.27$
Reynolds number	$Re = 5,000$, based on $U_{\text{in}} H_{\text{port}} \rho / \mu$ with $H_{\text{port}} = 0.2L$
Density and viscosity	$\rho = 1.0$, $\mu = 4.0 \times 10^{-5}$
Inlet condition	Uniform inlet velocity $\mathbf{u} = (1, 0)$; SA inlet estimated from the shared turbulence-intensity settings in Table E.1
Outlet condition	Pressure outlet $p = 0$ on the lower left port, with the component constraint used by the FEniCS configuration
Wall conditions	No-slip velocity and $\tilde{\nu} = 0$ on external walls and the fixed separator; topology-created solids enter through the design-density penalties
Validation metrics	Average inlet pressure, static pressure drop, Φ_D , and relative changes between FEniCS-SA, Ansys-SA, and Ansys SST k - ω

E.4.1 Turbulent FEniCS Optimization Diagnostics

Figure E.4 records the available convergence history for the turbulent U-bend topology through 275 global MMA iterations. The pressure objective is normalized by its initial value $J_{p,0}$, and the volume trace uses the same relative residual definition as Eq. (E.3). The run should be interpreted as an optimization diagnostic rather than a converged final result: the normalized objective decreases smoothly over the available iterations, while the volume residual shows continuation-related excursions before returning toward the active constraint.

The next step for this benchmark is therefore a more robust turbulent-design U-bend optimization campaign, with solver and continuation settings tuned for the forced 180° return flow and run to a defined convergence criterion.

TABLE E.10: Numerical FEniCS solver and topology-optimization parameters for the U-bend benchmark.

Parameter	Symbol / option	Setting
Flow finite elements	$(\mathbf{u}, p)$	Taylor–Hood P_2/P_1 continuous Galerkin elements
SA working-variable space	$\tilde{\nu}$	P_1 continuous Galerkin
Design-density space	γ	DG_0 cellwise constant density
Reference FEniCS mesh	–	<code>mesh_yplus1.xdmf</code> ; wall-resolved triangular Gmsh mesh with 45,280 vertices, 85,780 triangles, near-wall refinement based on $y^+ \approx 1$, and far-field $h_{\max} = 0.007L$
Objective	J_p	Average inlet pressure
Initial design density	γ_0	$V_f = 0.27$, matched to the filtered volume
Brinkman penalty	$\alpha(\bar{\gamma})$	RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 10^3$ for the active J_p path
RAMP continuation	q	[0.02, 0.04, 0.08, 0.12, 0.18]
Projection continuation	β	[1.5, 3, 5, 8, 12] with threshold $\eta = 0.50$
MMA move limits	–	[0.012, 0.010, 0.007, 0.005, 0.003]
Maximum optimization iterations	–	Five continuation stages with [60, 70, 90, 110, 140] MMA iterations; stop if objective change is below 10^{-5} for ten iterations
Filter radius	r_f	$4h_{\max} = 0.028L$
SA and wall-distance penalty continuation	–	Shared amplitude schedule [10 ³ , 3 × 10 ³ , 10 ⁴ , 3 × 10 ⁴ , 10 ⁵ , 10 ⁵ , 10 ⁵]; the first five entries are used by the active five-stage continuation
Forward flow solver	–	IPCS flow solves for frozen-SA Picard updates, followed by a warm-started SNES Newton line-search final flow solve
Frozen NS-SA coupling	–	Three Picard updates per design iteration with SA relaxation factor 0.04 for the active J_p path
Linear solver	–	MUMPS for SNES/s-tate and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves
Adjoint treatment	–	Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and G are frozen during gradient assembly

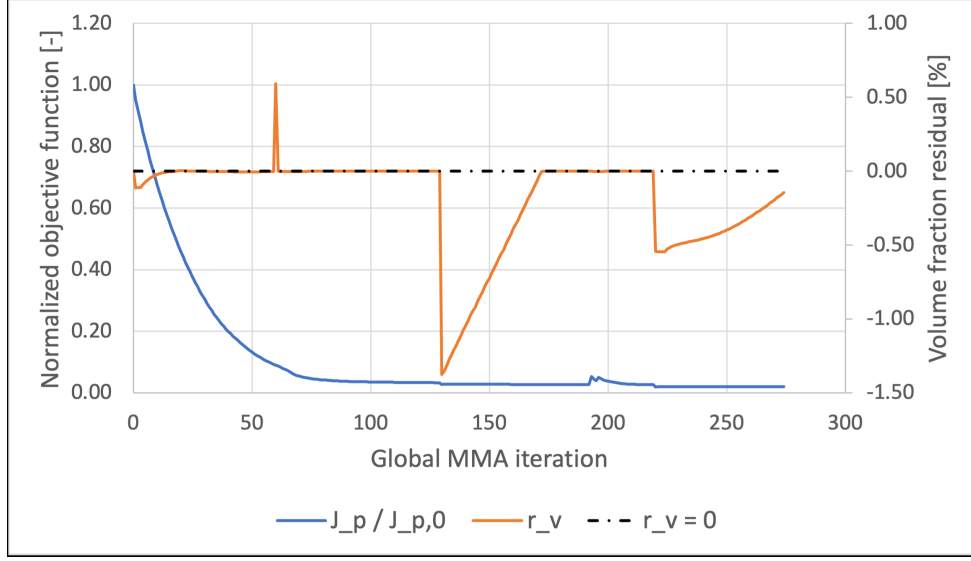


FIGURE E.4: Convergence history for the 275-iteration turbulent U-bend J_p run, showing $J_p/J_{p,0}$ and the relative volume-fraction residual r_V .

E.4.2 Turbulent-Design Mesh Independence

Table E.11 documents the Ansys validation mesh hierarchy for the extracted turbulent U-bend design. The coarse and fine mesh-sizing settings match those used for the laminar-design U-bend mesh-independence table in Table E.13, while the medium grid is refined for the turbulent-design geometry. The extracted sharp-wall geometry, final node and element counts, and resulting validation metrics therefore differ between the two studies.

TABLE E.11: Ansys mesh-independence table for the extracted turbulent U-bend design.

Parameter	Coarse	Medium	Fine
<i>Mesh settings</i>			
Mesh type	Tri.	Tri.	Tri.
h_{bulk} [m]	6.00×10^{-3}	3.25×10^{-3}	2.75×10^{-3}
Curvature angle	8°	6°	4°
Δy_1 [m]	6.00×10^{-4}	6.00×10^{-4}	6.00×10^{-4}
Inflation layers	12	18	20
Growth rate	1.10	1.07	1.06
t_{infl} [m]	1.28×10^{-2}	2.04×10^{-2}	2.21×10^{-2}
Nodes	15,250	44,627	59,252
Elements	22,878	68,769	91,806
<i>Validation metrics</i>			
Φ_D^{SA} [W/m]	0.0791	0.0785	0.0785
$\epsilon_{\text{rel}}(\Phi_D^{\text{SA}})$ [%]	–	0.76	0.00
$\Phi_D^{\text{SST } k-\omega}$ [W/m]	0.0961	0.1009	0.1016
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST } k-\omega})$ [%]	–	4.99	0.69
Static pressure drop Δp_s^{SA} [Pa]	0.8725	0.8123	0.8071
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SA}})$ [%]	–	6.90	0.64
Static pressure drop $\Delta p_s^{\text{SST } k-\omega}$ [Pa]	1.2139	1.1936	1.1973
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SST } k-\omega})$ [%]	–	1.67	0.31
Selected comparison mesh	Fine_Ansys		

Note. Φ_D = viscous dissipation; Δp_s = static pressure drop. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

E.4.3 Laminar FEniCS Configuration Differences

The laminar U-bend reference is a completed FEniCS topology-optimization run from `Config_UBendAlexandersen_Laminar.py`. It reuses the turbulent U-bend setup in Tables E.9 and E.10 wherever not stated otherwise: the geometry, fixed separator, wall-resolved mesh, non-design ports, uniform inlet velocity, pressure-outlet treatment, $V_f = 0.27$ volume constraint, average inlet-pressure objective J_p , filter radius, and projection threshold are unchanged. The Brinkman resistance and continuation controls are laminar-specific and are listed in Table E.12.

E.4.4 Laminar FEniCS Optimization Diagnostics

Figure E.5 records the completed laminar U-bend optimization history. As in the pipe-bend case, the objective reduction occurs mainly at the start of the run, after which the optimizer preserves a nearly active volume constraint while only small continuation-related corrections remain.

E. TOPOLOGY-OPTIMIZATION BENCHMARK DETAILS

TABLE E.12: Laminar-specific differences from the turbulent U-bend FEniCS setup above.

Parameter	Laminar setting
Flow model	Laminar incompressible Navier–Stokes; the SA working variable, reciprocal wall-distance equation, topology-dependent SA penalties, and frozen-turbulence coupling are omitted
Reynolds number	$Re = 1$, based on $U_{\text{in}} H_{\text{port}} \rho / \mu$ instead of the turbulent $Re = 5,000$ setup
Density and viscosity	$\rho = 1.0$ and $\mu = 0.2$, with the same $U_{\text{in}} = 1$ and $H_{\text{port}} = 0.2L$ as the turbulent case
Brinkman penalty	RAMP form with $\alpha_{\text{fluid}} = 0$ and $\alpha_{\text{solid}} = 10^5$
Flow finite elements	Taylor–Hood P_2/P_1 for $(\mathbf{u}, p)$; no separate P_1 SA working-variable space is used
Forward and adjoint solves	SNES/MUMPS laminar Newton forward solves with relative tolerance 10^{-6} , absolute tolerance 10^{-8} , and maximum forward iteration count 260; the adjoint is solved as a linear system with MUMPS
Continuation and optimizer controls	Seven-stage schedule $q = \{0.01, 0.02, 0.04, 0.08, 0.08, 0.12, 0.20\}$, $\beta = \{1, 2, 4, 8, 8, 12, 16\}$, move limits $\{0.010, 0.008, 0.006, 0.004, 0.003, 0.002, 0.0015\}$, and inner-iteration budgets $\{12, 25, 45, 80, 120, 160, 220\}$; stop if objective change is below 10^{-6} for ten iterations

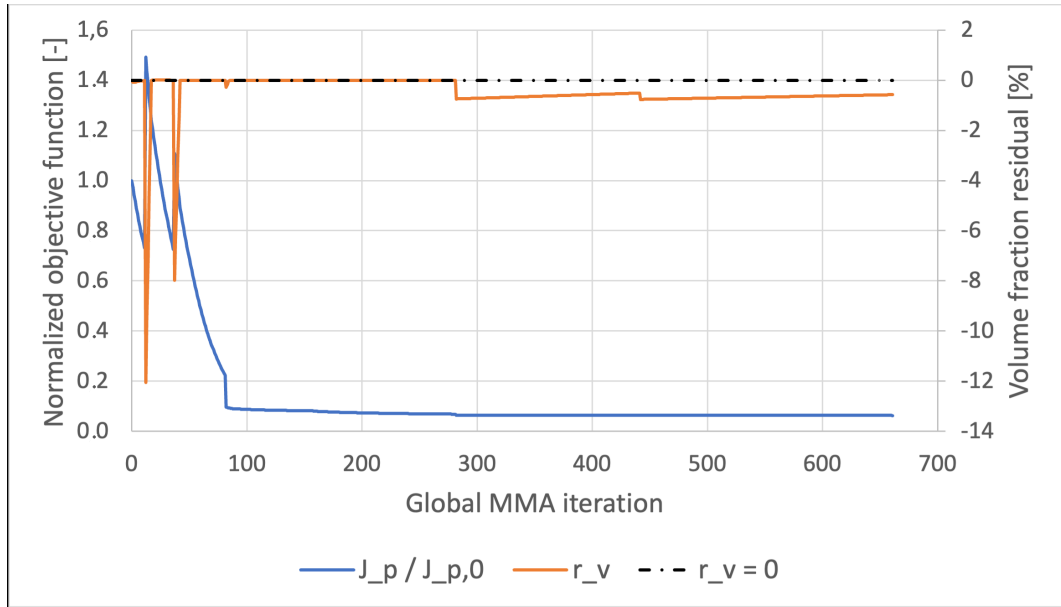


FIGURE E.5: FEniCS convergence history for the laminar U-bend topology-optimization run, showing $J_p/J_{p,0}$ and the relative volume-fraction residual.

E.4.5 Laminar-Design Mesh Independence

Table E.13 reports the Ansys mesh-independence study for the extracted laminar U-bend design. The turbulent-design validation table in Table E.11 uses the same coarse and fine sizing settings, but a refined medium grid. For the $Re = 5,000$ U-bend operating point, the inlet-port height is $0.2L$ and $U_{\text{in}} = 1 \text{ m s}^{-1}$, giving $\nu = 4.0 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$. Using the same smooth-wall estimate as the FEniCS mesh generator gives $u_\tau \approx 6.59 \times 10^{-2} \text{ m s}^{-1}$ and a wall distance of $y_+ = 1$ at $6.07 \times 10^{-4} \text{ m}$. The Ansys mesh hierarchy therefore keeps the first layer fixed at $6.00 \times 10^{-4} \text{ m}$ and refines the bulk triangular mesh, curvature resolution, and inflation-layer distribution around the separator tip and 180° return wall.

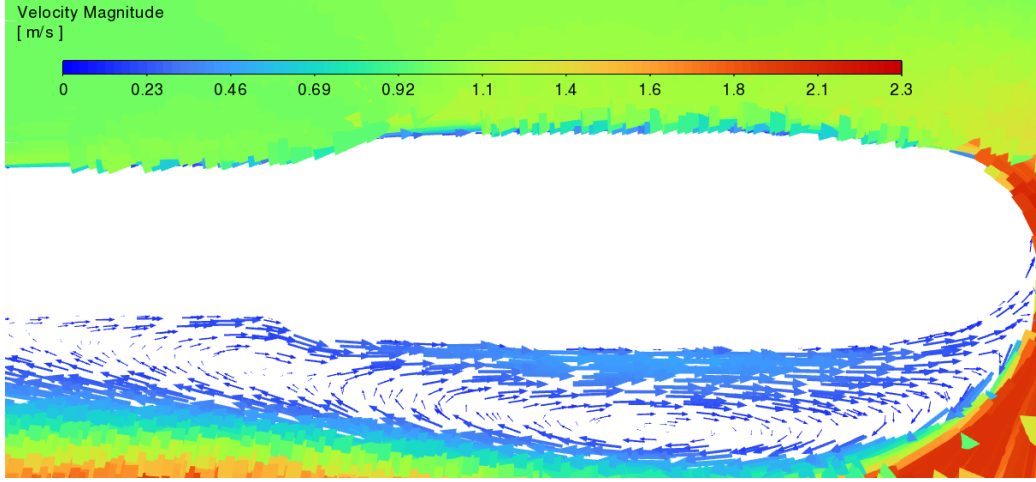
TABLE E.13: Ansys mesh-independence table for the extracted laminar U-bend design.

Parameter	Coarse	Medium	Fine
<i>Mesh settings</i>			
Mesh type	Tri.	Tri.	Tri.
h_{bulk} [m]	6.00×10^{-3}	4.00×10^{-3}	2.75×10^{-3}
Curvature angle	8°	6°	4°
Δy_1 [m]	6.00×10^{-4}	6.00×10^{-4}	6.00×10^{-4}
Inflation layers	12	16	20
Growth rate	1.10	1.08	1.06
t_{infl} [m]	1.28×10^{-2}	1.82×10^{-2}	2.21×10^{-2}
Nodes	16,106	32,580	62,622
Elements	24,460	50,067	98,084
<i>Validation metrics</i>			
$\Phi_D^{\text{SST } k-\omega}$ [W/m]	0.101	0.105	0.107
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST } k-\omega})$ [%]	–	3.96	1.90
Static pressure drop $\Delta p_s^{\text{SST } k-\omega}$ [Pa]	1.125	1.141	1.150
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SST } k-\omega})$ [%]	–	1.42	0.79
Selected comparison mesh	Fine_Ansys		

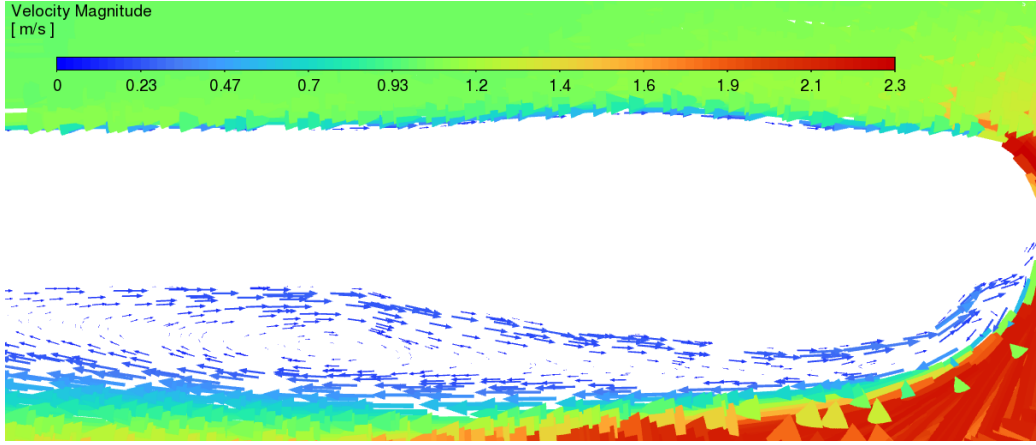
Note. Φ_D = viscous dissipation; Δp_s = static pressure drop. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

E.4.6 Recirculation Diagnostic

Figure E.6 gives a close-up of the SST $k - \omega$ velocity field near the separator and inner return path for the extracted laminar and turbulent U-bend designs. The vectors show local recirculation in the low-momentum region downstream of the separator, supporting the qualitative separation-and-recovery interpretation used in Chapter 6. The performance ranking is still taken from the integrated pressure-drop and dissipation metrics.



(A) Laminar-optimized design.



(B) Turbulent-optimized design.

FIGURE E.6: SST $k - \omega$ velocity-vector close-up near the U-bend separator and inner return path.

E.5 Diffuser Benchmark

The diffuser case follows Fig. 20–22 of Yoon [61]. The original velocity-outlet formulation prescribes parabolic velocity profiles at both the inlet and the outlet and uses the dissipation objective J_D . Chapter 6 uses that case as a boundary-condition sensitivity baseline, then treats the pressure-outlet variant as the primary diffuser result. The active turbulent FEniCS setup is summarized from `Config_DiffuserYoon_Frozen.py`. Both variants use the shared `design`, `reciprocal_penalized`, power-law, and frozen-adjoint conventions summarized in Table E.1.

For the body-fitted Ansys Fluent evaluations, the inlet parabolic profile is shifted to the extracted inlet segment. In the velocity-outlet baseline, the outlet profile is also shifted to the extracted outlet segment and its maximum velocity is adjusted

TABLE E.14: Geometric, physical, and boundary-condition parameters for the diffuser benchmark.

Parameter	Setting
Reference case	Diffuser benchmark from Yoon [61]
Reference figures for setup and topology	Yoon Fig. 20–22
Reference turbulence model	Spalart-Allmaras
Turbulence model in this thesis	Spalart-Allmaras with frozen-turbulence adjoint
Design-domain dimensions	Square design region $L \times L$ with $L = 1$ m
Full computational domain	$x/L \in [0, 1]$, $y/L \in [0, 1]$
Inlet geometry	FEniCS inlet is the full left boundary, $x/L = 0$ and $y/L \in [0, 1]$
Outlet geometry	FEniCS pressure outlet is the center third of the right boundary, $x/L = 1$ and $y/L \in [1/3, 2/3]$; right boundary outside the outlet is wall
Volume fraction	$V_f = 0.30$ fluid, corresponding to Yoon’s 70% solid mass usage
Reynolds number	$Re = 3,000$, based on $U_{\max, \text{in}} L \rho / \mu$
Density and viscosity	$\rho = 1000 \text{ kg/m}^3$, $\mu = 1.0 \text{ Pa s}$
Inlet condition	Parabolic velocity profile with $U_{\max, \text{in}} = 3 \text{ m/s}$; $\tilde{\nu} = 1$ for the FEniCS-SA solve
Outlet condition	Velocity-outlet baseline: parabolic outflow profile with maximum velocity set by incompressible flow-rate conservation and pressure pinned at $(0, 0)$ for gauge fixing. Pressure-outlet primary case: fixed static pressure $p = 0 \text{ Pa}$ on the center-third outlet with no prescribed outlet velocity.
Wall conditions	No-slip velocity and $\tilde{\nu} = 0$ on the top and bottom walls and on the right boundary outside the outlet
Brinkman solid resistance	$\alpha_{\text{solid}} = 10^9$, $\alpha_{\text{fluid}} = 0$
Optimization objective	Dissipation objective J_D , including physical viscous dissipation and Brinkman drag over the design domain
Validation metrics	J_D , static pressure drop, Φ_D , and relative changes between FEniCS-SA, Ansys-SA, and Ansys SST $k - \omega$

by incompressible flow-rate conservation. In the pressure-outlet variant, the outlet velocity distribution is solved rather than prescribed.

E.5.1 Velocity-Outlet Boundary-Condition Baseline

The geometry and boundary conditions follow Fig. 20 of Yoon [61]. The design domain is a square of side length $L = 1$ m. A parabolic inflow enters through the full left boundary with maximum speed $U_{\max, \text{in}} = 3$ m/s, while a parabolic outflow is prescribed on the center third of the right boundary with maximum speed $U_{\max, \text{out}} = 9$ m/s. The remaining external boundaries are treated as no-slip walls. The Reynolds number is $Re = 3,000$ based on $U_{\max, \text{in}} L \rho / \mu$, and the present density convention converts Yoon’s 70% solid mass usage into a maximum fluid volume fraction of $V_f = 0.30$. Figure E.7 shows the corresponding local geometry and boundary-condition layout used for the velocity-outlet baseline.

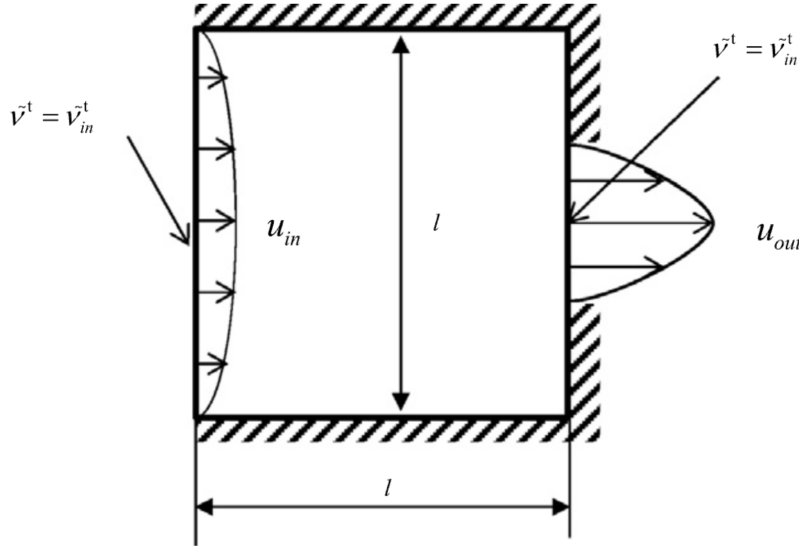


FIGURE E.7: Velocity-outlet diffuser baseline geometry and boundary conditions. Adapted from Yoon [61].

Figure E.8 reports the FEniCS-SA topology and velocity field for the velocity-outlet baseline. The outlet region widens before the prescribed outlet segment. This feature should not be interpreted as a purely geometry-driven diffuser optimum: the optimizer must deliver the imposed parabolic outflow over the fixed outlet boundary, and opening the channel near that boundary reduces the shear and redistribution needed to satisfy the Dirichlet profile.

Figure E.9 shows the corresponding Ansys-SA velocity contour for the extracted velocity-outlet design. The contour is used here only as a diagnostic of the baseline geometry: the widened outlet region contains low-speed separated or recovering flow,

TABLE E.15: Numerical FEniCS solver and topology-optimization parameters for the diffuser benchmark.

Parameter	Symbol / option	Setting
Flow finite elements	$(\mathbf{u}, p)$	Taylor–Hood P_2/P_1 continuous Galerkin elements
SA working-variable space	$\tilde{\nu}$	P_1 continuous Galerkin
Design-density space	γ	DG_0 cellwise constant density
Reference FEniCS mesh	–	<code>mesh_yoon_yplus1.xdmf</code> ; wall-resolved triangular mesh with 38,745 vertices, 76,704 triangles, far-field $h_{\max} = L/180 = 5.56 \times 10^{-3}L$, and first-cell width $1.82 \times 10^{-3}L$
Objective	J_D	Dissipation objective over the design domain
Initial design density	γ_0	$V_f = 0.30$, matched to the filtered volume
Brinkman penalty	$\alpha(\bar{\gamma})$	RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 10^9$
RAMP continuation	q	8-stage schedule: 0.01, 0.02, 0.04, 0.08, 0.08, 0.12, 0.20, 0.35
Projection continuation	β	8-stage schedule: 1, 2, 4, 8, 8, 12, 16, 32, with threshold $\eta = 0.50$
MMA move limits	–	8-stage schedule: 0.05, 0.04, 0.03, 0.02, 0.012, 0.010, 0.0075, 0.005
Maximum optimization iterations	–	8-stage schedule: 50, 60, 80, 100, 200, 180, 220, 260 MMA iterations; stop if objective change is below 10^{-6} for ten iterations
Filter radius	r_f	$4h_{\max} = 2.22 \times 10^{-2}L$
SA working-variable penalty	–	Power-law solid indicator with amplitude 10^5 and exponent $n = 3$
Wall-distance penalty	–	Power-law solid indicator with amplitude 10^5 and exponent $n = 3$
Wall-distance equation	–	Reciprocal penalized relaxed Eikonal equation with $\sigma_w = 0.1$ and $G_0 = 20$
Forward flow solver	–	IPCS flow solves for frozen-SA Picard updates, followed by an SNES Newton line-search final flow solve
Frozen NS-SA coupling	–	Three Picard updates per design iteration with SA relaxation factor 0.15
Linear solver	–	MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS Picard updates
Adjoint treatment	–	Frozen-turbulence adjoint $\mathbf{h}^3(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and G are frozen during gradient assembly
Sensitivity verification samples	–	Five active DG_0 cells, seed 13
Sensitivity verification step	Δh	10^{-6} , central finite difference
Sensitivity verification fields held fixed	–	SA working variable $\tilde{\nu}^0$ and reciprocal wall-distance field G^0

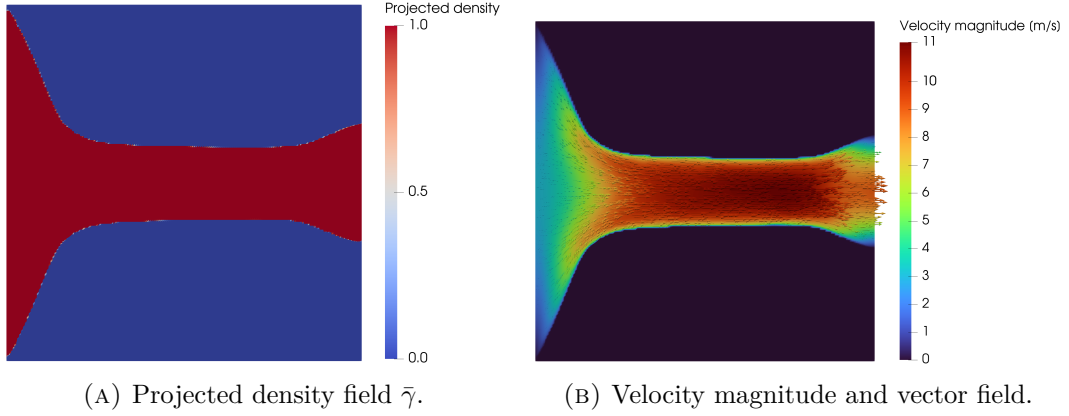


FIGURE E.8: FEniCS-SA result for the velocity-outlet diffuser baseline at $V_f = 0.30$. Final objective value $J_D = 3.1164 \times 10^5$ W/m.

which makes the original diffuser result strongly sensitive to the outflow treatment and turbulence closure.

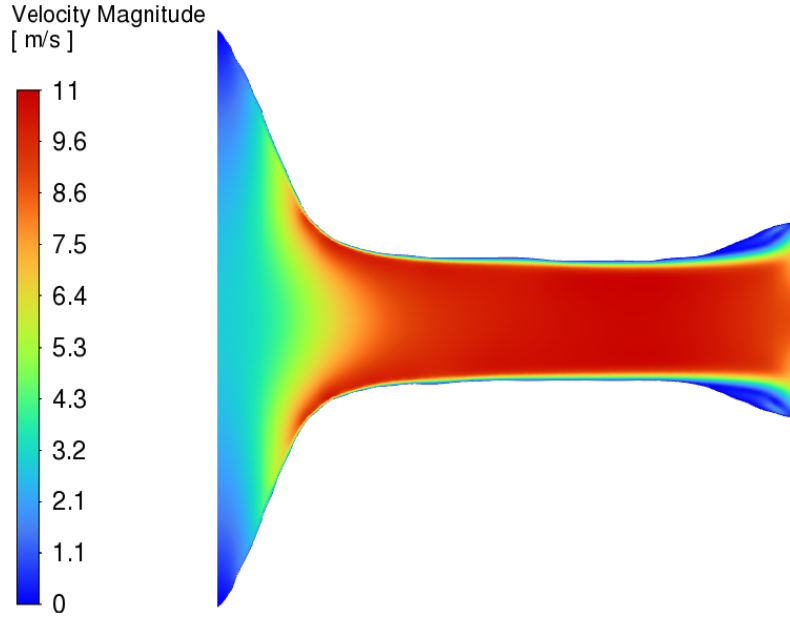


FIGURE E.9: Ansys-SA velocity-magnitude contour for the extracted velocity-outlet diffuser baseline.

E.5.2 Pressure-Outlet Turbulent FEniCS Optimization Diagnostics

Figure E.10 gives the pressure-outlet diffuser optimization history through the common 1150-iteration comparison point used in Chapter 6. The dissipation objective is

normalized by its initial value, while the volume trace is plotted as the relative residual $100(v_f^{(k)} - V_f)/V_f$ so that constraint activity can be read on the same iteration axis.

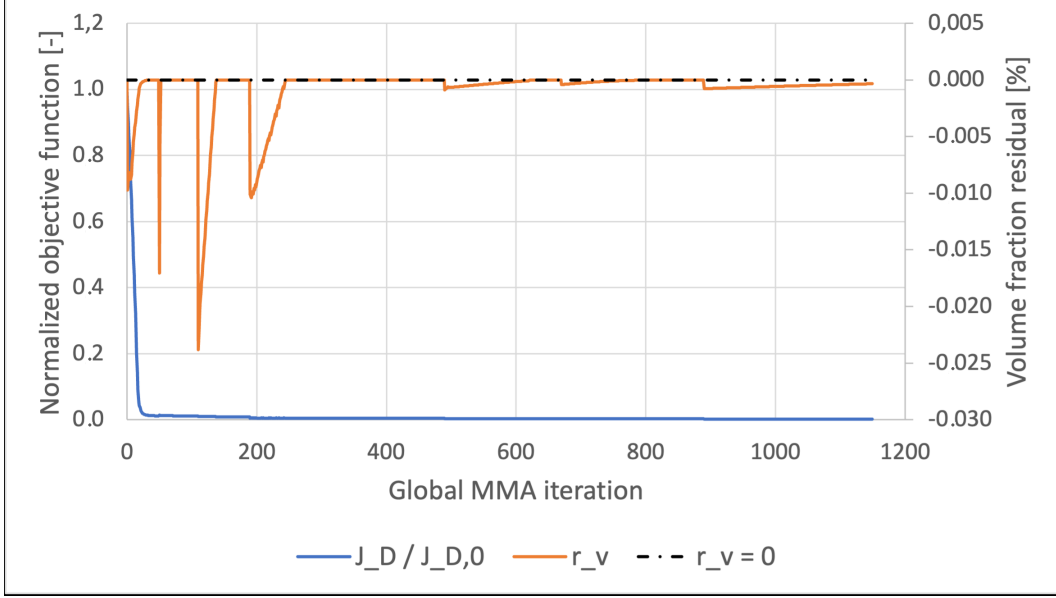


FIGURE E.10: Optimization convergence history for the turbulent pressure-outlet diffuser benchmark, showing $J_D/J_{D,0}$ and the relative volume-fraction residual.

E.5.3 Frozen Sensitivity Verification

The frozen adjoint is checked directly on the pressure-outlet diffuser sensitivity configuration in `Config_DiffuserYoon_Frozen_Sensitivity.py`. That file inherits the current pressure-outlet diffuser setup, including the dissipation objective J_D , the `design` wall-density source, the `reciprocal_penalized` wall-distance convention, and the same power-law topology penalties used by the production diffuser run. The purpose of the check is therefore narrow and explicit: compare the frozen derivative supplied to MMA with objective changes obtained by re-solving the perturbed velocity-pressure state while holding the SA working variable and reciprocal wall-distance field fixed.

The checked-in coordinate finite-difference result is performed at the initial diffuser design state, with base objective $J_0 = 1.1365575605 \times 10^8$. The sensitivity configuration evaluates five active DG_0 samples with seed 13 and perturbation size $\Delta h = 10^{-6}$. For sampled cell i , the raw density vector γ is perturbed by $\pm \Delta h \mathbf{e}_i$, where $\mathbf{e}_i$ is the unit vector for that active design variable, and the central finite-difference derivative is compared against the frozen-adjoint value.

The coordinate checks show consistent sign agreement and small relative errors across all five samples. The maximum relative discrepancy is 1.31×10^{-4} , while the largest absolute discrepancy is 1.35. That absolute value is not used as the

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TABLE E.16: Coordinate sensitivity check for the frozen pressure-outlet diffuser configuration. Values are taken from `SensitivityVerificationTable.tsv`.

Cell	Finite difference S_i^{FD}	Frozen adjoint S_i^{fr}	Absolute error Δ_i^{abs}	Relative error ϵ_i^{fr}
CV_1	-2.1542×10^4	-2.1541×10^4	6.7475×10^{-1}	3.1322×10^{-5}
CV_2	-3.9061×10^3	-3.9059×10^3	2.4168×10^{-1}	6.1872×10^{-5}
CV_3	-1.4988×10^4	-1.4990×10^4	1.3499×10^0	9.0058×10^{-5}
CV_4	-7.6433×10^3	-7.6430×10^3	2.3322×10^{-1}	3.0513×10^{-5}
CV_5	-4.3901×10^3	-4.3896×10^3	5.7420×10^{-1}	1.3079×10^{-4}

primary acceptance metric here: the sampled diffuser sensitivities themselves are of order 10^4 , and the base dissipation objective is of order 10^8 , so order-unity absolute derivative mismatches still correspond to small relative discrepancies in the actual derivative scale being checked.

The same sampled active cells are also used for a directional Taylor check with steps $(10^{-3}, 3 \times 10^{-4}, 10^{-4}, 3 \times 10^{-5})$ and seed 29. The directional derivative is $-1.3711285431 \times 10^4$, and the central finite differences remain within 1.09×10^{-6} relative error over the tested step range. The symmetric Taylor remainder decays with approximately second-order behavior over the first three step reductions, as reported in Table E.17.

TABLE E.17: Directional Taylor check for the frozen pressure-outlet diffuser sensitivity configuration. Values are taken from `TaylorCheckLog.tsv`.

Step ϵ	Frozen directional derivative	Central finite difference	Directional relative error	Symmetric Taylor order
1.0×10^{-3}	$-1.3711285431 \times 10^4$	$-1.3711284466 \times 10^4$	7.0351×10^{-8}	—
3.0×10^{-4}	$-1.3711285431 \times 10^4$	$-1.3711284200 \times 10^4$	8.9732×10^{-8}	2.0079
1.0×10^{-4}	$-1.3711285431 \times 10^4$	$-1.3711295649 \times 10^4$	7.4528×10^{-7}	2.0128
3.0×10^{-5}	$-1.3711285431 \times 10^4$	$-1.3711300244 \times 10^4$	1.0804×10^{-6}	2.7943

This check therefore verifies the implemented frozen-turbulence derivative for the pressure-outlet diffuser benchmark under the dissipation objective J_D . As with the bend cases, the conclusion is deliberately limited to that derivative: the SA working variable and reciprocal wall-distance field are held fixed during the checks, matching the optimization adjoint. It should not be interpreted as validation of a fully coupled turbulence sensitivity.

The extracted turbulent diffuser geometry is meshed with the same wall-resolved triangular strategy as the laminar diffuser comparison in Table E.20, but with a distinct coarse–medium–fine hierarchy. The first-layer height is kept fixed at $\Delta y_1 = 1.50 \times 10^{-3}$ m to match the laminar-design diffuser meshes, while the bulk size, curvature angle, and inflation-layer distribution are refined independently for the turbulent topology.

TABLE E.18: Ansys Fluent mesh-independence table for the extracted turbulent diffuser benchmark geometry.

Parameter	Coarse	Medium	Fine
<i>Mesh settings</i>			
Mesh type	Tri.	Tri.	Tri.
Near-wall	WR infl.	WR infl.	WR ST infl.
h_{bulk} [m]	7.50×10^{-3}	5.00×10^{-3}	4.00×10^{-3}
Curvature angle	9°	6°	5°
Δy_1 [m]	1.50×10^{-3}	1.50×10^{-3}	1.50×10^{-3}
Inflation layers	10	14	16
Growth rate	1.11	1.09	1.08
t_{infl} [m]	2.51×10^{-2}	3.90×10^{-2}	4.55×10^{-2}
Nodes	8,104	16,166	21,238
Elements	12,330	24,570	40,268
<i>Validation metrics</i>			
Φ_D^{SA} [W/m]	1.2921×10^5	1.2966×10^5	1.2995×10^5
$\epsilon_{\text{rel}}(\Phi_D^{\text{SA}})$ [%]	–	0.35	0.22
$\Phi_D^{\text{SST } k-\omega}$ [W/m]	1.5296×10^4	1.5570×10^4	1.5725×10^4
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST } k-\omega})$ [%]	–	1.79	0.99
Static pressure drop Δp_s^{SA} [Pa]	1.2460×10^5	1.2449×10^5	1.2422×10^5
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SA}})$ [%]	–	0.09	0.22
Static pressure drop $\Delta p_s^{\text{SST } k-\omega}$ [Pa]	4.2412×10^4	4.2034×10^4	4.1833×10^4
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SST } k-\omega})$ [%]	–	0.89	0.48
Selected validation mesh	Fine_Ansys		
<i>Note.</i> WR = wall-resolved; ST = smooth-transition; Φ_D = viscous dissipation; Δp_s = static pressure drop. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.			

E.5.4 Turbulent-Design Wall-Shear Diagnostic

Figure E.11 summarizes the downstream wall-shear sign changes for the extracted pressure-outlet turbulent diffuser on the fine Ansys mesh. The diagnostic is included only to support the separated-flow interpretation in Chapter 6; the performance ranking itself is taken from the integrated pressure-drop and dissipation metrics.

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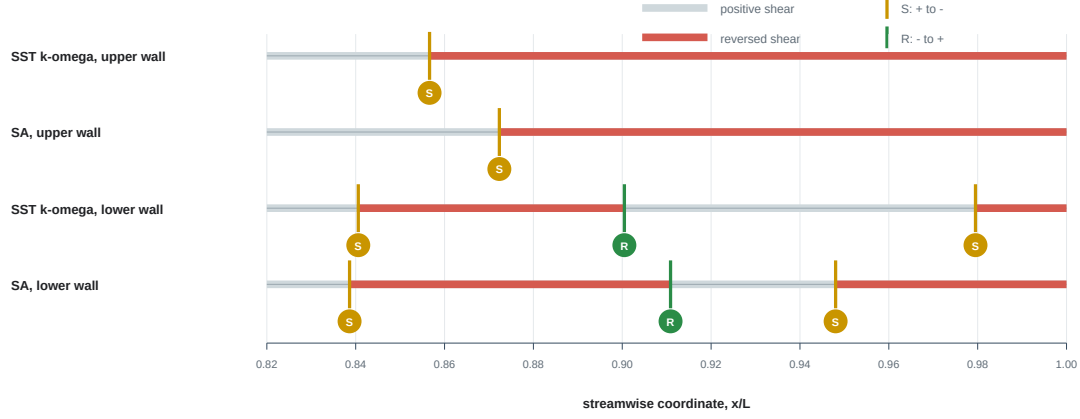
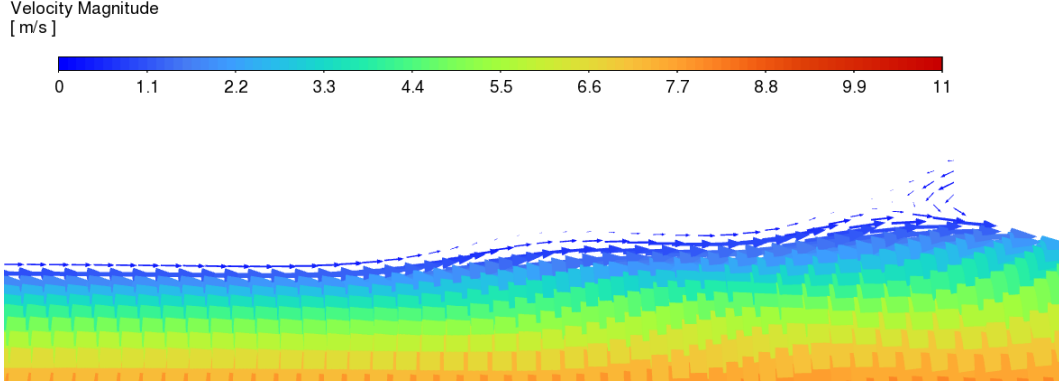


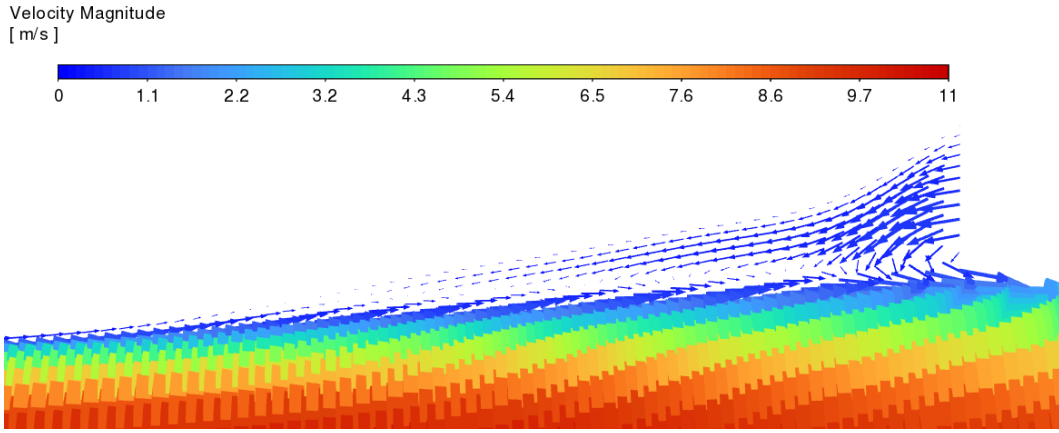
FIGURE E.11: Downstream wall-shear sign map for the extracted pressure-outlet turbulent diffuser. Reversed wall shear denotes $\tau_{w,x} < 0$. Interpolated locations: upper wall $x/L = 0.8566$ (SST $k - \omega$), 0.8723 (SA); lower wall $x/L = 0.8406$, 0.9005 , 0.9795 (SST $k - \omega$) and 0.8386 , 0.9109 , 0.9480 (SA). (ChatGPT, 2026).

E.5.5 Additional Diffuser Separation Diagnostic

Figure E.12 gives a close-up of the SST $k - \omega$ velocity vectors near the diffuser outlet for the extracted laminar and turbulent pressure-outlet designs. This figure is used as supporting visual evidence for the separated or recovering near-wall region discussed in Chapter 6; the wall-shear sign map in Figure E.11 remains the primary separation diagnostic because it directly indicates local flow reversal at the wall.



(A) Laminar pressure-outlet design.



(B) Turbulent pressure-outlet design.

FIGURE E.12: SST $k - \omega$ velocity-vector close-up near the outlet of the extracted pressure-outlet diffuser designs.

E.5.6 Laminar FEniCS Configuration Differences

The existing laminar diffuser reference is a completed FEniCS topology-optimization run from `Config_DiffuserYoon_Laminar.py`. It reuses the pressure-outlet turbulent diffuser setup in Tables E.14 and E.15 wherever not stated otherwise: the square $L = 1$ m domain, full-height parabolic inlet, center-third pressure outlet with $p = 0$ Pa, no prescribed outlet velocity, $V_f = 0.30$ volume constraint, dissipation objective J_D , Brinkman limits, mesh, filter radius, and projection threshold are unchanged. The laminar Reynolds number, solver settings, and continuation controls are listed in Table E.19.

E.5.7 Laminar FEniCS Optimization Diagnostics

Figure E.13 records the completed pressure-outlet laminar diffuser optimization history. The normalized objective drops rapidly during the early continuation stages and then remains nearly flat, while the volume-fraction residual stays close to the

E. TOPOLOGY-OPTIMIZATION BENCHMARK DETAILS

TABLE E.19: Laminar-specific differences from the pressure-outlet turbulent diffuser FEniCS setup above.

Parameter	Laminar setting
Flow model	Laminar incompressible Navier–Stokes; the SA working variable, SA wall-distance equation, and frozen-turbulence coupling are omitted
Reynolds number	$Re = 1$, based on $U_{\max,\text{in}} L \rho / \mu$ instead of the turbulent $Re = 3,000$ setup
Density and viscosity	$\rho = 1000 \text{ kg/m}^3$ and $\mu = 3000 \text{ Pas}$, with the same $U_{\max,\text{in}} = 3 \text{ m/s}$ as the turbulent pressure-outlet case; no outlet velocity is prescribed
Flow finite elements	Taylor–Hood P_2/P_1 for $(\mathbf{u}, p)$; no separate P_1 SA working-variable space is used
Forward and adjoint solves	SNES/MUMPS laminar Newton forward solves with method <code>newtontr</code> , relative tolerance 10^{-6} , absolute tolerance 10^{-9} , and maximum forward iteration count 80; the adjoint is solved as a linear system with MUMPS
Continuation and optimizer controls	Seven-stage schedule $q = \{0.01, 0.02, 0.04, 0.08, 0.08, 0.12, 0.20\}$, $\beta = \{1, 2, 4, 8, 8, 12, 16\}$, move limits $\{0.010, 0.008, 0.006, 0.004, 0.003, 0.002, 0.0015\}$, and inner-iteration budgets $\{12, 25, 45, 80, 120, 160, 220\}$; stop if objective change is below 10^{-6} for ten iterations

active constraint. This figure is kept in the appendix because it documents the laminar reference run rather than the turbulent performance comparison itself.

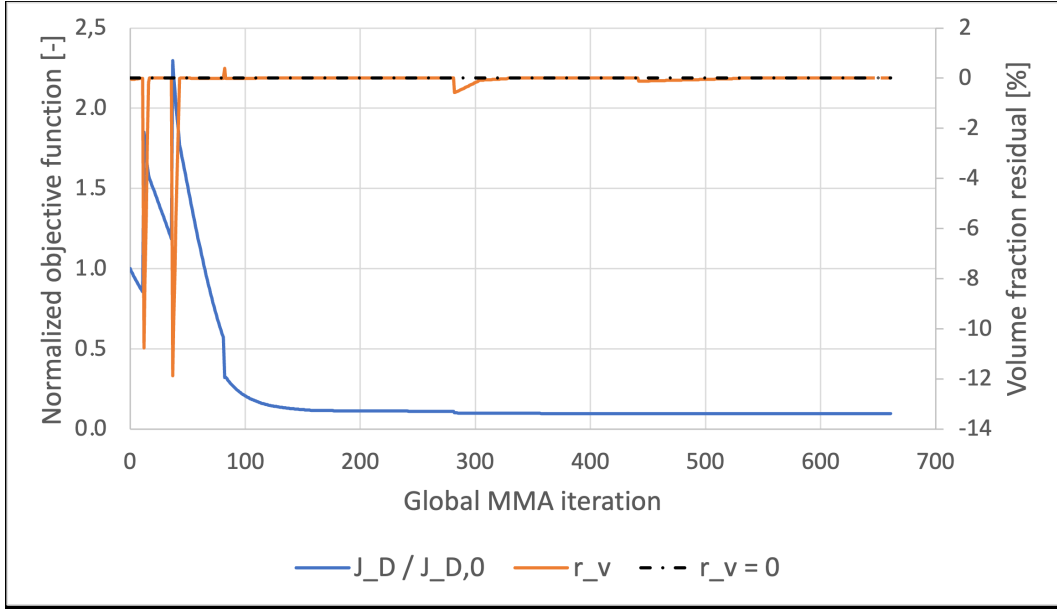


FIGURE E.13: FEniCS convergence history for the pressure-outlet laminar diffuser topology-optimization run, showing $J_D/J_{D,0}$ and the relative volume-fraction residual.

E.5.8 Laminar-Design Mesh Independence

Table E.20 reports the Ansys mesh-independence study for the extracted pressure-outlet laminar diffuser design. The turbulent diffuser mesh study is reported separately in Table E.18 of Section E.5. The Ansys Fluent solves use the same shifted inlet-profile and pressure-outlet convention as the turbulent diffuser validation setup.

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TABLE E.20: Ansys mesh-independence table for the extracted pressure-outlet laminar diffuser design.

Parameter	Coarse	Medium	Fine
<i>Mesh settings</i>			
Mesh type	Tri.	Tri.	Tri.
Near-wall	WR infl.	WR infl.	WR infl.
h_{bulk} [m]	1.20×10^{-2}	5.50×10^{-3}	3.75×10^{-3}
Curvature angle	15°	7°	5°
Δy_1 [m]	1.50×10^{-3}	1.50×10^{-3}	1.50×10^{-3}
Inflation layers	6	12	16
Growth rate	1.15	1.10	1.08
t_{infl} [m]	1.31×10^{-2}	3.19×10^{-2}	4.55×10^{-2}
Nodes	3,317	13,444	25,485
Elements	5,085	20,888	39,582
<i>Validation metrics</i>			
$\Phi_D^{\text{SST } k-\omega}$ [W/m]	1.2643×10^4	1.2983×10^4	1.3077×10^4
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST } k-\omega})$ [%]	–	2.69	0.72
Static pressure drop $\Delta p_s^{\text{SST } k-\omega}$ [Pa]	3.6628×10^4	3.6693×10^4	3.6464×10^4
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SST } k-\omega})$ [%]	–	0.18	0.62
Φ_D^{SA} [W/m]	1.4296×10^5	1.4315×10^5	1.4333×10^5
$\epsilon_{\text{rel}}(\Phi_D^{\text{SA}})$ [%]	–	0.13	0.13
Static pressure drop Δp_s^{SA} [Pa]	1.4191×10^5	1.4276×10^5	1.4200×10^5
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SA}})$ [%]	–	0.60	0.53
Selected comparison mesh	Fine_Ansys		

Note. WR = wall-resolved; Φ_D = viscous dissipation; Δp_s = static pressure drop. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change. The SA rows are re-evaluations for turbulence-model sensitivity.

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Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

Student name: Tiebert Lefebure

Student number: r0887630

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Course name: **Course number:**

This form is related to my ☐ **Bachelor's** or ☒ **Master's thesis**.

Title: Topology optimization of flow configurations operating in the turbulent regime

Supervisors: Prof. Dr. Ir. Niels Horsten & Prof. Dr. Ir. Maarten Blommaert

Daily supervisor: Esteban Foronda Obando

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